**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD**
**CENTRAL VALLEY REGION**

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# ORDER R5-2012-0083
# NPDES NO. CA0078930

# WASTE DISCHARGE REQUIREMENTS FOR THE CITY OF BIGGS BIGGS WASTEWATER TREATMENT PLANT BUTTE COUNTY

The following Discharger is subject to waste discharge requirements as set forth in this Order:

**Table 1. Discharger Information**

| | |
|---|---|
| **Discharger** | **City of Biggs** |
| **Name of Facility** | **Biggs Wastewater Treatment Plant** |
| **Facility Address** | **3016 Sixth Street** |
| | **Biggs, CA 95917** |
| | **Butte County** |
| The U.S. Environmental Protection Agency (USEPA) and the Regional Water Quality Control Board have classified this discharge as a **minor** discharge. | |

The discharge by the City of Biggs from the discharge points identified below is subject to waste discharge requirements as set forth in this Order:

**Table 2. Discharge Location**

| Discharge Point | Effluent Description | Discharge Point Latitude | Discharge Point Longitude | Receiving Water |
|---|---|---|---|---|
| EFF-001 | Secondary treated municipal wastewater | 39 º 24' 28" N | 121 º 43' 32" W | Lateral K (agricultural drain – Reclamation District #833) |

**Table 3. Administrative Information**

| | |
|---|---|
| This Order was adopted by the Regional Water Quality Control Board on: | **4 October 2012** |
| This Order shall become effective on: | **23 November 2012** |
| This Order shall expire on: | **1 November 2017** |
| The Discharger shall file a Report of Waste Discharge in accordance with title 23, California Code of Regulations, as application for issuance of new waste discharge requirements no later than: | **180 days prior to the Order expiration date OR 5 May 2017** |

I, **Pamela C. Creedon**, Executive Officer, do hereby certify that this Order with all attachments is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 4 October 2012.

*Original signed by*

**PAMELA C. CREEDON**, Executive Officer

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# I. FACILITY INFORMATION

The following Discharger is subject to waste discharge requirements as set forth in this Order:

**Table 4. Facility Information**

| | |
|---|---|
| **Discharger** | **City of Biggs** |
| **Name of Facility** | **Biggs Wastewater Treatment Facility** |
| **Facility Address** | **3016 Sixth Street** |
| | **Biggs, CA 95917** |
| | **Butte County** |
| **Facility Contact, Title, and Phone** | **Mr. Peter Carr, City Administrator (530) 868-0100** |
| **Mailing Address** | **P.O. Box 307, Biggs, CA 95917** |
| **Type of Facility** | **Publicly Owned Treatment Works (POTW)** |
| **Facility Design Flow** | **Dry weather flow = 0.38 million gallons per day (mgd), Peak wet weather flow = 1.0 mgd** |

# II. FINDINGS

The California Regional Water Quality Control Board, Central Valley Region (hereinafter Regional Water Board), finds:

**A. Background.** The City of Biggs (hereinafter Discharger) is currently discharging treated wastewater pursuant to Order No. **R5-2007-0032** and National Pollutant Discharge Elimination System (NPDES) Permit No. **CA0078930**. The Discharger submitted a Report of Waste Discharge, dated 20 December 2011, for renewal of Waste Discharge Requirements and continue discharging up to 0.38 million gallons per day (mgd) of treated wastewater; however the application was deemed incomplete on 30 December 2011. Supplemental information was submitted on **25 January 2012,** and the application was deemed complete on **3 February 2012.**

For the purposes of this Order, references to the “discharger” or “permittee” in applicable federal and state laws, regulations, plans, or policy are held to be equivalent to references to the Discharger herein.

**B. Facility Description**. The Discharger owns and operates the City of Biggs Wastewater Treatment Plant. The treatment system consists of two aerated lagoons, a ballast pond, three plug-flow rock filters in parallel, and chlorination/dechlorination facilities. Wastewater is discharged from Discharge Point No. EFF-001 to the Lateral K agricultural drain (Reclamation District #833). Lateral K was constructed to drain or convey excess agricultural flows away from fields. Lateral K flows into the Cherokee Canal and ultimately may flow to Butte Creek near the Butte Sink Wetland. Butte Creek is a water of the United States, and is tributary to the Sacramento River within Butte Basin Hydrologic Area. Attachment B provides a map of the area around the Facility. Attachment C provides a flow schematic of the Facility.

**C. Legal Authorities.** This Order is issued pursuant to section 402 of the Clean Water Act (CWA) and implementing regulations adopted by USEPA and chapter 5.5, division 7 of the California Water Code (CWC; commencing with section 13370). It shall serve as a NPDES permit for point source discharges from this facility to surface waters. This Order also serves as Waste Discharge Requirements (WDRs) pursuant to article 4, chapter 4, division 7 of the CWC (commencing with section 13260).

**D. Background and Rationale for Requirements.** The Regional Water Board developed the requirements in this Order based on information submitted as part of the application, through monitoring and reporting programs, and other available information. The Fact Sheet (Attachment F), which contains background information and rationale for Order requirements, is hereby incorporated into this Order and constitutes part of the Findings for this Order. Attachments A through E and G through J are also incorporated into this Order.

**E. California Environmental Quality Act (CEQA).** Under CWC section 13389, this action to adopt an NPDES permit is exempt from the provisions of CEQA, Public Resources Code sections 21100-21177.

**F. Technology-based Effluent Limitations.** Section 301(b) of the CWA and implementing USEPA permit regulations at section 122.44, title 40 of the Code of Federal Regulations (40 CFR 122.44), require that permits include conditions meeting applicable technology-based requirements at a minimum, and any more stringent effluent limitations necessary to meet applicable water quality standards. The discharge authorized by this Order must meet minimum federal technology-based requirements based on Secondary Treatment Standards at 40 CFR Part 133 AND/OR Effluent Limitations Guidelines and Standards for Alternative State Requirements Category in 40 CFR Part 133.105 AND/OR Best Professional Judgment (BPJ) in accordance with 40 CFR 125.3. A detailed discussion of the technology-based effluent limitations development is included in the Fact Sheet.

**G. Water Quality-based Effluent Limitations (WQBELs).** Section 301(b) of the CWA and 40 CFR 122.44(d) require that permits include limitations more stringent than applicable federal technology-based requirements where necessary to achieve applicable water quality standards.

40 CFR 122.44(d)(1)(i) mandates that permits include effluent limitations for all pollutants that are or may be discharged at levels that have the reasonable potential to cause or contribute to an exceedance of a water quality standard, including numeric and narrative objectives within a standard. Where reasonable potential has been established for a pollutant, but there is no numeric criterion or objective for the pollutant, WQBELs must be established using: (1) USEPA criteria guidance under CWA section 304(a), supplemented where necessary by other relevant information; (2) an indicator parameter for the pollutant of concern; or (3) a calculated numeric water quality criterion, such as a proposed state criterion or policy interpreting the state's narrative criterion, supplemented with other relevant information, as provided in 40 CFR 122.44(d)(1)(vi).

**H. Water Quality Control Plans.** The Regional Water Board adopted a *Water Quality Control Plan, Fourth Edition (Revised October 2011)*, for the Sacramento and San Joaquin River Basins (hereinafter Basin Plan) that designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve those objectives for all waters addressed through the plan. The Basin Plan at page II-2.00 that the "…*beneficial uses of any specifically identified water body generally apply to its tributary streams*." The Basin Plan does not specifically identify beneficial uses for Lateral K, but does identify present and potential uses for Sacramento River to which Lateral K via Cherokee Canal via Butte Creek is tributary. Lateral K intersects the 833 main line running south, which intersects the Cherokee Canal running southwest, which enters Butte Creek in Sutter County.

The beneficial uses for Butte Creek are specifically identified in the Sacramento-San Joaquin Basin Plan as shown in Table 5 below. The beneficial uses of Lateral K, Main Canal, and Cherokee Canal constructed agricultural drains, are based on demonstrated existing uses, as also shown in Table 5 below. In addition, the Basin Plan implements State Water Resources Control Board (State Water Board) Resolution No. 88-63, establishing the municipal or domestic supply (MUN) use on Lateral K, Main Canal and the Cherokee Canal. Butte Creek is specifically identified in the Basin Plan to not have the MUN use. The applicable beneficial uses for the receiving waters are as follows:

**Table 5. Basin Plan Beneficial Uses**

| Discharge Point | Receiving Water Name | Beneficial Use(s) |
|---|---|---|
| EFF-001 | Lateral K, Main Canal (Reclamation District #833) and Cherokee Canal (agricultural drains) | ***Surface Water***<br>**Constructed Agricultural Drain:**<br>Agricultural supply, including irrigation and stock watering (AGR);<br><br>Water Contact Recreation and Non-Contact Water Recreation (REC-1) & (REC-2);<br>Warm and cold freshwater habitat (WARM) & (COLD);<br>Cold water migration (MIGR);<br>Warm water spawning (SPWN); and<br>Wildlife habitat (WILD).<br><br>**State Water Board Resolution No. 88-63:**<br>Municipal and domestic supply (MUN) |
| | Butte Creek | **Basin Plan, including Table II-1:**<br>***Surface Water:***<br>Agricultural supply, including irrigation and stock watering (AGR);<br>Water Contact Recreation (REC-1);<br>Warm and cold freshwater habitat (WARM) & (COLD);<br>Cold water migration (MIGR);<br>Warm water spawning (SPWN); and<br>Wildlife habitat (WILD).<br><br>***Groundwater:***<br>Municipal and domestic water supply (MUN);<br>Industrial service supply (IND),<br>Industrial process supply (PRO), and<br>Agricultural supply (AGR). |

The Basin Plan implements State Water Resources Control Board (State Water Board) Resolution No. 88-63, which established state policy that all waters, with certain exceptions, should be considered suitable or potentially suitable for municipal and domestic supply. One exception is if the water is in systems designed or modified for the primary purpose of conveying or holding agricultural drainage waters, provided that the discharge from such systems is monitored to assure compliance with all relevant water quality objectives as required by the Central Valley Water Boards. In accordance with Chapter IV of the Basin Plan, the Central Valley Water Board must adopt a formal Basin Plan Amendment to grant an exception to Resolution No. 88-63. Therefore, until the Central Valley Water Board adopts a Basin Plan Amendment for an exception, and the State Water Board and Office of Administrative Law approve the Basin Plan Amendment, Lateral K, Main Canal, and Cherokee Canal are considered to be suitable or potentially suitable for municipal or domestic supply in accordance with State Water Board Resolution No. 88-63.

**I. National Toxics Rule (NTR) and California Toxics Rule (CTR).** USEPA adopted the NTR on 22 December 1992, and later amended it on 4 May 1995 and 9 November 1999. About 40 criteria in the NTR applied in California. On 18 May 2000, USEPA adopted the CTR. The CTR promulgated new toxics criteria for California and, in addition, incorporated the previously adopted NTR criteria that were applicable in the state. The CTR was amended on 13 February 2001. These rules contain water quality criteria for priority pollutants.

**J. State Implementation Policy.** On 2 March 2000, the State Water Board adopted the *Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California* (State Implementation Policy or SIP). The SIP became effective on 28 April 2000 with respect to the priority pollutant criteria promulgated for California by USEPA through the NTR and to the priority pollutant objectives established by the Regional Water Board in the Basin Plan. The SIP became effective on 18 May 2000 with respect to the priority pollutant criteria promulgated by USEPA through the CTR. The State Water Board adopted amendments to the SIP on 24 February 2005 that became effective on 13 July 2005. The SIP establishes implementation provisions for priority pollutant criteria and objectives and provisions for chronic toxicity control. Requirements of this Order implement the SIP.

**K. Compliance Schedules and Interim Requirements.** In general, an NPDES permit must include final effluent limitations that are consistent with CWA section 301 and with 40 CFR 122.44(d). There are exceptions to this general rule. The State Water Board's *Policy for Compliance Schedules in National Pollutant Discharge Elimination System Permits* (Compliance Schedule Policy) allows compliance schedules for new, revised, or newly interpreted water quality objectives or criteria, or in accordance with a TMDL. All compliance schedules must be as short as possible, and may not exceed ten years from the effective date of the adoption, revision, or new interpretation of the applicable water quality objective or criterion, unless a TMDL allows a longer schedule. The Regional Water Board, however, is not required to include a compliance schedule, but may issue a Time Schedule Order pursuant to CWC section 13300 or a Cease and Desist Order pursuant to CWC section 13301 where it finds that the discharger is violating or threatening to violate the permit. The Regional Water Board will consider the merits of each case in determining whether it is appropriate to include a compliance schedule in a permit, and, consistent with the Compliance Schedule Policy, should consider feasibility of achieving compliance, and must impose a schedule that is as short as possible to achieve compliance with the effluent limit based on the objective or criteria.

The Compliance Schedule Policy and the SIP do not allow compliance schedules for priority pollutants beyond 18 May 2010, except for new or more stringent priority pollutant criteria adopted by USEPA after 17 December 2008.

Where a compliance schedule for a final effluent limitation exceeds one year, the Order must include interim numeric limitations for that constituent or parameter, interim milestones and compliance reporting within 14 days after each interim milestone. The permit may also include interim requirements to control the pollutant, such as pollutant minimization and source control measures. This Order does not include any compliance schedules or interim effluent limitations

**L. Alaska Rule.** On 30 March 2000, USEPA revised its regulation that specifies when new and revised state and tribal water quality standards become effective for CWA purposes. (40 CFR 131.21 and 65 FR 24641 (27 April 2000).) Under the revised regulation (also known as the Alaska rule), new and revised standards submitted to USEPA after 30 May 2000, must be approved by USEPA before being used for CWA purposes. The final rule also provides that standards already in effect and submitted to USEPA by 30 May 2000 may be used for CWA purposes, whether or not approved by USEPA.

**M. Stringency of Requirements for Individual Pollutants.** This Order contains both technology-based effluent limitations and WQBELs for individual pollutants. The technology-based effluent limitations consist of restrictions on electrical conductivity, biochemical oxygen demand and total suspended solids. The WQBELs consist of restrictions on pathogens, ammonia, copper, pH and total chlorine residual. This Order's technology-based pollutant restrictions implement the minimum, applicable federal technology-based requirements. In addition, this Order includes new effluent limitations for ammonia and copper to meet numeric objectives or protect beneficial uses.

WQBELs have been scientifically derived to implement water quality objectives that protect beneficial uses. Both the beneficial uses and the water quality objectives have been approved pursuant to federal law and are the applicable federal water quality standards. To the extent that toxic pollutant WQBELs were derived from the CTR, the CTR is the applicable standard pursuant to 40 CFR 131.38. The scientific procedures for calculating the individual WQBELs for priority pollutants are based on the CTR-SIP, which was approved by USEPA on 18 May 2000. All beneficial uses and water quality objectives contained in the Basin Plan were approved under state law and submitted to and approved by USEPA prior to 30 May 2000. Any water quality objectives and beneficial uses submitted to USEPA prior to 30 May 2000, but not approved by USEPA before that date, are nonetheless "*applicable water quality standards for purposes of the [Clean Water] Act*" pursuant to 40 CFR 131.21(c)(1). Collectively, this Order's restrictions on individual pollutants are no more stringent than required to implement the technology-based requirements of the CWA and the applicable water quality standards for purposes of the CWA.

**N. Antidegradation Policy**. 40 CFR 131.12 requires that the state water quality standards include an antidegradation policy consistent with the federal policy. The State Water Board established California's antidegradation policy in State Water Board Resolution No. 68-16. Resolution No. 68-16 incorporates the federal antidegradation policy where the federal policy applies under federal law. Resolution No. 68-16 requires that existing quality of waters be maintained unless degradation is justified based on specific findings. The Regional Water Board's Basin Plan implements, and incorporates by reference, both the state and federal antidegradation policies. As discussed in detail in the Fact Sheet, the permitted discharge is consistent with the antidegradation provision of 40 CFR 131.12 and Resolution No. 68-16.

**O. Anti-Backsliding Requirements.** Sections 303(d)(4) and 402(o)(2) of the CWA and federal regulations at 40 CFR 122.44(l) prohibit backsliding in NPDES permits. These anti-backsliding provisions require effluent limitations in a reissued permit to be as stringent as those in the previous permit, with some exceptions.

**P. Endangered Species Act.** This Order does not authorize any act that results in the taking of a threatened or endangered species or any act that is now prohibited, or becomes prohibited in the future, under either the California Endangered Species Act (Fish and Game Code sections 2050 to 2097) or the Federal Endangered Species Act (16 U.S.C.A. sections 1531 to 1544). This Order requires compliance with effluent limits, receiving water limits, and other requirements to protect the beneficial uses of waters of the state. The discharger is responsible for meeting all requirements of the applicable Endangered Species Act.

**Q. Monitoring and Reporting.** 40 CFR 122.48 requires that all NPDES permits specify requirements for recording and reporting monitoring results. CWC sections 13267 and 13383 authorize the Regional Water Board to require technical and monitoring reports. The Monitoring and Reporting Program establishes monitoring and reporting requirements to implement federal and State requirements. The Monitoring and Reporting Program is provided in Attachment E.

**R. Standard and Special Provisions.** Standard Provisions, which apply to all NPDES permits in accordance with 40 CFR 122.41, and additional conditions applicable to specified categories of permits in accordance with 40 CFR 122.42, are provided in Attachment D. The discharger must comply with all standard provisions and with those additional conditions that are applicable under 40 CFR 122.42. The Regional Water Board has also included in this Order special provisions applicable to the Discharger. A rationale for the special provisions contained in this Order is provided in the Fact Sheet.

**S. Provisions and Requirements Implementing State Law.** The provisions/requirements in sections IV.B, IV.C, V.B, and VI.C of this Order are included to implement state law only. These provisions/requirements are not required or authorized under the federal CWA; consequently, violations of these provisions/requirements are not subject to the enforcement remedies that are available for NPDES violations.

**T. Notification of Interested Parties.** The Regional Water Board has notified the Discharger and interested agencies and persons of its intent to prescribe WDRs for the discharge and has provided them with an opportunity to submit their written comments and recommendations. Details of notification are provided in the Fact Sheet of this Order.

**U. Consideration of Public Comment.** The Regional Water Board, in a public meeting, heard and considered all comments pertaining to the discharge. Details of the Public Hearing are provided in the Fact Sheet.

THEREFORE, IT IS HEREBY ORDERED, that Order No. R5-2007-0032 is rescinded upon the effective date of this Order except for enforcement purposes, and, in order to meet the provisions contained in division 7 of the CWC (commencing with section 13000) and regulations adopted thereunder, and the provisions of the federal CWA and regulations and guidelines adopted thereunder, the Discharger shall comply with the requirements in this Order.

## III. DISCHARGE PROHIBITIONS

**A.** Discharge of wastewater at a location or in a manner different from that described in the Findings is prohibited.

**B.** The by-pass or overflow of wastes to surface waters is prohibited, except as allowed by Federal Standard Provisions I.G. and I.H. (Attachment D).

**C.** Neither the discharge nor its treatment shall create a nuisance as defined in section 13050 of the CWC.

**D.** The Discharger shall not allow pollutant-free wastewater to be discharged into the collection, treatment, and disposal system in amounts that significantly diminish the system's capability to comply with this Order. Pollutant-free wastewater means rainfall, groundwater, cooling waters, and condensates that are essentially free of pollutants.

## IV. EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

### A. Effluent Limitations – Discharge Point No. EFF-001

#### 1. Final Effluent Limitations – Discharge Point No. EFF-001

**a.** The Discharger shall maintain compliance with the following effluent limitations at Discharge Point No. EFF-001 with compliance measured at Monitoring Location EFF-001 as described in the Monitoring and Reporting Program:

**Table 6. Effluent Limitations**

| Parameter | Units | Effluent Limitations | | | | |
|---|---|---|---|---|---|---|
| | | Average Monthly | Average Weekly | Maximum Daily | Instantaneous Minimum | Instantaneous Maximum |
| Flow | MGD | 0.38 | | | | |
| Biochemical Oxygen Demand 5-day @ 20°C | mg/L | 30 | 45 | 60 | | |
| | lbs/day[1] | 95 | 143 | 190 | | |
| Total Suspended Solids | mg/L | 30 | 45 | 60 | | |
| | lbs/day[1] | 95 | 143 | 190 | | |
| pH | standard units | | | | 6.5 | 8.5 |
| Ammonia | mg/L | 1.23 | | 2.15 | | |

| Parameter | Units | Effluent Limitations | | | | |
|---|---|---|---|---|---|---|
| | | Average Monthly | Average Weekly | Maximum Daily | Instantaneous Minimum | Instantaneous Maximum |
| Electrical Conductivity (25° C) | µmhos/cm | 900 **annual average** | | | | |
| Copper, Total Recoverable | µg/L | 15.1 | | 28.0 | | |

[1]**Based on design treatment capacity of 0.38 mgd**

b. **Percent Removal.** The average monthly percent removal of 5-day biochemical oxygen demand ($BOD_5$) and total suspended solids (TSS) shall not be less than 65 percent.

c. **Acute Whole Effluent Toxicity.** Survival of aquatic organisms in 96-hour bioassays of undiluted waste shall be no less than:

**i.** 70%, minimum for any one bioassay; and
**ii.** 90%, median for any three consecutive bioassays.

d. **Total Residual Chlorine.** Effluent total residual chlorine shall not exceed:

**i.** 0.011 mg/L, as a 4-day average; and
**ii.** 0.019 mg/L, as a 1-hour average.

e. **Total Coliform Organisms.** When the wastewater receives dilution of less than 20:1, effluent total coliform organisms shall not exceed:

**i.** 2.2 most probable number (MPN) per 100 mL, as a 7-day median;
**ii.** 23 MPN/100mL no more than once in any 30-day period; and
**iii.** 240 MPN/100 mL, as an instantaneous maximum.

When the wastewater receives dilution of 20:1 or more, effluent total coliform organisms shall not exceed:

**i.** 23 MPN/100mL, as a 7-day median, and
**ii.** 240 MPN/100 mL, no more than once in any 30-day period.

f. **Average Dry Weather Flow.** The average dry weather discharge flow shall not exceed **0.38 MGD**

**B. Land Discharge Specifications – Not Applicable**

**C. Reclamation Specifications – Not Applicable**

**1.** All uses of reclaimed water shall be in accordance with a Master Reclamation Permit issued in accordance with Title 22 and the CWC.

# V. RECEIVING WATER LIMITATIONS

## A. Surface Water Limitations

Receiving water limitations are based on water quality objectives contained in the Basin Plan and are a required part of this Order. The discharge shall not cause the following in Lateral K Agricultural Drain.

1. **Bacteria.** The fecal coliform concentration, based on a minimum of not less than five samples for any 30-day period, to exceed a geometric mean of 200 MPN/100 mL, nor more than 10 percent of the total number of fecal coliform samples taken during any 30-day period to exceed 400 MPN/100 mL.
2. **Biostimulatory Substances.** Water to contain biostimulatory substances which promote aquatic growths in concentrations that cause nuisance or adversely affect beneficial uses.
3. **Chemical Constituents.** Chemical constituents to be present in concentrations that adversely affect beneficial uses.
4. **Color.** Discoloration that causes nuisance or adversely affects beneficial uses.
5. **Dissolved Oxygen:**
    - **a.** The 95 percentile dissolved oxygen concentration to fall below 75 percent of saturation; nor
    - **b.** The dissolved oxygen concentration to be reduced below 5.0 mg/L at any time.
6. **Floating Material.** Floating material to be present in amounts that cause nuisance or adversely affect beneficial uses.
7. **Oil and Grease.** Oils, greases, waxes, or other materials to be present in concentrations that cause nuisance, result in a visible film or coating on the surface of the water or on objects in the water, or otherwise adversely affect beneficial uses.
8. **pH.** The pH to be depressed below 6.5 nor raised above 8.5.
9. **Pesticides:**
    - **a.** Pesticides to be present, individually or in combination, in concentrations that adversely affect beneficial uses;
    - **b.** Pesticides to be present in bottom sediments or aquatic life in concentrations that adversely affect beneficial uses;
    - **c.** Total identifiable persistent chlorinated hydrocarbon pesticides to be present in the water column at concentrations detectable within the accuracy of

analytical methods approved by USEPA or the Executive Officer prescribed in Standard Methods for the Examination of Water and Wastewater, 18th Edition;

**d.** Pesticide concentrations to exceed those allowable by applicable antidegradation policies (see State Water Board Resolution No. 68-16 and 40 CFR 131.12.);

**e.** Pesticide concentrations to exceed the lowest levels technically and economically achievable;

**f.** Pesticides to be present in concentration in excess of the maximum contaminant levels set forth in CCR, Title 22, division 4, chapter 15; nor

**g.** Thiobencarb to be present in excess of 1.0 µg/L.

**10. Radioactivity:**

**a.** Radionuclides to be present in concentrations that are harmful to human, plant, animal, or aquatic life nor that result in the accumulation of radionuclides in the food web to an extent that presents a hazard to human, plant, animal, or aquatic life.

**b.** Radionuclides to be present in excess of the maximum contaminant levels specified in Table 4 (MCL Radioactivity) of section 64443 of Title 22 of the California Code of Regulations.

**11. Suspended Sediments.** The suspended sediment load and suspended sediment discharge rate of surface waters to be altered in such a manner as to cause nuisance or adversely affect beneficial uses.

**12. Settleable Substances.** Substances to be present in concentrations that result in the deposition of material that causes nuisance or adversely affects beneficial uses.

**13. Suspended Material.** Suspended material to be present in concentrations that cause nuisance or adversely affect beneficial uses.

**14. Taste and Odors.** Taste- or odor-producing substances to be present in concentrations that impart undesirable tastes or odors to fish flesh or other edible products of aquatic origin, or that cause nuisance, or otherwise adversely affect beneficial uses.

**15. Temperature.** The natural temperature to be increased by more than 5°F.

**16. Toxicity.** Toxic substances to be present, individually or in combination, in concentrations that produce detrimental physiological responses in human, plant, animal, or aquatic life.

**17. Turbidity.**

**a.** Shall not exceed 2 Nephelometric Turbidity Units (NTU) where natural turbidity is less than 1 NTU;

**b.** Shall not increase more than 1 NTU where natural turbidity is between 1 and 5 NTUs;

**c.** Shall not increase more than 20 percent where natural turbidity is between 5 and 50 NTUs;

**d.** Shall not increase more than 10 NTU where natural turbidity is between 50 and 100 NTUs; nor

**e.** Shall not increase more than 10 percent where natural turbidity is greater than 100 NTUs.

## B. Groundwater Limitations

**1.** Release of waste constituents from any portion of the Facility shall not cause or contribute to, in combination with other sources of waste constituents, groundwater within influence of the Facility to contain:

**a.** Taste or odor-producing constituents, toxic substances, or any other constituents, in concentrations that cause nuisance or adversely affect beneficial uses;

**b.** Waste constituent concentrations in excess of water quality objectives or background water quality, whichever is greater;

**c.** Waste constituent concentrations in excess of the concentrations specified below or background water quality, whichever is greater:

**i.** The most probable number of coliform organisms over any seven-day period shall be less than 2.2 MPN/100.

**ii.** Nitrate plus nitrite (as N) shall not exceed 10 mg/L.

**d.** Exhibit a pH of less than 6.5 or greater than 8.5 pH units.

# VI. PROVISIONS

## A. Standard Provisions

**1.** The Discharger shall comply with all (federal NPDES standard conditions from 40 CFR Part 122) Standard Provisions included in Attachment D of this Order.

**2.** The Discharger shall comply with the following provisions:

**a.** If the Discharger's wastewater treatment plant is publicly owned or subject to regulation by California Public Utilities Commission, it shall be supervised and operated by persons possessing certificates of appropriate grade according to Title 23, CCR, division 3, chapter 26.

**b.** After notice and opportunity for a hearing, this Order may be terminated or modified for cause, including, but not limited to:

**i.** violation of any term or condition contained in this Order;

**ii.** obtaining this Order by misrepresentation or by failing to disclose fully all relevant facts;

**iii.** a change in any condition that requires either a temporary or permanent reduction or elimination of the authorized discharge; and

**iv.** a material change in the character, location, or volume of discharge.

The causes for modification include:

- *New regulations.* New regulations have been promulgated under section 405(d) of the CWA, or the standards or regulations on which the permit was based have been changed by promulgation of amended standards or regulations or by judicial decision after the permit was issued.

- *Land application plans.* When required by a permit condition to incorporate a land application plan for beneficial reuse of sewage sludge, to revise an existing land application plan, or to add a land application plan.

- *Change in sludge use or disposal practice.* Under 40 CFR 122.62(a)(1), a change in the Discharger's sludge use or disposal practice is a cause for modification of the permit. It is cause for revocation and reissuance if the Discharger requests or agrees.

The Regional Water Board may review and revise this Order at any time upon application of any affected person or the Regional Water Board's own motion.

**c.** If a toxic effluent standard or prohibition (including any scheduled compliance specified in such effluent standard or prohibition) is established under section 307(a) of the CWA, or amendments thereto, for a toxic pollutant that is present in the discharge authorized herein, and such standard or prohibition is more stringent than any limitation upon such pollutant in this Order, the Regional Water Board will revise or modify this Order in accordance with such toxic effluent standard or prohibition.

The Discharger shall comply with effluent standards and prohibitions within the time provided in the regulations that establish those standards or prohibitions, even if this Order has not yet been modified.

**d.** This Order shall be modified, or alternately revoked and reissued, to comply with any applicable effluent standard or limitation issued or approved under sections 301(b)(2)(C) and (D), 304(b)(2), and 307(a)(2) of the CWA, if the effluent standard or limitation so issued or approved:

**i.** contains different conditions or is otherwise more stringent than any effluent limitation in the Order; or

**ii.** controls any pollutant limited in the Order.

The Order, as modified or reissued under this paragraph, shall also contain any other requirements of the CWA then applicable.

**e.** The provisions of this Order are severable. If any provision of this Order is found invalid, the remainder of this Order shall not be affected.

**f.** The Discharger shall take all reasonable steps to minimize any adverse effects to waters of the State or users of those waters resulting from any discharge or sludge use or disposal in violation of this Order. Reasonable steps shall include such accelerated or additional monitoring as necessary to determine the nature and impact of the non-complying discharge or sludge use or disposal.

**g.** The Discharger shall ensure compliance with any existing or future pretreatment standard promulgated by USEPA under section 307 of the CWA, or amendment thereto, for any discharge to the municipal system.

**h.** A copy of this Order shall be maintained at the discharge facility and be available at all times to operating personnel. Key operating personnel shall be familiar with its content.

**i.** Safeguard to electric power failure:

**i.** The Discharger shall provide safeguards to assure that, should there be reduction, loss, or failure of electric power, the discharge shall comply with the terms and conditions of this Order.

**ii.** Upon written request by the Regional Water Board the Discharger shall submit a written description of safeguards. Such safeguards may include alternate power sources, standby generators, retention capacity, operating procedures, or other means. A description of the safeguards provided shall include an analysis of the frequency, duration, and impact of power failures experienced over the past 5 years on effluent quality and on the capability of the Discharger to comply with the terms and conditions of the Order. The adequacy of the safeguards is subject to the approval of the Regional Water Board.

**iii.** Should the treatment works not include safeguards against reduction, loss, or failure of electric power, or should the Regional Water Board not approve the existing safeguards, the Discharger shall, within 90 days of having been advised in writing by the Regional Water Board that the existing safeguards are inadequate, provide to the Regional Water Board and USEPA a schedule of compliance for providing safeguards such that in the event of reduction, loss, or failure of electric power, the Discharger shall comply with the terms and conditions of this Order. The schedule of compliance shall, upon approval of the Regional Water Board, become a condition of this Order.

**j.** The Discharger, upon written request of the Regional Water Board, shall file with the Board a technical report on its preventive (failsafe) and contingency (cleanup) plans for controlling accidental discharges, and for minimizing the effect of such events. This report may be combined with that required under Regional Water Board Standard Provision contained in section VI.A.2.i. of this Order.

The technical report shall:

**i.** Identify the possible sources of spills, leaks, untreated waste by-pass, and contaminated drainage. Loading and storage areas, power outage, waste treatment unit outage, and failure of process equipment, tanks and pipes should be considered.

**ii.** Evaluate the effectiveness of present facilities and procedures and state when they became operational.

**iii.** Predict the effectiveness of the proposed facilities and procedures and provide an implementation schedule containing interim and final dates when they will be constructed, implemented, or operational.

The Regional Water Board, after review of the technical report, may establish conditions which it deems necessary to control accidental discharges and to minimize the effects of such events. Such conditions shall be incorporated as part of this Order, upon notice to the Discharger.

**k.** A publicly owned treatment works whose waste flow has been increasing, or is projected to increase, shall estimate when flows will reach hydraulic and treatment capacities of its treatment and disposal facilities. The projections shall be made in January, based on the last 3 years' average dry weather flows, peak wet weather flows and total annual flows, as appropriate. When any projection shows that capacity of any part of the facilities may be exceeded in 4 years, the Discharger shall notify the Regional Water Board by 31 January. A copy of the notification shall be sent to appropriate local elected officials, local permitting agencies and the press. Within 120 days of the notification, the Discharger shall submit a technical report showing how it will prevent flow volumes from exceeding capacity or how it will increase capacity to handle the larger flows. The Regional Water Board may extend the time for submitting the report.

**l.** The Discharger shall submit technical reports as directed by the Executive Officer. All technical reports required herein that involve planning, investigation, evaluation, or design, or other work requiring interpretation and proper application of engineering or geologic sciences, shall be prepared by or under the direction of persons registered to practice in California pursuant to California Business and Professions Code, sections 6735, 7835, and 7835.1. To demonstrate compliance with Title 16, CCR, sections 415 and 3065, all technical reports must contain a statement of the qualifications of the responsible registered professional(s). As required by these laws, completed technical reports must bear the signature(s) and seal(s) of the registered professional(s) in a manner such that all work can be clearly attributed to the professional responsible for the work.

**m.** The Regional Water Board is authorized to enforce the terms of this permit under several provisions of the CWC, including, but not limited to, sections 13385, 13386, and 13387.

**n.** For publicly owned treatment works, prior to making any change in the point of discharge, place of use, or purpose of use of treated wastewater that results in a decrease of flow in any portion of a watercourse, the Discharger must file a petition with the State Water Board, Division of Water Rights, and receive approval for such a change. (CWC section 1211).

**o.** In the event the Discharger does not comply or will be unable to comply for any reason, with any prohibition, maximum daily effluent limitation, 1-hour average effluent limitation, or receiving water limitation contained in this Order, the Discharger shall notify the Regional Water Board by telephone (916) 464-3291 within 24 hours of having knowledge of such noncompliance, and shall confirm this notification in writing within 5 days, unless the Regional Water Board waives confirmation. The written notification shall include the information required by the Standard Provision contained in Attachment D section V.E.1. [40 CFR 122.41(l)(6)(i)].

**p.** Failure to comply with provisions or requirements of this Order, or violation of other applicable laws or regulations governing discharges from this facility, may subject the Discharger to administrative or civil liabilities, criminal penalties, and/or other enforcement remedies to ensure compliance. Additionally, certain violations may subject the Discharger to civil or criminal enforcement from appropriate local, state, or federal law enforcement entities.

**q.** In the event of any change in control or ownership of land or waste discharge facilities presently owned or controlled by the Discharger, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to the Regional Water Board.

To assume operation under this Order, the succeeding owner or operator must apply in writing to the Executive Officer requesting transfer of the Order. The request must contain the requesting entity's full legal name, the state of incorporation if a corporation, address and telephone number of the persons responsible for contact with the Regional Water Board and a statement. The statement shall comply with the signatory and certification requirements in the federal Standard Provisions (Attachment D, section V.B) and state that the new owner or operator assumes full responsibility for compliance with this Order. Failure to submit the request shall be considered a discharge without requirements, a violation of the CWC. Transfer shall be approved or disapproved in writing by the Executive Officer.

## B. Monitoring and Reporting Program Requirements

The Discharger shall comply with the Monitoring and Reporting Program, and future revisions thereto, in Attachment E of this Order.

## C. Special Provisions

### 1. Reopener Provisions

**a.** Conditions that necessitate a major modification of a permit are described in 40 CFR 122.62, including:

**i.** If new or amended applicable water quality standards are promulgated or approved pursuant to section 303 of the CWA, or amendments thereto, this permit may be reopened and modified in accordance with the new or amended standards.

**ii.** When new information, that was not available at the time of permit issuance, would have justified different permit conditions at the time of issuance.

**b.** This Order may be reopened for modification, or revocation and reissuance, as a result of the detection of a reportable priority pollutant generated by special conditions included in this Order. These special conditions may be, but are not limited to, fish tissue sampling, whole effluent toxicity, monitoring requirements on internal waste stream(s), and monitoring for surrogate parameters. Additional requirements may be included in this Order as a result of the special condition monitoring data.

**c. Mercury.** If mercury is found to be causing toxicity based on acute or chronic toxicity test results, or if a TMDL program is adopted, this Order shall be reopened and the interim mass effluent limitation modified (higher or lower) or an effluent concentration limitation imposed. If the Regional Water Board determines that a mercury offset program is feasible for Dischargers subject to a NPDES permit, then this Order may be reopened to reevaluate the interim mercury mass loading limitation(s) and the need for a mercury offset program for the Discharger.

d. **Whole Effluent Toxicity.** As a result of a Toxicity Reduction Evaluation (TRE), this Order may be reopened to include a chronic toxicity limitation, a new acute toxicity limitation, and/or a limitation for a specific toxicant identified in the TRE. Additionally, if the State Water Board revises the SIP's toxicity control provisions that would require the establishment of numeric chronic toxicity effluent limitations, this Order may be reopened to include a numeric chronic toxicity effluent limitation based on the new provisions.

e. **Water Effects Ratios (WER) and Metal Translators.** A default WER of 1.0 has been used in this Order for calculating CTR criteria for applicable priority pollutant inorganic constituents. If the Discharger performs studies to determine site-specific WERs and/or site-specific dissolved-to-total metal translators, this Order may be reopened to modify the effluent limitations for the applicable inorganic constituents.

f. **Disinfection Requirements.** This reopener provision allows the Regional Water Board to reopen this Order to include a compliance schedule if the discharge does not meet the new coliform bacteria effluent limitations when dilution is <20:1.

**2. Special Studies, Technical Reports and Additional Monitoring Requirements**

a. **Chronic Whole Effluent Toxicity.** For compliance with the Basin Plan's narrative toxicity objective, this Order requires the Discharger to conduct chronic whole effluent toxicity (WET) testing, as specified in the Monitoring and Reporting Program (Attachment E, section V). Furthermore, this Provision requires the Discharger to investigate the causes of, and identify corrective actions to reduce or eliminate effluent toxicity. If the discharge exhibits toxicity, as described in subsection ii below, the Discharger is required to initiate a TRE in accordance with an approved TRE Workplan, and take actions to mitigate the impact of the discharge and prevent recurrence of toxicity. A TRE is a site-specific study conducted in a stepwise process to identify the source(s) of toxicity and the effective control measures for effluent toxicity. TREs are designed to identify the causative agents and sources of effluent toxicity, evaluate the effectiveness of the toxicity control options, and confirm the reduction in effluent toxicity. This Provision includes requirements for the Discharger to develop and submit a TRE Workplan and includes procedures for accelerated chronic toxicity monitoring and TRE initiation.

i. **Initial Investigative TRE Workplan. Within 90 days** of the effective date of this Order, the Discharger shall submit to the Regional Water Board an Initial Investigative TRE Workplan for approval by the Executive Officer. This should be a one to two page document including, at a minimum:

**(a)** A description of the investigation and evaluation techniques that will be used to identify potential causes and sources of effluent toxicity, effluent variability, and treatment system efficiency;

**(b)** A description of the facility's methods of maximizing in-house treatment efficiency and good housekeeping practices, and a list of all chemicals used in operation of the facility; and

**(c)** A discussion of who will conduct the Toxicity Identification Evaluation (TIE), if necessary (e.g., an in-house expert or outside contractor).

**ii. Accelerated Monitoring and TRE Initiation.** When the numeric toxicity monitoring trigger is exceeded during regular chronic toxicity monitoring, the Discharger shall initiate accelerated monitoring as required in the Accelerated Monitoring Specifications. The Discharger shall initiate a TRE to address effluent toxicity if any WET testing results exceed the numeric toxicity monitoring trigger during accelerated monitoring.

**iii. Numeric Toxicity Monitoring Trigger.** The numeric toxicity monitoring trigger to initiate a TRE is > **1 $TU_C$** (where $TU_C$ = 100/NOEC). The monitoring trigger is not an effluent limitation; it is the toxicity threshold at which the Discharger is required to begin accelerated monitoring and initiate a TRE when the effluent exhibits toxicity.

**iv. Accelerated Monitoring Specifications.** If the numeric toxicity monitoring trigger is exceeded during regular chronic toxicity testing, the Discharger shall initiate accelerated monitoring within 14 days of notification by the laboratory of the exceedance. Accelerated monitoring shall consist of four (4) chronic toxicity tests conducted once every 2 weeks using the species that exhibited toxicity. The following protocol shall be used for accelerated monitoring and TRE initiation:

**(a)** If the results of four (4) consecutive accelerated monitoring tests do not exceed the monitoring trigger, the Discharger may cease accelerated monitoring and resume regular chronic toxicity monitoring. However, notwithstanding the accelerated monitoring results, if there is adequate evidence of effluent toxicity, the Executive Officer may require that the Discharger initiate a TRE.

**(b)** If the source(s) of the toxicity is easily identified (e.g., temporary plant upset), the Discharger shall make necessary corrections to the facility and shall continue accelerated monitoring until four (4) consecutive accelerated tests do not exceed the monitoring trigger. Upon confirmation that the effluent toxicity has been removed, the Discharger may cease accelerated monitoring and resume regular chronic toxicity monitoring.

**(c)** If the result of any accelerated toxicity test exceeds the monitoring trigger, the Discharger shall cease accelerated monitoring and begin a TRE to investigate the cause(s) of, and identify corrective actions to reduce or eliminate effluent toxicity. Within thirty (30) days of notification by the laboratory of any test result exceeding the monitoring trigger during accelerated monitoring, the Discharger shall submit a TRE Action Plan to the Regional Water Board including, at minimum:

**(1)** Specific actions the Discharger will take to investigate and identify the cause(s) of toxicity, including a TRE WET monitoring schedule;

**(2)** Specific actions the Discharger will take to mitigate the impact of the discharge and prevent the recurrence of toxicity; and

**(3)** A schedule for these actions.

**b. Effluent and Receiving Water Characterization Study.** An effluent and receiving water monitoring study is required to ensure adequate information is available for the next permit renewal. During the third year of this permit term, the Discharger shall conduct **quarterly** monitoring of the effluent at EFF-001 and of the receiving water at RSW-001 for all priority pollutants and other constituents of concern as described in Attachment I. The report shall be completed in conformance with the following schedule.

**Table 7. Effluent and Receiving Water Characterization Study**

| Task | Compliance Date |
|---|---|
| i. Submit Work Plan and Time Schedule | No later than 2 years 6 months from adoption of this Order |
| ii. Conduct quarterly monitoring | During third year of permit term |
| iii. Submit Final Report | 6 months following completion of final monitoring event |

**3. Best Management Practices and Pollution Prevention**

**a. Pollutant Minimization Program (PMP)**

The Discharger shall develop and conduct a PMP as further described below when there is evidence (e.g., sample results reported as DNQ when the effluent limitation is less than the MDL, sample results from analytical methods more sensitive than those methods required by this Order, presence of whole effluent toxicity, health advisories for fish consumption, results of benthic or aquatic organism tissue sampling) that a priority pollutant is present in the effluent above an effluent limitation and either: (1) A sample result is reported as DNQ and the effluent limitation is less than the RL; or (2) A sample result is reported as ND and the effluent limitation is less than the MDL, using definitions described in Attachment A and reporting protocols described in the Monitoring and Reporting Program (Attachment E, section X.B.4).

The PMP shall include, but not be limited to, the following actions and submittals acceptable to the Regional Water Board:

- **i.** An annual review and semi-annual monitoring of potential sources of the reportable priority pollutant(s), which may include fish tissue monitoring and other bio-uptake sampling;
- **ii.** Quarterly monitoring (during the third year of the permit term) for the reportable priority pollutant(s) in the influent to the wastewater treatment system;
- **iii.** Submittal of a control strategy designed to proceed toward the goal of maintaining concentrations of the reportable priority pollutant(s) in the effluent at or below the effluent limitation;
- **iv.** Implementation of appropriate cost-effective control measures for the reportable priority pollutant(s), consistent with the control strategy; and
- **v.** An annual status report that shall be sent to the Regional Water Board including:
  - **(a)** All PMP monitoring results for the previous year;
  - **(b)** A list of potential sources of the reportable priority pollutant(s);
  - **(c)** A summary of all actions undertaken pursuant to the control strategy; and
  - **(d)** A description of actions to be taken in the following year.

**b. Salinity Evaluation and Minimization Plan.** The Discharger shall prepare a salinity evaluation and minimization plan to identify and address sources of salinity from the Facility. The plan shall be completed and submitted to the Central Valley Water Board **within 9 months of the adoption date of this Order** for the approval by the Executive Officer.

**4. Construction, Operation and Maintenance Specifications**

**b. Treatment Pond Operating Requirements.**

- **i.** The treatment facilities shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
- **ii.** Public contact with wastewater shall be precluded through such means as fences, signs, and other acceptable alternatives.
- **iii.** Ponds shall be managed to prevent breeding of mosquitoes. In particular,

**(a)** An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.

**(b)** Weeds shall be minimized.

**(c)** Dead algae, vegetation, and debris shall not accumulate on the water surface.

**iv.** Ponds shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration during the non-irrigation season. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with historical rainfall patterns. Freeboard shall never be less than 2 feet (measured vertically to the lowest point of overflow).

**v.** Prior to the onset of the rainy season of each year, available pond storage capacity shall at least equal the volume necessary to comply with the Land Discharge Specification at section IV.C.4.a.v., above.

**vi.** The monthly average discharge flow shall not exceed 0.38 mgd.

**vii.** The discharge of waste classified as "hazardous" as defined in section 2521(a) of Title 23, California Code of Regulations (CCR), or "designated", as defined in section 13173 of the CWC, to the treatment ponds is prohibited.

**viii.** Objectionable odors originating at this Facility shall not be perceivable beyond the limits of the wastewater treatment and disposal areas (or property owned by the Discharger).

**ix.** The dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds shall not be less than 1.0 mg/L.

**x.** Ponds shall not have a pH less than 6.5 or greater than 8.5.

**c. Turbidity Operational Requirements**. When dilution is <20:1, the Discharger shall operate the treatment system to ensure that the turbidity measured at EFF-001, as described in the MRP (Attachment E), shall not exceed:

i. 2 NTU as a daily average,
ii. 5 NTU not be exceeded more than 5% of the time within a 24-hour period, and
iii. 10 NTU as an instantaneous maximum.

**5. Special Provisions for Municipal Facilities (POTWs Only)**

**a. Pretreatment Requirements.**

**i.** The Discharger shall implement, as more completely set forth in 40 CFR 403.5, the necessary legal authorities, programs, and controls to ensure that the following incompatible wastes are not introduced to the treatment system, where incompatible wastes are:

**(a)** Wastes which create a fire or explosion hazard in the treatment works;

**(b)** Wastes which will cause corrosive structural damage to treatment works, but in no case wastes with a pH lower than 5.0, unless the works is specially designed to accommodate such wastes;

**(c)** Solid or viscous wastes in amounts which cause obstruction to flow in sewers, or which cause other interference with proper operation or treatment works;

**(d)** Any waste, including oxygen demanding pollutants (BOD, etc.), released in such volume or strength as to cause inhibition or disruption in the treatment works, and subsequent treatment process upset and loss of treatment efficiency;

**(e)** Heat in amounts that inhibit or disrupt biological activity in the treatment works, or that raise influent temperatures above 40°C (104°F), unless the Regional Water Board approves alternate temperature limits;

**(f)** Petroleum oil, non-biodegradable cutting oil, or products of mineral oil origin in amounts that will cause interference or pass through;

**(g)** Pollutants which result in the presence of toxic gases, vapors, or fumes within the treatment works in a quantity that may cause acute worker health and safety problems; and:

**(h)** Any trucked or hauled pollutants, except at points predesignated by the Discharger.

**ii.** The Discharger shall implement, as more completely set forth in 40 CFR 403.5, the legal authorities, programs, and controls necessary to ensure that indirect discharges do not introduce pollutants into the sewerage system that, either alone or in conjunction with a discharge or discharges from other sources:

**(a)** Flow through the system to the receiving water in quantities or concentrations that cause a violation of this Order, or:

**(b)** Inhibit or disrupt treatment processes, treatment system operations, or sludge processes, use, or disposal and either cause a violation of this Order or prevent sludge use or disposal in accordance with this Order.

**b. Sludge/Biosolids Discharge Specifications**

**i.** Collected screenings, residual sludge, biosolids, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer, and consistent with Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste, as set forth in Title 27, CCR, division 2, subdivision 1, section 20005, et seq. Removal for further treatment, disposal, or reuse at sites (e.g., landfill, composting sites, soil amendment sites) that are operated in accordance with valid waste discharge requirements issued by a Regional Water Board will satisfy these specifications.

**ii.** Sludge and solid waste shall be removed from screens, sumps, ponds, clarifiers, etc. as needed to ensure optimal plant performance.

**iii.** The treatment of sludge generated at the Facility shall be confined to the Facility property and conducted in a manner that precludes infiltration of waste constituents into soils in a mass or concentration that will violate groundwater limitations in section V.B. of this Order. In addition, the storage of residual sludge, solid waste, and biosolids on Facility property shall be temporary and controlled, and contained in a manner that minimizes leachate formation and precludes infiltration of waste constituents into soils in a mass or concentration that will violate groundwater limitations included in section V.B. of this Order.

**iv.** The use and disposal of biosolids shall comply with existing federal and state laws and regulations, including permitting requirements and technical standards included in 40 CFR Part 503. If the State Water Board and the Regional Water Board are given the authority to implement regulations contained in 40 CFR Part 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger must comply with the standards and time schedules contained in 40 CFR Part 503 whether or not they have been incorporated into this Order.

**c. Biosolids Disposal Requirements**

**i.** The Discharger shall comply with the Monitoring and Reporting Program for biosolids disposal contained in Attachment E.

**ii.** Any proposed change in biosolids use or disposal practice from a previously approved practice shall be reported to the Executive Officer and USEPA Regional Administrator at least 90 days in advance of the change.

**iii.** The Discharger is encouraged to comply with the “Manual of Good Practice for Agricultural Land Application of Biosolids” developed by the California Water Environment Association.

**d. Biosolids Storage Requirements**

**i.** Facilities for the storage of Class B biosolids shall be located, designed and maintained to restrict public access to biosolids.

**ii.** Biosolids storage facilities shall be designed and maintained to prevent washout or inundation from a storm or flood with a return frequency of 100 years.

**iii.** Biosolids storage facilities, which contain biosolids, shall be designed and maintained to contain all storm water falling on the biosolids storage area during a rainfall year with a return frequency of 100 years.

**iv.** Biosolids storage facilities shall be designed, maintained and operated to minimize the generation of leachate.

**e. Collection System.** On 2 May 2006, the State Water Board adopted State Water Board Order No. 2006-0003, a Statewide General WDR for Sanitary Sewer Systems. The Discharger shall be subject to the requirements of Order No. 2006-0003 and any future revisions thereto. Order No. 2006-0003 requires that all public agencies that currently own or operate sanitary sewer systems apply for coverage under the General WDR. The Discharger has applied for and has been approved for coverage under State Water Board Order 2006-0003 for operation of its wastewater collection system.

Regardless of the coverage obtained under Order No. 2006-0003, the Discharger's collection system is part of the treatment system that is subject to this Order. As such, pursuant to federal regulations, the Discharger must properly operate and maintain its collection system [40 CFR 122.41(e)], report any non-compliance [40 CFR 122.41(l)(6) and (7)], and mitigate any discharge from the collection system in violation of this Order [40 CFR 122.41(d)].

**f.** This permit, and the Monitoring and Reporting Program which is a part of this permit, requires that certain parameters be monitored on a continuous basis. The wastewater treatment plant is not staffed on a full time basis. Permit violations or system upsets can go undetected during this period. The Discharger is required to establish an electronic system for operator notification for continuous recording device alarms. For existing continuous monitoring systems, the electronic notification system shall be installed within 6 months of adoption of this permit. For systems installed following permit adoption, the notification system shall be installed simultaneously.

**6. Other Special Provisions**

**a.** When dilution is <20:1, the wastewater shall be filtered, and adequately disinfected pursuant to the Department of Public Health (DPH; formerly the Department of Health Services) reclamation criteria, CCR, Title 22, division 4, chapter 3, (Title 22), or equivalent**,** in accordance with the compliance schedule in Section VI.C.7.a, below. Adequately disinfected shall mean that the wastewater meets the total coliform bacterial effluent limits contained in this permit. Filtered means that the wastewater meets the definition of “Filtered Wastewater” contained in California Code of Regulations, title 22, section 60301.320.

**7. Compliance Schedules**

**a. Compliance Schedule for Filtration and Turbidity.** This permit requires that wastewater discharged to Lateral K during critical flow periods (<20:1 dilution in receiving water) shall be filtered and adequately disinfected. The effluent shall be disinfected in accordance with the total coliform organisms effluent limitations set forth in this Order. Because filtration is required, and the wastewater treatment plant does not currently provide filtration, this compliance schedule allows time for the Discharger to make necessary improvements. Until final compliance, the Discharger shall submit progress reports in accordance with the Monitoring and Reporting Program (Attachment E, section X.D.1). The Discharger shall comply with the following time schedule to ensure compliance with the requirements of this permit.

**Table 8. Compliance Schedule for Filtration and Turbidity**

| Task | Compliance Date |
|---|---|
| Submit Initial Workplan for Method of Compliance | 6 Months after effective Date of this Order |
| Submit Results of Proposed Method of Compliance Study | 18 Months after effective date of this Order |
| Submit record of Method of Compliance Decision and Project Work Plan with Time Schedule | 2 Years after effective date of this Order |
| Complete 25% project Work Plan and submit progress report[1] | 3 Years after effective date of this Order |
| Complete 50% project Work Plan and submit progress report[1] | 4 Years after effective date of this Order |
| Complete 100% project Work Plan and submit project completion report[1] | 5 Years after effective date of this Order |
| Achieve compliance with applicable final effluent limits | 5 years after effective date of this Order |

[1] Completion includes necessary project components such as studies, financing, environmental review, permitting, design, bidding, construction, testing, etc., relative to project work plan submitted

## VII. COMPLIANCE DETERMINATION

**A. $BOD_5$ and TSS Effluent Limitations (Section IV.A.1.a).** Compliance with the final effluent limitations for $BOD_5$ and TSS required in Limitations and Discharge Requirements shall be ascertained by 24-hour composite samples. Compliance with effluent limitations required in Limitations and Discharge Requirements for percent removal shall be calculated using the arithmetic mean of $BOD_5$ and TSS in effluent samples collected over a monthly period as a percentage of the arithmetic mean of the values for influent samples collected at approximately the same times during the same period.

**B. Average Dry Weather Flow Effluent Limitations (Section IV.A.1.a).** The average dry weather discharge flow represents the daily average flow when groundwater is at or near normal and runoff is not occurring. Compliance with the average dry weather flow effluent limitations will be determined annually based on the average daily flow over three consecutive dry weather months (e.g., July, August, and September).

**C. Total Coliform Organisms Effluent Limitations (Section IV.A.1.e.).** For each day that an effluent sample is collected and analyzed for total coliform organisms, the 7-day median shall be determined by calculating the median concentration of total coliform bacteria in the effluent utilizing the bacteriological results of the last 7 days. For example, if a sample is collected on a Wednesday, the result from that sampling event and all results from the previous 6 days (i.e., Tuesday, Monday, Sunday, Saturday, Friday, and Thursday) are used to calculate the 7-day median. If the 7-day median of total coliform organisms exceeds a most probable number (MPN) of 23 per 100 milliliters, the Discharger will be considered out of compliance.

**D. Instantaneous Maximum Effluent Limitation (Section IV.A.1.a).** The Discharger shall use USEPA standard analytical techniques for analyzing pH with a maximum reporting level not to exceed the minimum levels listed in Appendix 4 of the SIP (Table 2d) and if the constituents are priority pollutants or 0.05 µg/L for non-priority pollutants.

**E. Total Residual Chlorine Effluent Limitations (Section IV.A.1.d).** Continuous monitoring analyzers for chlorine residual or for dechlorination agent residual in the effluent are appropriate methods for compliance determination. A positive residual dechlorination agent in the effluent indicates that chlorine is not present in the discharge, which demonstrates compliance with the effluent limitations. This type of monitoring can also be used to prove that some chlorine residual exceedances are false positives. Continuous monitoring data showing either a positive dechlorination agent residual or a chlorine residual at or below the prescribed limit are sufficient to show compliance with the total residual chlorine effluent limitations, as long as the instruments are maintained and calibrated in accordance with the manufacturer's recommendations.

Any excursion above the 1-hour average or 4-day average total residual chlorine effluent limitations is a violation. If the Discharger conducts continuous monitoring and the Discharger can demonstrate, through data collected from a back-up monitoring system, that a chlorine spike recorded by the continuous monitor was not actually due to chlorine, then any excursion resulting from the recorded spike will not be considered an exceedance, but rather reported as a false positive. Records supporting validation of false positives shall be maintained in accordance with Section IV Standard Provisions (Attachment D).

**F. Volatile Organic Compounds (VOCs) Average Monthly Effluent Limitation.** VOCs include all constituents listed in USEPA Method 502.2 (Attachment I). The average monthly effluent limitation of less than 0.5 µg/L applies to each VOCs. When calculating the average monthly of each VOC, non-detect results shall be counted as one-half the detection level.

**G. Chronic Whole Effluent Toxicity Effluent Limitation.** Compliance with the accelerated monitoring and TRE/TIE provisions of Provision VI.C.2.a shall constitute compliance with the effluent limitation.

# ATTACHMENT A – DEFINITIONS

**Arithmetic Mean ($\mu$)**
Also called the average, is the sum of measured values divided by the number of samples. For ambient water concentrations, the arithmetic mean is calculated as follows:

Arithmetic mean = $\mu = \Sigma x / n$ where: $\Sigma x$ is the sum of the measured ambient water concentrations, and n is the number of samples.

**Average Monthly Effluent Limitation (AMEL)**
The highest allowable average of daily discharges over a calendar month, calculated as the sum of all daily discharges measured during a calendar month divided by the number of daily discharges measured during that month.

**Average Weekly Effluent Limitation (AWEL)**
The highest allowable average of daily discharges over a calendar week (Sunday through Saturday), calculated as the sum of all daily discharges measured during a calendar week divided by the number of daily discharges measured during that week.

**Bioaccumulative**
Those substances taken up by an organism from its surrounding medium through gill membranes, epithelial tissue, or from food and subsequently concentrated and retained in the body of the organism.

**Carcinogenic**
Pollutants are substances that are known to cause cancer in living organisms.

**Coefficient of Variation (CV)**
CV is a measure of the data variability and is calculated as the estimated standard deviation divided by the arithmetic mean of the observed values.

**Daily Discharge**
Daily Discharge is defined as either: (1) the total mass of the constituent discharged over the calendar day (12:00 am through 11:59 pm) or any 24-hour period that reasonably represents a calendar day for purposes of sampling (as specified in the permit), for a constituent with limitations expressed in units of mass or; (2) the unweighted arithmetic mean measurement of the constituent over the day for a constituent with limitations expressed in other units of measurement (e.g., concentration).

The daily discharge may be determined by the analytical results of a composite sample taken over the course of 1 day (a calendar day or other 24-hour period defined as a day) or by the arithmetic mean of analytical results from one or more grab samples taken over the course of the day.

For composite sampling, if 1 day is defined as a 24-hour period other than a calendar day, the analytical result for the 24-hour period will be considered as the result for the calendar day in which the 24-hour period ends.

**Detected, but Not Quantified (DNQ)**
DNQ are those sample results less than the RL, but greater than or equal to the laboratory's MDL.

**Dilution Credit**
Dilution Credit is the amount of dilution granted to a discharge in the calculation of a water quality-based effluent limitation, based on the allowance of a specified mixing zone. It is calculated from the dilution ratio or determined through conducting a mixing zone study or modeling of the discharge and receiving water.

**Effluent Concentration Allowance (ECA)**
ECA is a value derived from the water quality criterion/objective, dilution credit, and ambient background concentration that is used, in conjunction with the coefficient of variation for the effluent monitoring data, to calculate a long-term average (LTA) discharge concentration. The ECA has the same meaning as waste load allocation (WLA) as used in USEPA guidance (Technical Support Document For Water Quality-based Toxics Control, March 1991, second printing, EPA/505/2-90-001).

**Enclosed Bays**
Enclosed Bays means indentations along the coast that enclose an area of oceanic water within distinct headlands or harbor works. Enclosed bays include all bays where the narrowest distance between the headlands or outermost harbor works is less than 75 percent of the greatest dimension of the enclosed portion of the bay. Enclosed bays include, but are not limited to, Humboldt Bay, Bodega Harbor, Tomales Bay, Drake's Estero, San Francisco Bay, Morro Bay, Los Angeles-Long Beach Harbor, Upper and Lower Newport Bay, Mission Bay, and San Diego Bay. Enclosed bays do not include inland surface waters or ocean waters.

**Estimated Chemical Concentration**
The estimated chemical concentration that results from the confirmed detection of the substance by the analytical method below the ML value.

**Estuaries**
Estuaries means waters, including coastal lagoons, located at the mouths of streams that serve as areas of mixing for fresh and ocean waters. Coastal lagoons and mouths of streams that are temporarily separated from the ocean by sandbars shall be considered estuaries. Estuarine waters shall be considered to extend from a bay or the open ocean to a point upstream where there is no significant mixing of fresh water and seawater. Estuarine waters included, but are not limited to, the Sacramento-San Joaquin Delta, as defined in CWC section 12220, Suisun Bay, Carquinez Strait downstream to the Carquinez Bridge, and appropriate areas of the Smith, Mad, Eel, Noyo, Russian, Klamath, San Diego, and Otay rivers. Estuaries do not include inland surface waters or ocean waters.

**Inland Surface Waters**
All surface waters of the State that do not include the ocean, enclosed bays, or estuaries.

**Instantaneous Maximum Effluent Limitation**
The highest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous maximum limitation).

**Instantaneous Minimum Effluent Limitation**
The lowest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous minimum limitation).

**Maximum Daily Effluent Limitation (MDEL)**
The highest allowable daily discharge of a pollutant, over a calendar day (or 24-hour period). For pollutants with limitations expressed in units of mass, the daily discharge is calculated as the total mass of the pollutant discharged over the day. For pollutants with limitations expressed in other units of measurement, the daily discharge is calculated as the arithmetic mean measurement of the pollutant over the day.

**Median**
The middle measurement in a set of data. The median of a set of data is found by first arranging the measurements in order of magnitude (either increasing or decreasing order). If the number of measurements (*n*) is odd, then the median = $X_{(n+1)/2}$. If *n* is even, then the median = $(X_{n/2} + X_{(n/2)+1})/2$ (i.e., the midpoint between the *n*/2 and *n*/2+1).

**Method Detection Limit (MDL)**
MDL is the minimum concentration of a substance that can be measured and reported with 99 percent confidence that the analyte concentration is greater than zero, as defined in 40 CFR Part 136, Attachment B, revised as of 3 July 1999.

**Minimum Level (ML)**
ML is the concentration at which the entire analytical system must give a recognizable signal and acceptable calibration point. The ML is the concentration in a sample that is equivalent to the concentration of the lowest calibration standard analyzed by a specific analytical procedure, assuming that all the method specified sample weights, volumes, and processing steps have been followed.

**Mixing Zone**
Mixing Zone is a limited volume of receiving water that is allocated for mixing with a wastewater discharge where water quality criteria can be exceeded without causing adverse effects to the overall water body.

**Not Detected (ND)**
Sample results which are less than the laboratory's MDL.

**Ocean Waters**
The territorial marine waters of the State as defined by California law to the extent these waters are outside of enclosed bays, estuaries, and coastal lagoons. Discharges to ocean waters are regulated in accordance with the State Water Board's California Ocean Plan.

**Persistent Pollutants**
Persistent pollutants are substances for which degradation or decomposition in the environment is nonexistent or very slow.

**Pollutant Minimization Program (PMP)**
PMP means waste minimization and pollution prevention actions that include, but are not limited to, product substitution, waste stream recycling, alternative waste management

methods, and education of the public and businesses. The goal of the PMP shall be to reduce all potential sources of a priority pollutant(s) through pollutant minimization (control) strategies, including pollution prevention measures as appropriate, to maintain the effluent concentration at or below the water quality-based effluent limitation. Pollution prevention measures may be particularly appropriate for persistent bioaccumulative priority pollutants where there is evidence that beneficial uses are being impacted. The Regional Water Board may consider cost effectiveness when establishing the requirements of a PMP. The completion and implementation of a Pollution Prevention Plan, if required pursuant to CWC section 13263.3(d), shall be considered to fulfill the PMP requirements.

**Pollution Prevention**
Pollution Prevention means any action that causes a net reduction in the use or generation of a hazardous substance or other pollutant that is discharged into water and includes, but is not limited to, input change, operational improvement, production process change, and product reformulation (as defined in Water Code section 13263.3). Pollution prevention does not include actions that merely shift a pollutant in wastewater from one environmental medium to another environmental medium, unless clear environmental benefits of such an approach are identified to the satisfaction of the State or Regional Water Board.

**Reporting Level (RL)**
RL is the ML (and its associated analytical method) chosen by the Discharger for reporting and compliance determination from the MLs included in this Order. The MLs included in this Order correspond to approved analytical methods for reporting a sample result that are selected by the Regional Water Board either from Appendix 4 of the SIP in accordance with section 2.4.2 of the SIP or established in accordance with section 2.4.3 of the SIP. The ML is based on the proper application of method-based analytical procedures for sample preparation and the absence of any matrix interferences. Other factors may be applied to the ML depending on the specific sample preparation steps employed. For example, the treatment typically applied in cases where there are matrix-effects is to dilute the sample or sample aliquot by a factor of ten. In such cases, this additional factor must be applied to the ML in the computation of the RL.

**Satellite Collection System**
The portion, if any, of a sanitary sewer system owned or operated by a different public agency than the agency that owns and operates the wastewater treatment facility that a sanitary sewer system is tributary to.

**Source of Drinking Water**
Any water designated as municipal or domestic supply (MUN) in a Regional Water Board Basin Plan.

**Standard Deviation ($\sigma$)**
Standard Deviation is a measure of variability that is calculated as follows:

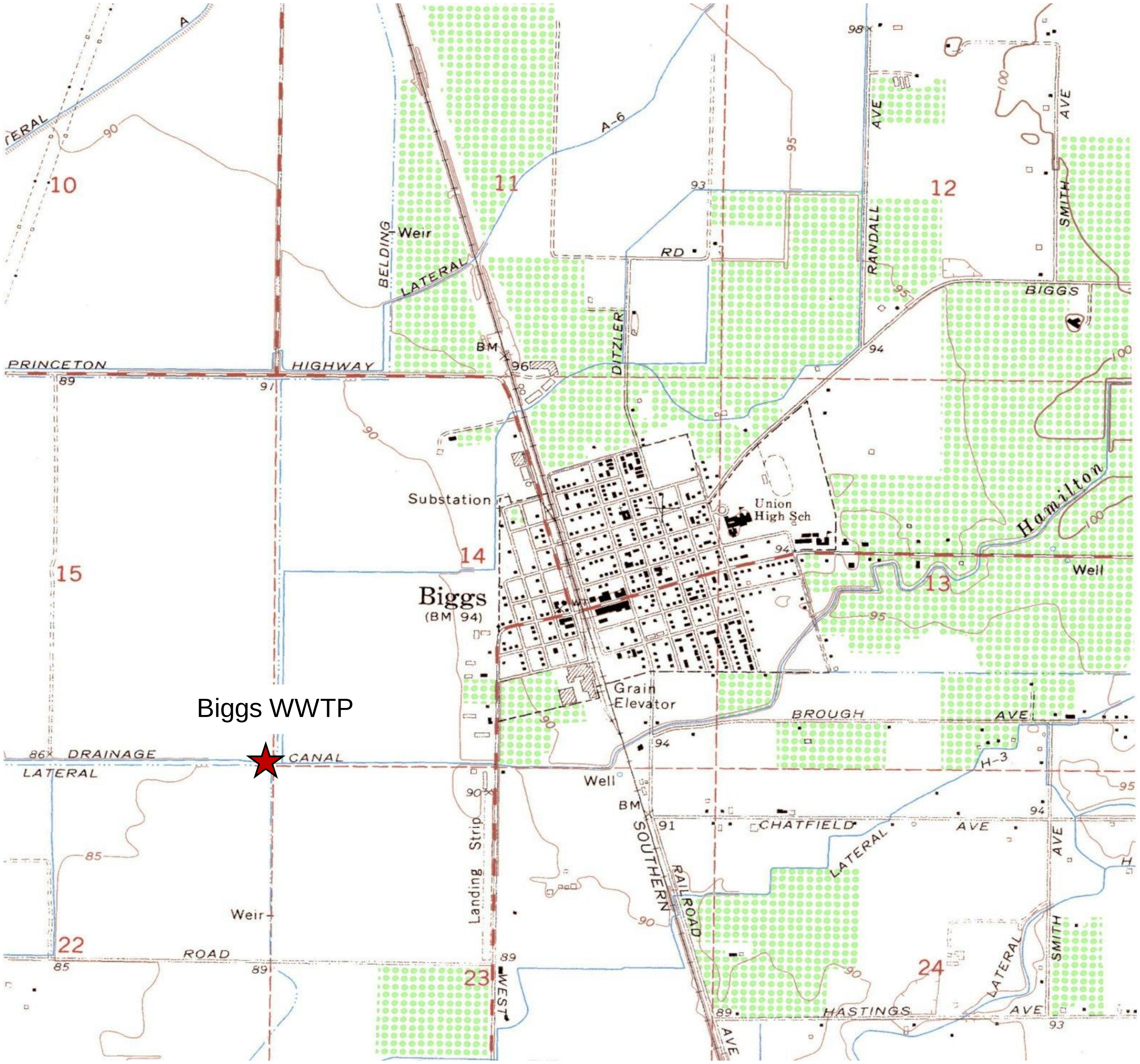
$$\sigma = (\sum[(x - \mu)^2]/(n - 1))^{0.5}$$

where:
x is the observed value;
$\mu$ is the arithmetic mean of the observed values; and

n is the number of samples.

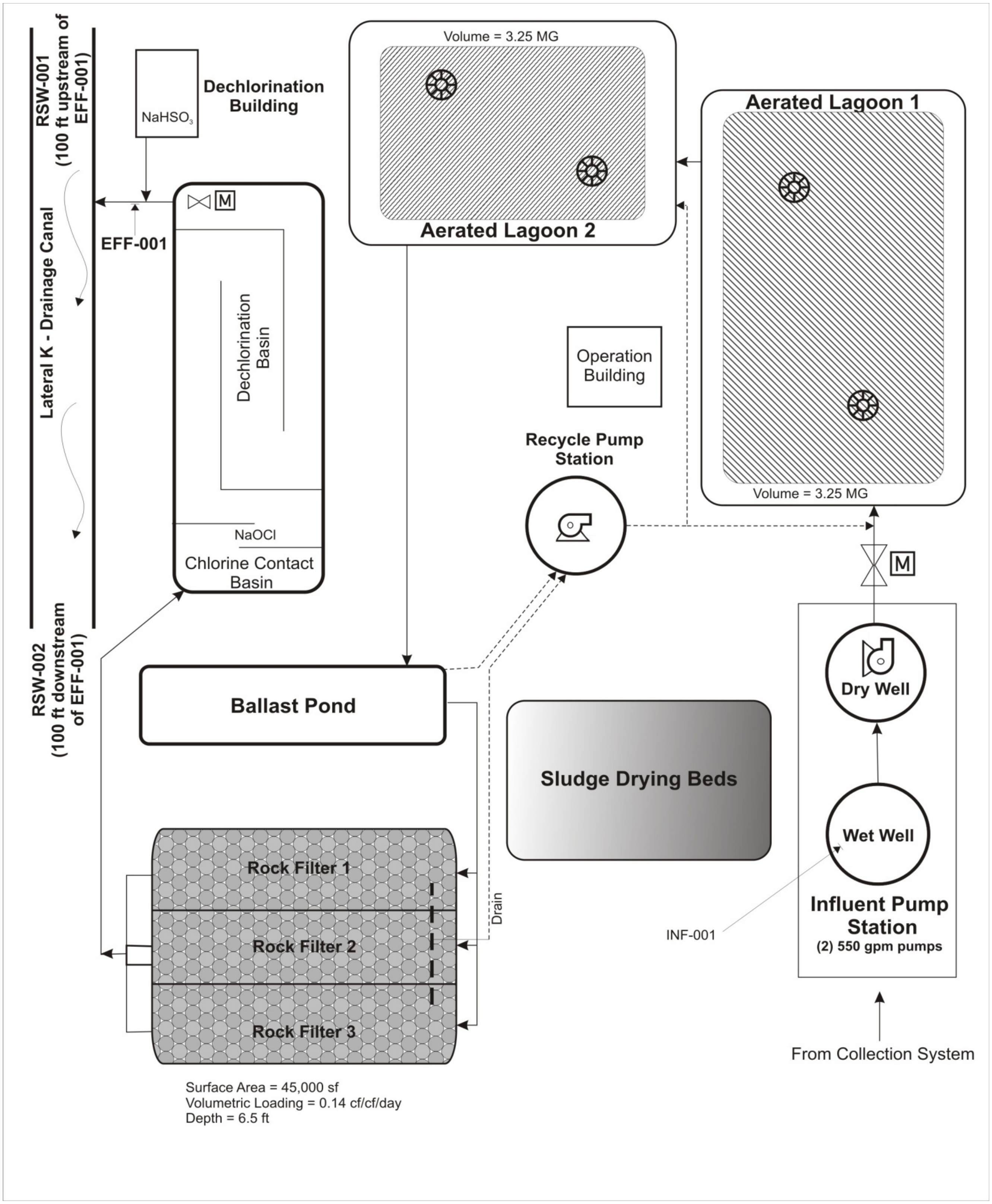
**Toxicity Reduction Evaluation (TRE)**
TRE is a study conducted in a step-wise process designed to identify the causative agents of effluent or ambient toxicity, isolate the sources of toxicity, evaluate the effectiveness of toxicity control options, and then confirm the reduction in toxicity. The first steps of the TRE consist of the collection of data relevant to the toxicity, including additional toxicity testing, and an evaluation of facility operations and maintenance practices, and best management practices. A Toxicity Identification Evaluation (TIE) may be required as part of the TRE, if appropriate. (A TIE is a set of procedures to identify the specific chemical(s) responsible for toxicity. These procedures are performed in three phases (characterization, identification, and confirmation) using aquatic organism toxicity tests.)

## ATTACHMENT B – MAP




| Drawing Reference: BIGGS U.S.G.S TOPOGRAPHIC MAP 7.5 MINUTE QUADRANGLE ***Not to scale*** | SITE LOCATION MAP<br>**CITY OF BIGGS**<br>**WASTEWATER TREATMENT FACILITY**<br>**BUTTE COUNTY** | N W E S |
|---|---|---|

## ATTACHMENT C – FLOW SCHEMATIC



# ATTACHMENT D – STANDARD PROVISIONS

## I. STANDARD PROVISIONS – PERMIT COMPLIANCE

### A. Duty to Comply

**1.** The Discharger must comply with all of the conditions of this Order. Any noncompliance constitutes a violation of the Clean Water Act (CWA) and the California Water Code (CWC) and is grounds for enforcement action, for permit termination, revocation and reissuance, or modification; or denial of a permit renewal application. (40 CFR 122.41(a).)

**2.** The Discharger shall comply with effluent standards or prohibitions established under section 307(a) of the CWA for toxic pollutants and with standards for sewage sludge use or disposal established under section 405(d) of the CWA within the time provided in the regulations that establish these standards or prohibitions, even if this Order has not yet been modified to incorporate the requirement. (40 CFR 122.41(a)(1).)

### B. Need to Halt or Reduce Activity Not a Defense

It shall not be a defense for a Discharger in an enforcement action that it would have been necessary to halt or reduce the permitted activity in order to maintain compliance with the conditions of this Order. (40 CFR 122.41(c).)

### C. Duty to Mitigate

The Discharger shall take all reasonable steps to minimize or prevent any discharge or sludge use or disposal in violation of this Order that has a reasonable likelihood of adversely affecting human health or the environment. (40 CFR 122.41(d).)

### D. Proper Operation and Maintenance

The Discharger shall at all times properly operate and maintain all facilities and systems of treatment and control (and related appurtenances) which are installed or used by the Discharger to achieve compliance with the conditions of this Order. Proper operation and maintenance also includes adequate laboratory controls and appropriate quality assurance procedures. This provision requires the operation of backup or auxiliary facilities or similar systems that are installed by a Discharger only when necessary to achieve compliance with the conditions of this Order. (40 CFR 122.41(e).)

### E. Property Rights

**1.** This Order does not convey any property rights of any sort or any exclusive privileges. (40 CFR 122.41(g).)

**2.** The issuance of this Order does not authorize any injury to persons or property or invasion of other private rights, or any infringement of state or local law or regulations. (40 CFR 122.5(c).)

## F. Inspection and Entry

The Discharger shall allow the Regional Water Board, State Water Board, United States Environmental Protection Agency (USEPA), and/or their authorized representatives (including an authorized contractor acting as their representative), upon the presentation of credentials and other documents, as may be required by law, to (40 CFR 122.41(i); CWC section 13383):

**1.** Enter upon the Discharger's premises where a regulated facility or activity is located or conducted, or where records are kept under the conditions of this Order (40 CFR 122.41(i)(1));

**2.** Have access to and copy, at reasonable times, any records that must be kept under the conditions of this Order (40 CFR 122.41(i)(2));

**3.** Inspect and photograph, at reasonable times, any facilities, equipment (including monitoring and control equipment), practices, or operations regulated or required under this Order (40 CFR 122.41(i)(3)); and

**4.** Sample or monitor, at reasonable times, for the purposes of assuring Order compliance or as otherwise authorized by the CWA or the CWC, any substances or parameters at any location. (40 CFR 122.41(i)(4).)

## G. Bypass

**1.** *Definitions*

**a.** *“Bypass” means the intentional diversion of waste streams from any portion of a treatment facility. (40 CFR 122.41(m)(1)(i).)*

**b.** *“Severe property damage” means substantial physical damage to property, damage* to the treatment facilities, which causes them to become inoperable, or substantial and permanent loss of natural resources that can reasonably be expected to occur in the absence of a bypass. Severe property damage does not mean economic loss caused by delays in production. (40 CFR 122.41(m)(1)(ii).)

**2.** Bypass not exceeding limitations. The Discharger may allow any bypass to occur which does not cause exceedances of effluent limitations, but only if it is for essential maintenance to assure efficient operation. These bypasses are not subject to the provisions listed in Standard Provisions – Permit Compliance I.G.3, I.G.4, and I.G.5 below. (40 CFR 122.41(m)(2).)

3. Prohibition of bypass. Bypass is prohibited, and the Regional Water Board may take enforcement action against a Discharger for bypass, unless (40 CFR 122.41(m)(4)(i)):

   a. Bypass was unavoidable to prevent loss of life, personal injury, or severe property damage (40 CFR 122.41(m)(4)(i)(A));

   b. There were no feasible alternatives to the bypass, such as the use of auxiliary treatment facilities, retention of untreated wastes, or maintenance during normal periods of equipment downtime. This condition is not satisfied if adequate back-up equipment should have been installed in the exercise of reasonable engineering judgment to prevent a bypass that occurred during normal periods of equipment downtime or preventive maintenance (40 CFR 122.41(m)(4)(i)(B)); and

   c. The Discharger submitted notice to the Regional Water Board as required under Standard Provisions – Permit Compliance I.G.5 below. (40 CFR 122.41(m)(4)(i)(C).)

4. The Regional Water Board may approve an anticipated bypass, after considering its adverse effects, if the Regional Water Board determines that it will meet the three conditions listed in Standard Provisions – Permit Compliance I.G.3 above. (40 CFR 122.41(m)(4)(ii).)

5. Notice

   a. Anticipated bypass. If the Discharger knows in advance of the need for a bypass, it shall submit a notice, if possible at least 10 days before the date of the bypass. (40 CFR 122.41(m)(3)(i).)

   b. Unanticipated bypass. The Discharger shall submit notice of an unanticipated bypass as required in Standard Provisions - Reporting V.E below (24-hour notice). (40 CFR 122.41(m)(3)(ii).)

## H. Upset

Upset means an exceptional incident in which there is unintentional and temporary noncompliance with technology based permit effluent limitations because of factors beyond the reasonable control of the Discharger. An upset does not include noncompliance to the extent caused by operational error, improperly designed treatment facilities, inadequate treatment facilities, lack of preventive maintenance, or careless or improper operation. (40 CFR 122.41(n)(1).)

1. Effect of an upset. An upset constitutes an affirmative defense to an action brought for noncompliance with such technology based permit effluent limitations if the requirements of Standard Provisions – Permit Compliance I.H.2 below are met. No determination made during administrative review of claims that noncompliance was

caused by upset, and before an action for noncompliance, is final administrative action subject to judicial review. (40 CFR 122.41(n)(2).)

**2.** Conditions necessary for a demonstration of upset. A Discharger who wishes to establish the affirmative defense of upset shall demonstrate, through properly signed, contemporaneous operating logs or other relevant evidence that (40 CFR 122.41(n)(3)):

**a.** An upset occurred and that the Discharger can identify the cause(s) of the upset (40 CFR 122.41(n)(3)(i));

**b.** The permitted facility was, at the time, being properly operated (40 CFR 122.41(n)(3)(ii));

**c.** The Discharger submitted notice of the upset as required in Standard Provisions – Reporting V.E.2.b below (24-hour notice) (40 CFR 122.41(n)(3)(iii)); and

**d.** The Discharger complied with any remedial measures required under Standard Provisions – Permit Compliance I.C above. (40 CFR 122.41(n)(3)(iv).)

**3.** Burden of proof. In any enforcement proceeding, the Discharger seeking to establish the occurrence of an upset has the burden of proof. (40 CFR 122.41(n)(4).)

## II. STANDARD PROVISIONS – PERMIT ACTION

### A. General

This Order may be modified, revoked and reissued, or terminated for cause. The filing of a request by the Discharger for modification, revocation and reissuance, or termination, or a notification of planned changes or anticipated noncompliance does not stay any Order condition. (40 CFR 122.41(f).)

### B. Duty to Reapply

If the Discharger wishes to continue an activity regulated by this Order after the expiration date of this Order, the Discharger must apply for and obtain a new permit. (40 CFR 122.41(b).)

### C. Transfers

This Order is not transferable to any person except after notice to the Regional Water Board. The Regional Water Board may require modification or revocation and reissuance of the Order to change the name of the Discharger and incorporate such other requirements as may be necessary under the CWA and the CWC. (40 CFR 122.41(l)(3) and 122.61.)

## III. STANDARD PROVISIONS – MONITORING

**A.** Samples and measurements taken for the purpose of monitoring shall be representative of the monitored activity. (40 CFR 122.41(j)(1).)

**B.** Monitoring results must be conducted according to test procedures under 40 CFR Part 136 or, in the case of sludge use or disposal, approved under 40 CFR Part 136 unless otherwise specified in 40 CFR Part 503 unless other test procedures have been specified in this Order. (40 CFR 122.41(j)(4) and 122.44(i)(1)(iv).)

## IV. STANDARD PROVISIONS – RECORDS

**C.** Except for records of monitoring information required by this Order related to the Discharger's sewage sludge use and disposal activities, which shall be retained for a period of at least 5 years (or longer as required by 40 CFR Part 503), the Discharger shall retain records of all monitoring information, including all calibration and maintenance records and all original strip chart recordings for continuous monitoring instrumentation, copies of all reports required by this Order, and records of all data used to complete the application for this Order, for a period of at least three (3) years from the date of the sample, measurement, report or application. This period may be extended by request of the Regional Water Board Executive Officer at any time. (40 CFR 122.41(j)(2).)

**D. Records of monitoring information shall include:**

1. The date, exact place, and time of sampling or measurements (40 CFR 122.41(j)(3)(i));
2. The individual(s) who performed the sampling or measurements (40 CFR 122.41(j)(3)(ii));
3. The date(s) analyses were performed (40 CFR 122.41(j)(3)(iii));
4. The individual(s) who performed the analyses (40 CFR 122.41(j)(3)(iv));
5. The analytical techniques or methods used (40 CFR 122.41(j)(3)(v)); and
6. The results of such analyses. (40 CFR 122.41(j)(3)(vi).)

**E. Claims of confidentiality for the following information will be denied (40 CFR 122.7(b)):**

1. The name and address of any permit applicant or Discharger (40 CFR 122.7(b)(1)); and
2. Permit applications and attachments, permits and effluent data. (40 CFR 122.7(b)(2).)

## V. STANDARD PROVISIONS – REPORTING

### E. Duty to Provide Information

The Discharger shall furnish to the Regional Water Board, State Water Board, or USEPA within a reasonable time, any information which the Regional Water Board, State Water Board, or USEPA may request to determine whether cause exists for modifying, revoking and reissuing, or terminating this Order or to determine compliance with this Order. Upon request, the Discharger shall also furnish to the Regional Water Board, State Water Board, or USEPA copies of records required to be kept by this Order. (40 CFR 122.41(h); Wat. Code, § 13267.)

### F. Signatory and Certification Requirements

**1.** All applications, reports, or information submitted to the Regional Water Board, State Water Board, and/or USEPA shall be signed and certified in accordance with Standard Provisions – Reporting V.B.2, V.B.3, V.B.4, and V.B.5 below. (40 CFR 122.41(k).)

**2.** All permit applications shall be signed by either a principal executive officer or ranking elected official. For purposes of this provision, a principal executive officer of a federal agency includes: (i) the chief executive officer of the agency, or (ii) a senior executive officer having responsibility for the overall operations of a principal geographic unit of the agency (e.g., Regional Administrators of USEPA). (40 CFR 122.22(a)(3).).

**3.** All reports required by this Order and other information requested by the Regional Water Board, State Water Board, or USEPA shall be signed by a person described in Standard Provisions – Reporting V.B.2 above, or by a duly authorized representative of that person. A person is a duly authorized representative only if:

- **a.** The authorization is made in writing by a person described in Standard Provisions – Reporting V.B.2 above (40 CFR 122.22(b)(1));
- **b.** The authorization specifies either an individual or a position having responsibility for the overall operation of the regulated facility or activity such as the position of plant manager, operator of a well or a well field, superintendent, position of equivalent responsibility, or an individual or position having overall responsibility for environmental matters for the company. (A duly authorized representative may thus be either a named individual or any individual occupying a named position.) (40 CFR 122.22(b)(2)); and
- **c.** The written authorization is submitted to the Regional Water Board and State Water Board. (40 CFR 122.22(b)(3).)

**4.** If an authorization under Standard Provisions – Reporting V.B.3 above is no longer accurate because a different individual or position has responsibility for the overall operation of the facility, a new authorization satisfying the requirements of Standard

Provisions – Reporting V.B.3 above must be submitted to the Regional Water Board and State Water Board prior to or together with any reports, information, or applications, to be signed by an authorized representative. (40 CFR 122.22(c).)

**5.** Any person signing a document under Standard Provisions – Reporting V.B.2 or V.B.3 above shall make the following certification:

"*I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.*" (40 CFR 122.22(d).)

## G. Monitoring Reports

**1.** Monitoring results shall be reported at the intervals specified in the Monitoring and Reporting Program (Attachment E) in this Order. (40 CFR 122.22(l)(4).)

**2.** Monitoring results must be reported on a Discharge Monitoring Report (DMR) form or forms provided or specified by the Regional Water Board or State Water Board for reporting results of monitoring of sludge use or disposal practices. (40 CFR 122.41(l)(4)(i).)

**3.** If the Discharger monitors any pollutant more frequently than required by this Order using test procedures approved under 40 CFR Part 136 or, in the case of sludge use or disposal, approved under 40 CFR Part 136 unless otherwise specified in 40 CFR Part 503, or as specified in this Order, the results of this monitoring shall be included in the calculation and reporting of the data submitted in the DMR or sludge reporting form specified by the Regional Water Board. (40 CFR 122.41(l)(4)(ii).)

**4.** Calculations for all limitations, which require averaging of measurements, shall utilize an arithmetic mean unless otherwise specified in this Order. (40 CFR 122.41(l)(4)(iii).)

## H. Compliance Schedules

Reports of compliance or noncompliance with, or any progress reports on, interim and final requirements contained in any compliance schedule of this Order, shall be submitted no later than 14 days following each schedule date. (40 CFR 122.41(l)(5).)

## I. Twenty-Four Hour Reporting

**1.** The Discharger shall report any noncompliance that may endanger health or the environment. Any information shall be provided orally within 24 hours from the time the Discharger becomes aware of the circumstances. A written submission shall also be provided within five (5) days of the time the Discharger becomes aware of the circumstances. The written submission shall contain a description of the noncompliance and its cause; the period of noncompliance, including exact dates and times, and if the noncompliance has not been corrected, the anticipated time it is expected to continue; and steps taken or planned to reduce, eliminate, and prevent reoccurrence of the noncompliance. (40 CFR 122.41(l)(6)(i).)

**2.** The following shall be included as information that must be reported within 24 hours under this paragraph (40 CFR 122.41(l)(6)(ii)):

**a.** Any unanticipated bypass that exceeds any effluent limitation in this Order. (40 CFR 122.41(l)(6)(ii)(A).)

**b.** Any upset that exceeds any effluent limitation in this Order. (40 CFR 122.41(l)(6)(ii)(B).)

**3.** The Regional Water Board may waive the above-required written report under this provision on a case-by-case basis if an oral report has been received within 24 hours. (40 CFR 122.41(l)(6)(iii).)

## J. Planned Changes

The Discharger shall give notice to the Regional Water Board as soon as possible of any planned physical alterations or additions to the permitted facility. Notice is required under this provision only when (40 CFR 122.41(l)(1)):

**1.** The alteration or addition to a permitted facility may meet one of the criteria for determining whether a facility is a new source in 40 CFR 122.29(b) (40 CFR 122.41(l)(1)(i)); or

**2.** The alteration or addition could significantly change the nature or increase the quantity of pollutants discharged. This notification applies to pollutants that are subject neither to effluent limitations in this Order nor to notification requirements under 40 CFR 122.42(a)(1) (see Additional Provisions—Notification Levels VII.A.1). (40 CFR 122.41(l)(1)(ii).)

**3.** The alteration or addition results in a significant change in the Discharger's sludge use or disposal practices, and such alteration, addition, or change may justify the application of permit conditions that are different from or absent in the existing permit, including notification of additional use or disposal sites not reported during the permit application process or not reported pursuant to an approved land application plan. (40 CFR 122.41(l)(1)(iii).)

## K. Anticipated Noncompliance

The Discharger shall give advance notice to the Regional Water Board or State Water Board of any planned changes in the permitted facility or activity that may result in noncompliance with General Order requirements. (40 CFR 122.41(l)(2).)

**L. Other Noncompliance**

The Discharger shall report all instances of noncompliance not reported under Standard Provisions – Reporting V.C, V.D, and V.E above at the time monitoring reports are submitted. The reports shall contain the information listed in Standard Provision – Reporting V.E above. (40 CFR 122.41(l)(7).)

**M. Other Information**

When the Discharger becomes aware that it failed to submit any relevant facts in a permit application, or submitted incorrect information in a permit application or in any report to the Regional Water Board, State Water Board, or USEPA, the Discharger shall promptly submit such facts or information. (40 CFR 122.41(l)(8).)

## VI. STANDARD PROVISIONS – ENFORCEMENT

**A.** The Regional Water Board is authorized to enforce the terms of this permit under several provisions of the CWC, including, but not limited to, sections 13385, 13386, and 13387

## VII. ADDITIONAL PROVISIONS – NOTIFICATION LEVELS

**A. Publicly-Owned Treatment Works (POTWs)**

All POTWs shall provide adequate notice to the Regional Water Board of the following (40 CFR 122.42(b)):

**1.** Any new introduction of pollutants into the POTW from an indirect discharger that would be subject to sections 301 or 306 of the CWA if it were directly discharging those pollutants (40 CFR 122.42(b)(1)); and

**2.** Any substantial change in the volume or character of pollutants being introduced into that POTW by a source introducing pollutants into the POTW at the time of adoption of the Order. (40 CFR 122.42(b)(2).)

**3.** Adequate notice shall include information on the quality and quantity of effluent introduced into the POTW as well as any anticipated impact of the change on the quantity or quality of effluent to be discharged from the POTW. (40 CFR 122.42(b)(3).)

# ATTACHMENT E – MONITORING AND REPORTING PROGRAM

## Table of Contents



## List of Tables

# ATTACHMENT E – MONITORING AND REPORTING PROGRAM

Title 40 of the Code of Federal Regulations (CFR), section 122.48 (40 CFR 122.48) requires that all NPDES permits specify monitoring and reporting requirements. California Water Code (CWC) sections 13267 and 13383 also authorize the Regional Water Quality Control Board (Regional Water Board) to require technical and monitoring reports. This Monitoring and Reporting Program establishes monitoring and reporting requirements, which implement the federal and California regulations.

## I. GENERAL MONITORING PROVISIONS

**A.** Samples and measurements taken as required herein shall be representative of the volume and nature of the monitored discharge. All samples shall be taken at the monitoring locations specified below and, unless otherwise specified, before the monitored flow joins or is diluted by any other waste stream, body of water, or substance. Monitoring locations shall not be changed without notification to and the approval of this Regional Water Board.

**B.** Effluent samples shall be taken downstream of the last addition of wastes to the treatment or discharge works where a representative sample may be obtained prior to mixing with the receiving waters. Samples shall be collected at such a point and in such a manner to ensure a representative sample of the discharge.

**C.** Chemical, bacteriological, and bioassay analyses of any material required by this Order shall be conducted by a laboratory certified for such analyses by the Department of Public Health (DPH; formerly the Department of Health Services). Laboratories that perform sample analyses must be identified in all monitoring reports submitted to the Regional Water Board. In the event a certified laboratory is not available to the Discharger for any onsite field measurements such as pH, turbidity, temperature and residual chlorine, such analyses performed by a noncertified laboratory will be accepted provided a Quality Assurance-Quality Control Program is instituted by the laboratory. A manual containing the steps followed in this program for any onsite field measurements such as pH, turbidity, temperature and residual chlorine must be kept onsite in the treatment facility laboratory and shall be available for inspection by Regional Water Board staff. The Quality Assurance-Quality Control Program must conform to USEPA guidelines or to procedures approved by the Regional Water Board.

**D.** All chemical, bacteriological and bioassay analyses of any material required by this Order shall be performed in a laboratory certified to perform such analyses by DPH. Laboratories that perform sample analyses must be identified in all monitoring reports submitted to the Regional Water Board. The Discharger shall institute a Quality Assurance-Quality Control Program for any onsite field measurements such as pH, turbidity, temperature and residual chlorine. A manual containing the steps followed in this program must be kept onsite and shall be available for inspection by Regional Water Board staff. The Discharger must demonstrate sufficient capability (qualified and trained employees, properly calibrated and maintained field instruments, etc.) to adequately perform these field measurements. The Quality Assurance-Quality Control

Program must conform to USEPA guidelines or to procedures approved by the Regional Water Board.

**E.** Appropriate flow measurement devices and methods consistent with accepted scientific practices shall be selected and used to ensure the accuracy and reliability of measurements of the volume of monitored discharges. All monitoring instruments and devices used by the Discharger to fulfill the prescribed monitoring program shall be properly maintained and calibrated as necessary, at least yearly, to ensure their continued accuracy. All flow measurement devices shall be calibrated at least once per year to ensure continued accuracy of the devices.

**F.** Monitoring results, including noncompliance, shall be reported at intervals and in a manner specified in this Monitoring and Reporting Program.

**G.** Laboratories analyzing monitoring samples shall be certified by DPH, in accordance with the provision of CWC section 13176, and must include quality assurance/quality control data with their reports.

**H.** The Discharger shall conduct analysis on any sample provided by USEPA as part of the Discharge Monitoring Quality Assurance (DMQA) program. The results of any such analysis shall be submitted to USEPA's DMQA manager.

**I.** The Discharger shall file with the Regional Water Board technical reports on self-monitoring performed according to the detailed specifications contained in this Monitoring and Reporting Program.

**J.** The results of all monitoring required by this Order shall be reported to the Regional Water Board, and shall be submitted in such a format as to allow direct comparison with the limitations and requirements of this Order. Unless otherwise specified, discharge flows shall be reported in terms of the monthly average and the daily maximum discharge flows.

## II. MONITORING LOCATIONS

The Discharger shall establish the following monitoring locations to demonstrate compliance with the effluent limitations, discharge specifications, and other requirements in this Order:

**Table E-1. Monitoring Station Locations**

| Discharge Point Name | Monitoring Location Description |
|---|---|
| INF-001 | Influent sample location |
| EFF-001 | Effluent sample location, last connection before treated wastewater enters receiving water |
| RSW-001 | Receiving water (Lateral K) sample location upstream of discharge point. |
| RSW-002 | Receiving water (Lateral K) downstream sample location, 100 feet downstream of discharge point. |
| BIO-001 | Biosolids sample location |
| SPL-001 | Municipal water supply sample location |

## III. INFLUENT MONITORING REQUIREMENTS

### A. Monitoring Location INF-001

1. The Discharger shall monitor influent to the facility at INF-001 as follows:

**Table E-2. Influent Monitoring**

| Parameter | Units | Sample Type | Minimum Sampling Frequency |
|---|---|---|---|
| pH | | Grab | 1/Week |
| Biochemical Oxygen Demand (5-day, 25 °C) | mg/L, lbs/day | 24-hr Composite | 1/Week |
| Total Suspended Solids | mg/L, lbs/day | 24-hr Composite | 1/Week |
| CA Title 22 Metals (CAM 17) | µg/L | 24-hr Composite | 1/Year |
| Flow | mgd | Meter | Continuous |

## IV. EFFLUENT MONITORING REQUIREMENTS

### A. Monitoring Location EFF-001

1. The Discharger shall monitor treated wastewater, just prior to discharge into Lateral K at EFF-001 as follows. If more than one analytical test method is listed for a given parameter, the Discharger must select from the listed methods and corresponding Minimum Level:

**Table E-3. Effluent Monitoring**

| Parameter | Units | Sample Type | Minimum Sampling Frequency | Notes |
|---|---|---|---|---|
| Flow | mgd | Meter | Continuous | 1 |
| Biochemical Oxygen Demand (BOD) (5-day @ 20 Deg. C) | mg/L | 24-hr Composite | 1/Week | 1 |
| | lbs/day | Calculate | 1/Week | 1 |
| Total Suspended Solids | mg/L | 24-hr Composite | 1/Week | 1 |
| | lbs/day | Calculate | 1/Week | 1 |
| Temperature | °F | Grab | 1/Month | |
| Hardness | mg/L | Grab | 1/Month | 1,6 |
| pH | | Grab | 1/Week | 6 |
| Turbidity | NTU | Grab/Meter | 1/Week | |
| Total Coliform Organisms | MPN/100 mL | Grab | 1/Week | |
| EC @ 25°C | µmhos/cm | Grab | 1/Month | |
| Chlorine, Total Residual | mg/L | Grab/Meter | Continuous | 1,8 |
| Ammonia Nitrogen, Total (as N) | mg/L | Grab | 1/Week | 1,6,9,10 |
| Unionized ammonia | mg/L | Grab | 1/Month | 1,6 |
| Nitrate (as N) | mg/L | Grab | 1/Month | 1 |
| Copper, Total | µg/L | Grab | 1/Month | 1 |
| Standard Minerals | mg/L | Grab | 1/Year | 1,11 |
| Priority Pollutants | µg/L | 24-hr Composite | 4/Permit Term | 1,2,3,4,5,7 |
| Acute Toxicity | -- | Comp or Grab | 2/Year | |
| Whole Effluent Toxicity (see Section V. below) | -- | -- | -- | -- |

1 Pollutants shall be analyzed using the analytical methods described in 40 CFR Part 136.

2 In order to verify if bis (2-ethylhexyl) phthalate is truly present in the effluent discharge, the Discharger shall take steps to assure that sample containers, sampling apparatus, and analytical equipment are not sources of the detected contaminant.

3 For priority pollutant constituents with effluent limitations, detection limits shall be below the effluent limitations. If the lowest minimum level (ML) published in Appendix 4 of the Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California (State Implementation Plan or SIP) is not below the effluent limitation, the detection limit shall be the lowest ML. For priority pollutant constituents without effluent limitations, the detection limits shall be equal to or less than the lowest ML published in Appendix 4 of the SIP. Sampling and analysis of Bis (2-ethylhexyl) phthalate shall be conducted using ultra-clean techniques that eliminate the possibility of sample contamination.

4 Persistent chlorinated hydrocarbon pesticides include: aldrin, dieldrin, chlordane, endrin, endrin aldehyde, heptachlor, heptachlor epoxide, hexachlorocyclohexane (alpha-BHC, beta-BHC, delta-BHC, and gamma-BHC or lindane), endosulfan (alpha and beta), endosulfan sulfate, toxaphene, 4,4'DDD, 4,4'DDE, and 4,4'DDT.

5 Volatile constituents shall be sampled in accordance with 40 CFR Part 136.

6 Concurrent with receiving surface water sampling.

**7** **Priority pollutants shall be sampled quarterly during the third year** following the date of permit adoption and shall be conducted concurrently with upstream receiving water monitoring for hardness (as $CaCO_3$) and pH.

**8** Total chlorine residual must be monitored with a method sensitive to and accurate at the permitted level of 0.01 mg/L.

[9] Concurrent with whole effluent toxicity monitoring.
[10] pH and temperature shall be recorded at the time of ammonia sample collection.
[11] Standard minerals shall include the following: boron, calcium, iron, magnesium, potassium, sodium, chloride, manganese, phosphorus, total alkalinity (including alkalinity series), and hardness, and include verification that the analysis is complete (i.e., cation/anion balance).

## V. WHOLE EFFLUENT TOXICITY TESTING REQUIREMENTS

**A. Acute Toxicity Testing.** The Discharger shall conduct acute toxicity testing to determine whether the effluent is contributing acute toxicity to the receiving water. The Discharger shall meet the following acute toxicity testing requirements:

1. *Monitoring Frequency* – The Discharger shall perform **semi-annual** acute toxicity testing, concurrent with effluent ammonia sampling.

2. *Sample Types* – For static non-renewal and static renewal testing, the samples shall be flow proportional 24-hour composites or grab samples and shall be representative of the volume and quality of the discharge. The effluent samples shall be taken at the effluent monitoring location EFF-001.

3. *Test Species* – Test species shall be fathead minnows (Pimephales promelas).

4. *Methods* – The acute toxicity testing samples shall be analyzed using EPA-821-R-02-012, Fifth Edition. Temperature, total residual chlorine, and pH shall be recorded at the time of sample collection. No pH adjustment may be made unless approved by the Executive Officer.

5. *Test Failure* – If an acute toxicity test does not meet all test acceptability criteria, as specified in the test method, the Discharger must re-sample and re-test as soon as possible, not to exceed 7 days following notification of test failure.

**B. Chronic Toxicity Testing.** The Discharger shall conduct three species chronic toxicity testing to determine whether the effluent is contributing chronic toxicity to the receiving water. The Discharger shall meet the following chronic toxicity testing requirements:

1. *Monitoring Frequency –* The Discharger shall perform **annual** three species chronic toxicity testing.

2. *Sample Types* – Effluent samples shall be flow proportional 24-hour composites or grab samples and shall be representative of the volume and quality of the discharge. The effluent samples shall be taken at the effluent monitoring location EFF-001. The receiving water control shall be a grab sample obtained from the RSW-001 Upstream and out of influence of the discharge, as identified in this Monitoring and Reporting Program.

3. *Sample Volumes* – Adequate sample volumes shall be collected to provide renewal water to complete the test in the event that the discharge is intermittent.

**4.** *Test Species* – Chronic toxicity testing measures sublethal (e.g., reduced growth, reproduction) and/or lethal effects to test organisms exposed to an effluent compared to that of the control organisms. The Discharger shall conduct chronic toxicity tests with:

- The cladoceran, water flea, *Ceriodaphnia dubia* (survival and reproduction test);
- The fathead minnow, *Pimephales promelas* (larval survival and growth test); and
- The green alga, *Selenastrum capricornutum* (growth test).

**5.** *Methods* – The presence of chronic toxicity shall be estimated as specified in *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms, Fourth Edition*, EPA/821-R-02-013, October 2002.

**6.** *Reference Toxicant* – As required by the SIP, all chronic toxicity tests shall be conducted with concurrent testing with a reference toxicant and shall be reported with the chronic toxicity test results.

**7.** *Dilutions* – The chronic toxicity testing shall be performed using the dilution series identified in the table, below. The receiving water control shall be used as the diluent (unless the receiving water is toxic).

**Table E-4. Chronic Toxicity Testing Dilution Series**

| Sample | Dilutions (%) | | | | | Controls | |
|---|---|---|---|---|---|---|---|
| | 100 | 75 | 50 | 25 | 12.5 | Receiving Water | Laboratory Water |
| % Effluent | 100 | 75 | 50 | 25 | 12.5 | 0 | 0 |
| % Receiving Water | 0 | 25 | 50 | 75 | 87.5 | 100 | 0 |
| % Laboratory Water | 0 | 0 | 0 | 0 | 0 | 0 | 100 |

**7.** *Test Failure* – The Discharger must re-sample and re-test as soon as possible, but no later than fourteen (14) days after receiving notification of a test failure. A test failure is defined as follows:

**a.** The reference toxicant test or the effluent test does not meet all test acceptability criteria as specified in the *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms, Fourth Edition*, EPA/821-R-02-013, October 2002 (Method Manual), and its subsequent amendments or revisions; or

**b.** The percent minimum significant difference (PMSD) measured for the test exceeds the upper PMSD bound variability criterion in Table 6 on page 52 of the Method Manual. (A retest is only required in this case if the test results do not exceed the monitoring trigger specified in the Special Provision at section VI. 2.a.iii. of the Order.)

**C. WET Testing Notification Requirements.** The Discharger shall notify the Regional Water Board within 24-hours after the receipt of test results exceeding the monitoring trigger during regular or accelerated monitoring, or an exceedance of the acute toxicity effluent limitation.

**D. WET Testing Reporting Requirements.** All toxicity test reports shall include the contracting laboratory's complete report provided to the Discharger and shall be in accordance with the appropriate "Report Preparation and Test Review" sections of the method manuals. At a minimum, whole effluent toxicity monitoring shall be reported as follows:

1. **Chronic WET Reporting.** Regular chronic toxicity monitoring results shall be reported to the Regional Water Board within 30 days following completion of the test, and shall contain, at minimum:

    **a.** The results expressed in TUc, measured as 100/NOEC, and also measured as 100/LC50, 100/EC25, 100/IC25, and 100/IC50, as appropriate.

    **b.** The statistical methods used to calculate endpoints;

    **c.** The statistical output page, which includes the calculation of the percent minimum significant difference (PMSD);

    **d.** The dates of sample collection and initiation of each toxicity test; and

    **e.** The results compared to the numeric toxicity monitoring trigger.

    Additionally, the monthly discharger self-monitoring reports shall contain an updated chronology of chronic toxicity test results expressed in TUc, and organized by test species, type of test (survival, growth or reproduction), and monitoring frequency, i.e., either quarterly, monthly, accelerated, or Toxicity Reduction Evaluation (TRE).

2. **Acute WET Reporting.** Acute toxicity test results shall be submitted with the monthly discharger self-monitoring reports and reported as percent survival.

3. **TRE Reporting.** Reports for TREs shall be submitted in accordance with the schedule contained in the Discharger's approved TRE Workplan.

4. **Quality Assurance (QA).** The Discharger must provide the following information for QA purposes :

    **a.** Results of the applicable reference toxicant data with the statistical output page giving the species, NOEC, LOEC, type of toxicant, dilution water used, concentrations used, PMSD, and dates tested.

    **b.** The reference toxicant control charts for each endpoint, which include summaries of reference toxicant tests performed by the contracting laboratory.

**c.** Any information on deviations or problems encountered and how they were dealt with.

## VI. LAND DISCHARGE MONITORING REQUIREMENTS

### A. Pond Monitoring

**1.** Pond/lagoon monitoring shall be conducted when water is present in the pond(s)/lagoon(s). All samples shall be grab samples. The Discharger shall monitor all ponds, at a minimum as follows**:**

**Table E-5. Summary of Pond Monitoring**

| Parameter | Units | Sample Type | Sampling Frequency |
|---|---|---|---|
| Freeboard | Feet | Measure | 1/Week |
| Dissolved Oxygen | mg/L | Grab/Meter | 1/Week |
| General Conditions | Visual | N/A | 1/Week |

## VII. RECLAMATION MONITORING REQUIREMENTS – NOT APPLICABLE

## VIII. RECEIVING WATER MONITORING REQUIREMENTS – SURFACE WATER

### A. Monitoring Location RSW-001 and RSW-002

**1.** The Discharger shall monitor Lateral K at RSW-001 and RSW-002 as follows:

**Table E-6. Receiving Water Monitoring Requirements**

| Parameter | Units | Sample Type | Minimum Sampling Frequency |
|---|---|---|---|
| pH | | Grab/Meter | 1/Week |
| Turbidity | NTU | Grab | 1/Week |
| Dissolved Oxygen | mg/L | Grab/Meter | 1/Month |
| Electrical Conductivity | µmhos/cm | Grab/Meter | 1/Month |
| Hardness (RSW-001 only) | mg/L | Grab | 1/Month (concurrent with effluent monitoring) |
| Priority Pollutants **(RSW-001 only)** | µg/L | 24-hr Composite | 4/Permit Term **(concurrent with effluent monitoring)** |

## IX. OTHER MONITORING REQUIREMENTS

### A. Biosolids

#### 1. Monitoring Location BIO-001

**a.** A composite sample of sludge shall be collected annually at Monitoring Location BIO-001 in accordance with USEPA's *POTW Sludge Sampling and Analysis Guidance Document*, August 1989, and tested for the metals listed in Title 22.

**b.** Sampling records shall be retained for a minimum of **5 years**. A log shall be maintained of sludge quantities generated and of handling and disposal activities. The frequency of entries is discretionary; however, the log must be complete enough to serve as a basis for part of the annual report.

### B. Municipal Water Supply

#### 1. Monitoring Location SPL-001

The Discharger shall monitor the municipal water supply at SPL-001 as follows. A sampling station shall be established where a representative sample of the municipal water supply can be obtained. Municipal water supply samples shall be collected at approximately the same time as effluent samples.

**Table E-7. Municipal Water Supply Monitoring Requirements**

| Parameter | Units | Sample Type | Minimum Sampling Frequency |
|---|---|---|---|
| Total Dissolved Solids | mg/L | Grab | 1/year |
| Electrical Conductivity @ 25°C | µmhos/cm | Grab | 1/year |
| Standard Minerals | mg/L | Grab | 1/year |

## X. REPORTING REQUIREMENTS

### A. General Monitoring and Reporting Requirements

**1.** The Discharger shall comply with all Standard Provisions (Attachment D) related to monitoring, reporting, and recordkeeping.

**2.** Upon written request of the Regional Water Board, the Discharger shall submit a summary monitoring report. The report shall contain both tabular and graphical summaries of the monitoring data obtained during the previous year(s).

**3. Compliance Time Schedules.** For compliance time schedules included in the Order, the Discharger shall submit to the Regional Water Board, on or before each compliance due date, the specified document or a written report detailing compliance or noncompliance with the specific date and task. If noncompliance is reported, the Discharger shall state the reasons for noncompliance and include an estimate of the date when the Discharger will be in compliance. The Discharger shall notify the Regional Water Board by letter when it returns to compliance with the compliance time schedule.

**4.** The Discharger shall report to the Regional Water Board any toxic chemical release data it reports to the State Emergency Response Commission within 15 days of reporting the data to the Commission pursuant to section 313 of the "*Emergency Planning and Community Right to Know Act*" of 1986.

**B. Self Monitoring Reports (SMRs)**

**1.** At any time during the term of this permit, the State Water Board or the Regional Water Board may notify the Discharger to electronically submit Self-Monitoring Reports (SMRs) using the State Water Board's California Integrated Water Quality System (CIWQS) Program Web site (http://www.waterboards.ca.gov/ciwqs/index.html). Until such notification is given, the Discharger shall submit hard copy SMRs. The CIWQS Web site will provide additional directions for SMR submittal in the event there will be service interruption for electronic submittal.

**2.** The Discharger shall report in the SMR the results for all monitoring specified in this Monitoring and Reporting Program under sections III through IX. The Discharger shall submit monthly SMRs including the results of all required monitoring using USEPA-approved test methods or other test methods specified in this Order. If the Discharger monitors any pollutant more frequently than required by this Order, the results of this monitoring shall be included in the calculations and reporting of the data submitted in the SMR.

**3.** Monitoring periods and reporting for all required monitoring shall be completed according to the following schedule:

**Table E-8. Monitoring Periods and Reporting Schedule**

| Sampling Frequency | Monitoring Period Begins On… | Monitoring Period | SMR Due Date |
|---|---|---|---|
| Continuous | **Permit effective date** | All | Submit with monthly SMR |
| 1/Hour | **Permit effective date** | Hourly | Submit with monthly SMR |
| 1/Day | **Permit effective date** | (Midnight through 11:59 PM) or any 24-hour period that reasonably represents a calendar day for purposes of sampling. | Submit with monthly SMR |
| 1/Week | **Permit effective date** | Sunday through Saturday | Submit with monthly SMR |
| 1/Month | **Permit effective date** | First day of calendar month through last day of calendar month | 32 days from the end of the monitoring period |
| 1/Quarter | **Permit effective date** | 1 January through 1 March<br>1 April through 30 June<br>1 July through 30 September<br>1 October through 31 December | 32 days from the end of the monitoring period |
| 2/Year | **Permit effective date** | 1 January through 30 June<br>1 July through 31 December | 32 days from the end of the monitoring period |
| 1/Year | **Permit effective date** | 1 January through 31 December | 32 days from the end of the monitoring period |

4. **Reporting Protocols.** The Discharger shall report with each sample result the applicable reported Minimum Level (ML) and the current Method Detection Limit (MDL), as determined by the procedure in 40 CFR Part 136.

   The Discharger shall report the results of analytical determinations for the presence of chemical constituents in a sample using the following reporting protocols:

   **a.** Sample results greater than or equal to the reported ML shall be reported as measured by the laboratory (i.e., the measured chemical concentration in the sample).

   **b.** Sample results less than the RL, but greater than or equal to the laboratory's MDL, shall be reported as "Detected, but Not Quantified," or DNQ. The estimated chemical concentration of the sample shall also be reported.

   For the purposes of data collection, the laboratory shall write the estimated chemical concentration next to DNQ as well as the words "Estimated Concentration" (may be shortened to "Est. Conc."). The laboratory may, if such information is available, include numerical estimates of the data quality for the reported result. Numerical estimates of data quality may be percent accuracy (+ a percentage of the reported value), numerical ranges (low to high), or any other means considered appropriate by the laboratory.

**c.** Sample results less than the laboratory's MDL shall be reported as "Not Detected," or ND.

**d.** Dischargers are to instruct laboratories to establish calibration standards so that the ML value (or its equivalent if there is differential treatment of samples relative to calibration standards) is the lowest calibration standard. At no time is the Discharger to use analytical data derived from extrapolation beyond the lowest point of the calibration curve.

**5. Compliance Determination.** Compliance with effluent limitations for priority pollutants shall be determined using sample reporting protocols defined above and in Attachment of this Order. For purposes of reporting and administrative enforcement by the Regional Water Board and the State Water Board, the Discharger shall be deemed out of compliance with effluent limitations if the concentration of the priority pollutant in the monitoring sample is greater than the effluent limitation and greater than or equal to the reporting level (RL).

**6. Multiple Sample Data.** When determining compliance with an AMEL, AWEL, or MDEL for priority pollutants and more than one sample result is available, the Discharger shall compute the arithmetic mean unless the data set contains one or more reported determinations of "Detected, but Not Quantified" (DNQ) or "Not Detected" (ND). In those cases, the Discharger shall compute the median in place of the arithmetic mean in accordance with the following procedure:

**a.** The data set shall be ranked from low to high, ranking the reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.

**b.** The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ.

**7.** The Discharger shall submit SMRs in accordance with the following requirements:

**a.** The Discharger shall arrange all reported data in a tabular format. The data shall be summarized to clearly illustrate whether the facility is operating in compliance with interim and/or final effluent limitations. The Discharger is not required to duplicate the submittal of data that is entered in a tabular format within CIWQS. When electronic submittal of data is required and CIWQS does not provide for entry into a tabular format within the system, the Discharger shall electronically submit the data in a tabular format as an attachment.

**b.** The Discharger shall attach a cover letter to the SMR. The information contained in the cover letter shall clearly identify violations of the WDRs; discuss corrective actions taken or planned; and the proposed time schedule for corrective actions.

Identified violations must include a description of the requirement that was violated and a description of the violation.

**c.** SMRs must be submitted to the Regional Water Board, signed and certified as required by the Standard Provisions (Attachment D), to the address listed below:

Regional Water Quality Control Board
Central Valley Region
NPDES Compliance and Enforcement Unit
364 Knollcrest Drive, Suite #205
Redding, CA 96002

**8.** Reports must clearly show when discharging to EFF-001 or other permitted discharge locations. Reports must show the date and time that the discharge started and stopped at each location.

## C. Discharge Monitoring Reports (DMRs)

**1.** As described in section X.B.1 above, at any time during the term of this permit, the State Water Board or Regional Water Board may notify the Discharger to electronically submit SMRs that will satisfy federal requirements for submittal of Discharge Monitoring Reports (DMRs). Until such notification is given, the Discharger shall submit DMRs in accordance with the requirements described below.

**2.** DMRs must be signed and certified as required by the standard provisions (Attachment D). The Discharger shall submit the original DMR and one copy of the DMR to the address listed below:

| STANDARD MAIL | FEDEX/UPS/<br>OTHER PRIVATE CARRIERS |
|---|---|
| State Water Resources Control Board<br>Division of Water Quality<br>c/o DMR Processing Center<br>PO Box 100<br>Sacramento, CA 95812-1000 | State Water Resources Control Board<br>Division of Water Quality<br>c/o DMR Processing Center<br>1001 I Street, 15$^{th}$ Floor<br>Sacramento, CA 95814 |

**3.** All discharge monitoring results must be reported on the official USEPA pre-printed DMR forms (EPA Form 3320-1). Forms that are self-generated will not be accepted unless they follow the exact same format of EPA Form 3320-1.

## D. Other Reports

**1.** **Annual Progress Reports.** As specified in the compliance time schedules required in the Special Provisions contained in section VI of the Order, progress reports shall be submitted in accordance with the following reporting requirements. At minimum, the progress reports shall include a discussion of the status of final compliance, whether the Discharger is on schedule to meet the final compliance date, and the remaining tasks to meet the final compliance date.

**Table E-9. Reporting Requirements for Special Provision Reports**

| **Special Provision** | **Reporting Requirements** |
|---|---|
| Filtration and Turbidity Compliance Schedule (Section VI.C.7.a.) | **1 February**, annually, until final compliance |

2. The Discharger shall report the results of any special studies, acute and chronic toxicity testing, TRE/TIE, PMP, and Pollution Prevention Plan required by Special Provisions VI.C. of this Order. The Discharger shall report the progress in satisfaction of compliance schedule dates specified in the Special Provision at section VI.C.7 of this Order. The Discharger shall submit reports with the first monthly SMR scheduled to be submitted on or immediately following the report due date and/or in compliance with SMR reporting requirements described in subsection X.B.5 above.

3. **Within 60 days of permit adoption**, the Discharger shall submit a report outlining minimum levels, method detection limits, and analytical methods for approval, with a goal to achieve detection levels below applicable water quality criteria. At a minimum, the Discharger shall comply with the monitoring requirements for CTR constituents as outlined in section 2.3 and 2.4 of the SIP.

4. The Discharger's sanitary sewer system collects wastewater using sewers, pipes, pumps, and/or other conveyance systems and directs the raw sewage to the wastewater treatment plant. A "sanitary sewer overflow" is defined as a discharge to ground or surface water from the sanitary sewer system at any point upstream of the wastewater treatment plant. Sanitary sewer overflows are prohibited by this Order. All violations must be reported as required in Standard Provisions. Facilities (such as wet wells, regulated impoundments, tanks, highlines, etc.) may be part of a sanitary sewer system and discharges to these facilities are not considered sanitary sewer overflows, provided that the waste is fully contained within these temporary storage facilities.

5. **Annual Operations Report. By 30 January of each year**, the Discharger shall submit a written report to the Executive Officer containing the following:

   a. The names, certificate grades, and general responsibilities of all persons employed at the Facility.

   b. The names and telephone numbers of persons to contact regarding the plant for emergency and routine situations.

**c.** A statement certifying when the flow meter(s) and other monitoring instruments and devices were last calibrated, including identification of who performed the calibration.

**d.** A statement certifying whether the current operation and maintenance manual, and contingency plan, reflect the wastewater treatment plant as currently constructed and operated, and the dates when these documents were last revised and last reviewed for adequacy.

**e.** The Discharger may also be requested to submit an annual report to the Regional Water Board with both tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss the compliance record. If violations have occurred, the report shall also discuss the corrective actions taken and planned to bring the discharge into full compliance with the waste discharge requirements.

# ATTACHMENT F – FACT SHEET

## Table of Contents

**List of Tables**

# ATTACHMENT F – FACT SHEET

As described in the Findings in section II of this Order, this Fact Sheet includes the legal requirements and technical rationale that serve as the basis for the requirements of this Order.

This Order has been prepared under a standardized format to accommodate a broad range of discharge requirements for Dischargers in California. Only those sections or subsections of this Order that are specifically identified as "not applicable" have been determined not to apply to this Discharger. Sections or subsections of this Order not specifically identified as "not applicable" are fully applicable to this Discharger.

## I. PERMIT INFORMATION

The following table summarizes administrative information related to the Facility.

**Table F-1. Facility Information**

| | |
|---|---|
| **WDID** | 5A040100001 |
| **Discharger** | City of Biggs |
| **Name of Facility** | City of Biggs Wastewater Treatment Facility - Biggs |
| **Facility Address** | 3016 Sixth Street |
| | Biggs, CA 95917 |
| | Butte County |
| **Facility Contact, Title and Phone** | Mr. Mark Sorensen, City Administrator, (530) 868-5493 |
| **Authorized Person to Sign and Submit Reports** | Mr. Hayden Wasser, Plant Operator, (530) 868-5685 |
| **Mailing Address** | P.O. Box 307, Biggs, CA 95917 |
| **Billing Address** | same |
| **Type of Facility** | POTW |
| **Major or Minor Facility** | Minor |
| **Threat to Water Quality** | 2 |
| **Complexity** | B |
| **Pretreatment Program** | N |
| **Reclamation Requirements** | Not Applicable |
| **Facility Permitted Flow** | 0.38 million gallons per day (mgd) |
| **Facility Design Flow** | 0.38 mgd |
| **Watershed** | Sacramento River |
| **Receiving Water** | Lateral K Agricultural Drain – Reclamation District #833 |
| **Receiving Water Type** | Agricultural Drain/Canal |

**A.** The City of Biggs (hereinafter Discharger) is the owner and operator of the City of Biggs Wastewater Treatment Facility (hereinafter Facility), a secondary treatment wastewater plant.

For the purposes of this Order, references to the "discharger" or "permittee" in applicable federal and state laws, regulations, plans, or policy are held to be equivalent to references to the Discharger herein.

**B.** The Facility discharges wastewater to Lateral K, an agricultural drain, and is currently regulated by Order No. R5-2007-0032 which was adopted on 3 May 2007 and expired on 1 June 2012. The terms and conditions of the current Order have been automatically continued and remain in effect until new Waste Discharge Requirements (WDRs) and National Pollutant Discharge Elimination System (NPDES) permit are adopted pursuant to this Order.

**C.** The Discharger filed a report of waste discharge and submitted an application for renewal of its WDRs and NPDES permit on 20 December 2011. Supplemental information was requested on 30 December 2011 and received on 25 January 2012. A site visit was conducted on 7 June 2012 to observe operations and collect additional data to develop permit limitations and conditions.

## II. FACILITY DESCRIPTION

The Discharger provides sewerage service for the community of Biggs and serves a population of approximately 1,800. The design daily average flow capacity of the Facility is 0.38 million gallons per day (mgd).

### A. Description of Wastewater and Biosolids Treatment or Controls

The treatment system at the Facility consists of two aerated lagoons, a ballast pond, three plug flow rock filters in parallel, chlorination/dechlorination facilities, and a sludge drying bed. Dried biosolids are hauled to a landfill.

### B. Discharge Points and Receiving Waters

1. The Facility is located in Section 14, T18N, R2E, MDB&M, as shown in Attachment B, a part of this Order.

2. Treated municipal wastewater is discharged at Discharge Point No. EFF-001 to Lateral K a tributary to Butte Creek, a downstream water body at a point latitude 39° 24’ 28” N and longitude 121° 43’ 32” W.

### C. Summary of Existing Requirements and Self-Monitoring Report (SMR) Data

**Effluent limitations and/or Discharge Specifications** contained in Order No. R5-2007-0032 for discharges from Discharge Point No. EFF-001 and representative monitoring data from the term of Order No. R5-2007-0032 are as follows:

**Table F-2. Historic Effluent Limitations and Monitoring Data**

| Parameter | Units | Effluent Limitation | | | | Monitoring Data From July 2007 To November 2011 | | |
|---|---|---|---|---|---|---|---|---|
| | | **Average Monthly** | **Average Weekly** | **Minimum Daily** | **Maximum Daily** | **Average Monthly Discharge** | **Average Weekly Discharge** | **Highest Daily Discharge** |
| Flow | mgd | 0.38 | | | | 0.3 | | 1.0 |
| $BOD_5$ | mg/L | 30 | 45 | | 90 | 17.6 | | 38.0 |
| | lbs/day | 95 | 143 | | 285 | 34.2 | | 127.0 |
| | % removal | 65 | | | | 78 | | 43 to 93 (range) |
| Total Suspended Solids | mgd | 30 | 45 | | 90 | 8.1 | | 22.0 |
| | lbs/day | 95 | 143 | | 285 | 17.3 | | 112.0 |
| | % removal | 65 | | | | 90 | | 73 to 98 (range) |
| pH | | | | 6.0 | 9.0 | 7.3 | | 7.8 |
| Ammonia | mg/L | 2.72 | | | 7.44 | 7.9 | | 18.0 |
| EC | µmhos/cm | 900 | | | | 735.2 (average of average monthly) | 566.5 - 920.0 (range of average monthly) | |
| | | | | | | **Monitoring Data From January 2010 to November 2011** | | |
| Total Coliform Organisms | MPN/100 mL | 23 Once in 30 days | | | 500 | <2.0 | <2.0 | 16 |

## D. Compliance Summary

The City of Biggs received a *Notice of Violation of Permit Conditions* on 20 May 2011 for violating ammonia effluent limitations in Waste Discharger Requirements Order No. R5-2007-0032, during the period from January 2009 to February 2011. The violations included 26 Group I violations, and 42 non-serious violations, which exceeded the average monthly and maximum daily effluent limits for ammonia. The mandatory minimum penalty for these violations is assessed at $201,000.

## E. Planned Changes

There are no immediate changes planned for the facility; however, in 2009, the City of Biggs prepared the *Revised Final City of Biggs Wastewater Treatment Plant Planning Alternatives Study (Alternatives Study)* that was prepared in response to compliance concerns, specifically with ammonia. The City of Biggs is in the process of evaluating a number of improvements to the facility to meet current and future permit requirements. Some of the necessary upgrades include: constructing a headworks, improving influent flow equalization, adding nitrification and denitrification to the treatment process, and improving the disinfection and sludge handling systems. The City of Biggs is also evaluating land disposal options.

# III. APPLICABLE PLANS, POLICIES, AND REGULATIONS

The requirements contained in this Order are based on the applicable plans, policies, and regulations identified in the Findings in section II of this Order. The applicable plans, policies, and regulations relevant to the discharge include the following:

## A. Legal Authorities

This Order is issued pursuant to regulations in the Clean Water Act (CWA) and the California Water Code (CWC) as specified in the Finding contained at section II.C of this Order.

## B. California Environmental Quality Act (CEQA)

This Order meets the requirements of CEQA as specified in the Finding contained at section II.E of this Order.

## C. State and Federal Regulations, Policies, and Plans

1. **Water Quality Control Plans.** This Order implements the following water quality control plans as specified in the Finding contained at section II.H of this Order.

    a. *Water Quality Control Plan, Fourth Edition (Revised February 2007), for the Sacramento and San Joaquin River Basins* (Basin Plan)

2. **National Toxics Rule (NTR) and California Toxics Rule (CTR).** This Order implements the NTR and CTR as specified in the Finding contained at section II.I of this Order.

3. **State Implementation Policy (SIP).** This Order implements the SIP as specified in the Finding contained at section II.I of this Order.

4. **Alaska Rule.** This Order is consistent with the Alaska Rule as specified in the Finding contained at section II.L of this Order.

5. **Antidegradation Policy.** As specified in the Finding contained at section II.N of this Order and as discussed in detail in the Fact Sheet (Attachment F, Section IV.D.4.), the discharge is consistent with the antidegradation provisions of 40 CFR section 131.12 and State Water Resources Control Board (State Water Board) Resolution 68-16.

6. **Anti-Backsliding Requirements.** This Order is consistent with anti-backsliding policies as specified in the Finding contained at section II.M of this Order. Compliance with the anti-backsliding requirements is discussed in the Fact Sheet (Attachment F, Section IV.D.3).

7. **Emergency Planning and Community Right to Know Act**

Section 13263.6(a) of the CWC, requires that "*the Regional Water Board shall prescribe effluent limitations as part of the waste discharge requirements of a POTW for all substances that the most recent toxic chemical release data reported to the state emergency response commission pursuant to Section 313 of the Emergency Planning and Community Right to Know Act of 1986 (42 U.S.C. Sec. 11023) (EPCRA) indicate as discharged into the POTW, for which the State Water Board or the Regional Water Board has established numeric water quality objectives, and has determined that the discharge is or may be discharged at a level which will cause, have the reasonable potential to cause, or contribute to, an excursion above any numeric water quality objective*".

The Regional Water Board has adopted numeric water quality objectives in the Basin Plan for the following constituents: ammonia, and copper, for which numeric water quality objectives have been adopted for the receiving waters involved in this discharge. As detailed elsewhere in this Permit, available effluent quality data indicate that none of these constituents have a reasonable potential to cause or contribute to an excursion above any numeric water quality objectives included within the Basin Plan or in any State Water Board plan, so no effluent limitations are included in this permit pursuant to CWC Section 13263.6(a).

8. **Storm Water Requirements**

USEPA promulgated federal regulations for storm water on 16 November 1990 in 40 CFR Parts 122, 123, and 124. The NPDES Industrial Storm Water Program regulates storm water discharges from wastewater treatment facilities. Wastewater treatment plants are applicable industries under the storm water program and are obligated to comply with the federal regulations.

9. **Endangered Species Act.** This Order is consistent with the Endangered Species Act as specified in the Finding contained at section II.P of this Order.

**D. Impaired Water Bodies on CWA 303(d) List**

1. Under section 303(d) of the 1972 CWA, states, territories and authorized tribes are required to develop lists of water quality limited segments. The waters on these lists do not meet water quality standards, even after point sources of pollution have installed the minimum required levels of pollution control technology. On 30 November 2006 USEPA gave final approval to California's 2006 section 303(d) List of Water Quality Limited Segments. The Basin Plan references this list of Water Quality Limited Segments (WQLSs), which are defined as "…*those sections of lakes, streams, rivers or other fresh water bodies where water quality does not meet (or is not expected to meet) water quality standards even after the application of appropriate limitations for point sources* (40 CFR Part 130, et seq.)." The Basin Plan also states, "*Additional treatment beyond minimum federal standards will be imposed on dischargers to [WQLSs]. Dischargers will be assigned or allocated a maximum allowable load of critical pollutants so that water quality objectives can be met in the segment*." Lateral K is not listed in the 303(d) list of impaired water

bodies, and is not currently scheduled for a Total Maximum Daily Limit analysis (TMDL).

2. **Total Maximum Daily Loads (TMDLs).** USEPA requires the Regional Water Board to develop TMDLs for each 303(d) listed pollutant and water body combination. The 303(d) listings and TMDLs have been considered in the development of the Order. A pollutant-by-pollutant evaluation of each pollutant of concern is described in section VI.C.3. of this Fact Sheet.

### E. Other Plans, Polices and Regulations

1. The discharge authorized herein and the treatment and storage facilities associated with the discharge of treated municipal wastewater, except for discharges of residual sludge and solid waste, are exempt from the requirements of Title 27, California Code of Regulations (CCR), section 20005 *et seq.* (hereafter Title 27). The exemption, pursuant to Title 27 CCR section 20090(a), is based on the following:

   **a.** The waste consists of domestic sewage and treated effluent;

   **b.** The waste discharge requirements are consistent with water quality objectives; and

   **c.** The treatment and storage facilities described herein are associated with a municipal wastewater treatment plant.

## IV. RATIONALE FOR EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

Effluent limitations and toxic and pretreatment effluent standards established pursuant to sections 301 (Effluent Limitations), 302 (Water Quality Related Effluent Limitations), 304 (Information and Guidelines), and 307 (Toxic and Pretreatment Effluent Standards) of the CWA and amendments thereto are applicable to the discharge.

The CWA mandates the implementation of effluent limitations that are as stringent as necessary to meet water quality standards established pursuant to state or federal law [33 U.S.C., §1311(b)(1)(C); 40 CFR 122.44(d)(1)]. NPDES permits must incorporate discharge limits necessary to ensure that water quality standards are met. This requirement applies to narrative criteria as well as to criteria specifying maximum amounts of particular pollutants. Pursuant to federal regulations, 40 CFR 122.44(d)(1)(i), NPDES permits must contain limits that control all pollutants that "*are or may be discharged at a level which will cause, have the reasonable potential to cause, or contribute to an excursion above any state water quality standard, including state narrative criteria for water quality*." Federal regulations, 40 CFR 122.44(d)(1)(vi), further provide that "[w]*here a state has not established a water quality criterion for a specific chemical pollutant that is present in an effluent at a concentration that causes, has the reasonable potential to cause, or contributes to an excursion above a narrative criterion within an applicable State water quality standard, the permitting authority must establish effluent limits*."

The CWA requires point source dischargers to control the amount of conventional, non-conventional, and toxic pollutants that are discharged into the waters of the United States. The control of pollutants discharged is established through effluent limitations and other requirements in NPDES permits. There are two principal bases for effluent limitations in the Code of Federal Regulations: 40 CFR 122.44(a) requires that permits include applicable technology-based limitations and standards; and 40 CFR 122.44(d) requires that permits include WQBELs to attain and maintain applicable numeric and narrative water quality criteria to protect the beneficial uses of the receiving water where numeric water quality objectives have not been established. The Basin Plan at page IV-17.00**,** contains an implementation policy, "*Policy for Application of Water Quality Objectives*", that specifies that the Regional Water Board "*will, on a case-by-case basis, adopt numerical limitations in orders which will implement the narrative objectives*." This Policy complies with 40 CFR 122.44(d)(1). With respect to narrative objectives, the Regional Water Board must establish effluent limitations using one or more of three specified sources, including: (1) USEPA's published water quality criteria, (2) a proposed state criterion (i.e., water quality objective) or an explicit state policy interpreting its narrative water quality criteria (i.e., the Regional Water Board's "*Policy for Application of Water Quality Objectives*")(40 CFR 122.44(d)(1)(vi)(A), (B) or (C)), or (3) an indicator parameter.

The Basin Plan includes numeric site-specific water quality objectives and narrative objectives for toxicity, chemical constituents, discoloration, radionuclides, and tastes and odors. The narrative toxicity objective states: "*All waters shall be maintained free of toxic substances in concentrations that produce detrimental physiological responses in human, plant, animal, or aquatic life*." (Basin Plan at III-8.00.) The Basin Plan states that material and relevant information, including numeric criteria, and recommendations from other agencies and scientific literature will be utilized in evaluating compliance with the narrative toxicity objective. The narrative chemical constituents objective states that waters shall not contain chemical constituents in concentrations that adversely affect beneficial uses. At minimum, "…*water designated for use as domestic or municipal supply (MUN) shall not contain concentrations of chemical constituents in excess of the maximum contaminant levels (MCLs)" in Title 22 of CCR.* The Basin Plan further states that, to protect all beneficial uses, the Regional Water Board may apply limits more stringent than MCLs. The narrative tastes and odors objective states: *"Water shall not contain taste- or odor-producing substances in concentrations that impart undesirable tastes or odors to domestic or municipal water supplies or to fish flesh or other edible products of aquatic origin, or that cause nuisance, or otherwise adversely affect beneficial uses*."

### A. Discharge Prohibitions

**1.** As stated in section I.G of Attachment D, Standard Provisions, this Order prohibits bypass from any portion of the treatment facility. Federal regulations, 40 CFR 122.41(m), define "bypass" as the intentional diversion of waste streams from any portion of a treatment facility. This section of the federal regulations, 40 CFR 122.41(m)(4), prohibits bypass unless it is unavoidable to prevent loss of life, personal injury, or severe property damage. In considering the Regional Water Board's prohibition of bypasses, the State Water Board adopted a precedential decision, Order No. WQO 2002-0015, which cites the federal regulations, 40 CFR 122.41(m), as allowing bypass only for essential maintenance to assure efficient operation.

## B. Technology-Based Effluent Limitations

### 1. Scope and Authority

Section 301(b) of the CWA and implementing USEPA permit regulations at 40 CFR 122.44 require that permits include conditions meeting applicable technology-based requirements at a minimum, and any more stringent effluent limitations necessary to meet applicable water quality standards. The discharge authorized by this Order must meet minimum federal technology-based requirements based on Secondary Treatment Standards at 40 CFR Part 133, Effluent Limitations Guidelines and Standards in 40 CFR Part and/or Best Professional Judgment (BPJ) in accordance with 40 CFR 125.3.

Following publication of the secondary treatment regulations, legislative history indicates that Congress was concerned that USEPA had not "sanctioned" the use of certain biological treatment techniques that were effective in achieving significant reductions in 5-day biochemical oxygen demand ($BOD_5$) and total suspended solids (TSS) for secondary treatment. Therefore to prevent unnecessary construction of costly new facilities, Congress included language in the 1981 amendment to the Construction Grants statues [Section 23 of Pub. L. 97-147] that required USEPA to provide allowance for alternative biological treatment technologies such as trickling filters or waste stabilization ponds. In response to this requirement, definition of secondary treatment was modified on 20 September 1984 and 3 June 1985, and published in the revised secondary treatment regulations contained in 40 CFR 133.105. These regulations allow alternative limitations for facilities using trickling filters and waste stabilization ponds that meet the requirements for "equivalent to secondary treatment." These "equivalent to secondary treatment" limitations are *up to* 45 mg/L (monthly average) and *up to* 65 mg/L (weekly average) for $BOD_5$ and TSS.

Therefore, POTWs that use waste stabilization ponds, identified in 40 CFR 133.103, as the principal process for secondary treatment and whose operation and maintenance data indicate that the TSS values specified in the equivalent-to-secondary regulations cannot be achieved, can qualify to have their minimum levels of effluent quality for TSS adjusted upwards.

Furthermore, in order to address the variations in facility performance due to geographic, climatic, or seasonal conditions in different States, the Alternative State Requirements (ASR) provision contained in 40 CFR 133.105(d) was written. ASR allows States the flexibility to set permit limitations above the maximum levels of 45 mg/L (monthly average) and 65 mg/L (weekly average) for TSS from lagoons. However, before ASR limitations for suspended solids can be set, the effluent must meet the $BOD_5$ limitations as prescribed by 40 CFR 133.102(a). Presently, the maximum TSS value set by the State of California for lagoon effluent is 95 mg/L.

This value corresponds to a 30-day consecutive average or an average over duration of less than 30 days.

In order to be eligible for equivalent-to-secondary limitations, a POTW must meet all of the following criteria:

**a.** The principal treatment process must be either a trickling filter or waste stabilization pond.

**b.** The effluent quality consistently achieved, despite proper operations and maintenance, is in excess of 30 mg/L $BOD_5$ and TSS.

**c.** Water quality is not adversely affected by the discharge. (40 CFR 133.101(g))

The treatment works as a whole provides significant biological treatment such that a minimum 65 percent reduction of $BOD_5$ is consistently attained (30-day average).

**2. Applicable Technology-Based Effluent Limitations**

**a.** **$BOD_5$ and TSS.** Federal regulations, 40 CFR Part 133, establish the minimum weekly and monthly average level of effluent quality attainable by secondary treatment for $BOD_5$ and TSS. $BOD_5$ is a measure of the amount of oxygen used in the biochemical oxidation of organic matter. The secondary and tertiary treatment standards for $BOD_5$ and TSS are indicators of the effectiveness of the treatment processes. The principal design parameter for wastewater treatment plants is the daily $BOD_5$ and TSS loading rates and the corresponding removal rate of the system. In addition to the average weekly and average monthly effluent limitations, a daily maximum effluent limitation for $BOD_5$ and TSS is included in the Order to ensure that the treatment works are not organically overloaded and operate in accordance with design capabilities. Secondary treatment requirements for $BOD_5$ and TSS are defined as meeting a AMEL of 30 mg/L and AWEL of 45 mg/L with a 30-day average removal of 85%. Facilities that qualify as “equivalent to secondary” must meet a removal requirement of 65%. This Order contains a limitation requiring an average of 65% removal of $BOD_5$ and TSS over each calendar month.

**b.** **Flow.** The Facility was designed to provide a secondary level treatment objective for flows up to of 0.38 mgd. Therefore, this Order contains an average dry weather discharge flow limit of 0.38 mgd.

**c.** **pH.** The secondary treatment regulations at 40 CFR Part 133 also require that pH be maintained between 6.0 and 9.0 standard units. However, as discussed in the Water Quality Based Effluent Limits Section IV.C.3.d.v , to meet Basin Plan objectives, this Order contains pH limitations of 6.5 to 8.5.

**Summary of Technology-based Effluent Limitations**
**Discharge Point No. EFF-001**

**Table F-3. Summary of Technology-based Effluent Limitations**

| Parameter | Units | Effluent Limitations | | | | |
|---|---|---|---|---|---|---|
| | | Average Monthly | Average Weekly | Maximum Daily | Instantaneous Minimum | Instantaneous Maximum |
| Biochemical Oxygen Demand, 5-day @ 25°C | mg/L | 30 | 45 | 60 | | |
| | lbs/day | 95 | 143 | 190 | | |
| Total Suspended Solids | mg/L | 30 | 45 | 60 | | |
| | lbs/day | 95 | 143 | 190 | | |
| pH | | | | | 6.0[1] | 9.0[1] |

[1] To meet Basin Plan objectives, this Order contains pH limitations of 6.5 to 8.5 as discussed in the Water Quality Based Effluent Limits Section IV.C.3.d.v.

## C. Water Quality-Based Effluent Limitations (WQBELs)

### 1. Scope and Authority

Section 301(b) of the CWA and 40 CFR 122.44(d) requires that permits include limitations more stringent than applicable federal technology-based requirements where necessary to achieve applicable water quality standards. This Order contains requirements, expressed as a technology equivalence requirement, more stringent than secondary treatment requirements that are necessary to meet applicable water quality standards. The rationale for these requirements, which consist of tertiary treatment or equivalent requirements or other provisions, is discussed in the following sections of this Fact Sheet.

40 CFR 122.44(d)(1)(i) mandates that permits include effluent limitations for all pollutants that are or may be discharged at levels that have the reasonable potential to cause or contribute to an exceedance of a water quality standard, including numeric and narrative objectives within a standard. Where reasonable potential has been established for a pollutant, but there is no numeric criterion or objective for the pollutant, WQBELs must be established using: (1) USEPA criteria guidance under CWA section 304(a), supplemented where necessary by other relevant information; (2) an indicator parameter for the pollutant of concern; or (3) a calculated numeric water quality criterion, such as a proposed state criterion or policy interpreting the state's narrative criterion, supplemented with other relevant information, as provided in 40 CFR 122.44(d)(1)(vi).

The process for determining reasonable potential and calculating WQBELs when necessary is intended to protect the designated uses of the receiving water as specified in the Basin Plan, and achieve applicable water quality objectives and criteria that are contained in other state plans and policies, or any applicable water quality criteria contained in the CTR and NTR.

### 2. Applicable Beneficial Uses and Water Quality Criteria and Objectives

The Basin Plan designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve those objectives for all waters addressed through the plan. In addition, the Basin Plan implements State Water Board Resolution No. 88-63, which established state policy that all waters, with certain exceptions, should be considered suitable or potentially suitable for municipal or domestic supply.

The Basin Plan on page II-1.00 states: "*Protection and enhancement of existing and potential beneficial uses are primary goals of water quality planning…*" and with respect to disposal of wastewaters states that "...*disposal of wastewaters is [not] a prohibited use of waters of the State; it is merely a use which cannot be satisfied to the detriment of beneficial uses.*"

The federal CWA section 101(a)(2), states: "*it is the national goal that wherever attainable, an interim goal of water quality which provides for the protection and propagation of fish, shellfish, and wildlife, and for recreation in and on the water be achieved by July 1, 1983.*" Federal Regulations, developed to implement the requirements of the CWA, create a rebuttable presumption that all waters be designated as fishable and swimmable. Federal Regulations, 40 CFR sections 131.2 and 131.10, require that all waters of the State regulated to protect the beneficial uses of public water supply, protection and propagation of fish, shell fish and wildlife, recreation in and on the water, agricultural, industrial and other purposes including navigation. Section 131.3(e), 40 CFR, defines existing beneficial uses as those uses actually attained after 28 November 1975, whether or not they are included in the water quality standards. Federal Regulation, 40 CFR section 131.10 requires that uses be obtained by implementing effluent limitations, requires that all downstream uses be protected and states that in no case shall a state adopt waste transport or waste assimilation as a beneficial use for any waters of the United States.

The previous Order No. R5 2007-0032, established secondary level effluent limitations for protection of beneficial uses of the receiving water. The previous permit, however, did not recognize the MUN beneficial use to the receiving water. Although the receiving waters consist of modified agricultural drains (Lateral K), which is specifically not designated with the MUN beneficial use in Table II-1 in the Basin Plan, this Order correctly interprets the beneficial uses of the receiving waters to include the beneficial use of MUN through implementation of State Water Board Resolution No. 88-63. As stated in Chapter II of the Basin Plan, "Water Bodies within the basins that do not have beneficial uses designated in Table II-1 are assigned MUN designations in accordance with the provisions of State Water Board Resolution No. 88-63 which is, by reference, a part of the Basin Plan" except for two non-applicable exceptions. Furthermore, as specified in Chapter IV of the Basin Plan, an exception to Resolution No. 88-63, and removal of the MUN beneficial use designation for the receiving waters, is effective after a Basin Plan Amendment is adopted by the Central Valley Water Board and approved by the State Water Board and Office of Administrative Law. Therefore, this Order contains new effluent limitations necessary to protect the municipal and domestic supply use of the receiving waters.

**a. Receiving Water and Beneficial Uses. Lateral K agricultural drain.**

The beneficial uses for Butte Creek are specifically identified in the Sacramento-San Joaquin Basin Plan as shown in Table F-4 below. The beneficial uses for Lateral K, Main Canal, and Cherokee Canal constructed agricultural drains, are based on demonstrated existing uses, as also shown in Table F-4 below. In addition, the Basin Plan implements State Water Resources Control Board (State Water Board) Resolution No. 88-63, establishing the municipal or domestic supply (MUN) use on Lateral K, Main Canal and the Cherokee Canal. Butte Creek is specifically identified in the Basin Plan to not have the MUN use. The applicable beneficial uses for the receiving waters are as follows:

**Table F-4. Basin Plan Beneficial Uses**

| Discharge Point | Receiving Water Name | Beneficial Use(s) |
|---|---|---|
| 001 | Lateral K, Main Canal (Reclamation District #833) and Cherokee Canal (agricultural drains) | Surface Water<br>**Constructed Agricultural Drain:**<br>Agricultural supply, including irrigation and stock watering (AGR);<br><br>Water Contact Recreation and Non-Contact Water Recreation (REC-1) & (REC-2);<br>Warm and cold freshwater habitat (WARM) & (COLD);<br>Cold water migration (MIGR);<br>Warm water spawning (SPWN); and<br>Wildlife habitat (WILD).<br><br>**State Water Board Resolution No. 88-63:**<br>Municipal and domestic supply (MUN) |
| | Butte Creek | **Basin Plan, including Table II-1:**<br>***Surface Water:***<br>Agricultural supply, including irrigation and stock watering (AGR);<br>Water Contact Recreation (REC-1);<br>Warm and cold freshwater habitat (WARM) & (COLD);<br>Cold water migration (MIGR);<br>Warm water spawning (SPWN); and<br>Wildlife habitat (WILD).<br><br>***Groundwater:***<br>Municipal and domestic water supply (MUN);<br>Industrial service supply (IND),<br>Industrial process supply (PRO), and<br>Agricultural supply (AGR). |

b. **Effluent and Ambient Background Data.** The reasonable potential analysis (RPA), as described in section IV.C.3 of this Fact Sheet, was based on data from July 2007 through November 2011, which includes effluent and ambient background data submitted in SMRs and the Report of Waste Discharge (ROWD).

c. **Priority Pollutant Metals**

i. **Hardness-Dependent CTR Metals Criteria.** The *California Toxics Rule* and the *National Toxics Rule* contain water quality criteria for seven metals that vary as a function of hardness. The lower the hardness the lower the water quality criteria. The metals with hardness-dependent criteria include cadmium, copper, chromium III, lead, nickel, silver, and zinc.

This Order has established the criteria for hardness-dependent metals based on the reasonable worst-case ambient hardness as required by the SIP[1], the CTR[2] and State Water Board Order No. WQO 2008-0008 (City of Davis). The SIP and the CTR require the use of "receiving water" or "actual ambient" hardness, respectively, to determine effluent limitations for these metals. (SIP, § 1.2; 40 CFR § 131.38(c)(4), Table 4, note 4.) The CTR does not define whether the term "ambient," as applied in the regulations, necessarily requires the consideration of upstream as opposed to downstream hardness conditions. Therefore, where reliable, representative data are available, the hardness value for calculating criteria can be the downstream receiving water hardness, after mixing with the effluent (Order WQO 2008-0008, p. 11). The Regional Water Board thus has considerable discretion in determining ambient hardness (*Id.*, p.10.).

The hardness values must also be protective under all flow conditions (*Id.*, pp. 10-11). As discussed below, scientific literature provides a reliable method for calculating protective hardness-dependent CTR criteria, considering all discharge conditions. This methodology produces criteria that ensure these metals do not cause receiving water toxicity, while avoiding criteria that are unnecessarily stringent.

**Reasonable Potential Analysis (RPA).** The SIP in Section 1.3 states, "The RWQCB shall…determine whether a discharge may: (1) cause, (2) have a reasonable potential to cause, or (3) contribute to an excursion above any applicable priority pollutant criterion or objective." Section 1.3 provides a step-by-step procedure for conducting the RPA. The procedure requires the

[1] The SIP does not address how to determine the hardness for application to the equations for the protection of aquatic life when using hardness-dependent metals criteria. It simply states, in Section 1.2, that the criteria shall be properly adjusted for hardness using the hardness of the receiving water.

[2] The CTR requires that, for waters with a hardness of 400 mg/L (as $CaCO_3$), or less, the actual ambient hardness of the surface water must be used. It further requires that the hardness values used must be consistent with the design discharge conditions for design flows and mixing zones.

comparison of the Maximum Effluent Concentration (MEC) and Maximum Ambient Background Concentration to the applicable criterion that has been properly adjusted for hardness. Unless otherwise noted, for the hardness-dependent CTR metals criteria the following procedures were followed for properly adjusting the criterion for hardness when conducting the RPA.

- For comparing the MEC to the applicable criterion, in accordance with the SIP, CTR, and Order WQO 2008-0008, the reasonable worst-case downstream hardness was used to adjust the criterion. In this evaluation the portion of the receiving water affected by the discharge is analyzed. For hardness-dependent criteria, the hardness of the effluent has an impact on the determination of the applicable criterion in areas in the receiving water affected by the discharge. Therefore, for this situation it is necessary to consider the hardness of the effluent in determining the applicable hardness to adjust the criterion. The procedures for determining the applicable criterion after proper adjustment using the reasonable worst-case downstream hardness is outlined in subsection ii, below.

- For comparing the Maximum Ambient Background Concentration to the applicable criterion, in accordance with the SIP, CTR, and Order WQO 2008-0008, the reasonable worst-case upstream hardness was used to adjust the criterion. In this evaluation the area outside the influence of the discharge is analyzed. For this situation, the discharge does not impact the upstream hardness. Therefore, the effect of the effluent hardness was not included in this evaluation.

**Calculation of Water Quality-Based Effluent Limitations.** The remaining discussion in this section relates to the development of water quality-based effluent limits when it has been determined that the discharge has reasonable potential to cause or contribute to an exceedance of the CTR hardness-dependent metals criteria in the receiving water.

A 2006 Study[1] developed procedures for calculating the effluent concentration allowance (ECA)[2] for CTR hardness-dependent metals. The 2006 Study demonstrated that it is necessary to evaluate all discharge conditions (e.g. high and low flow conditions) and the hardness and metals concentrations of the effluent and receiving water when determining the appropriate ECA for these hardness-dependent metals. Simply using the lowest recorded upstream receiving water hardness to calculate the ECA may result in over or under protective water quality-based effluent limitations.

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[1] Emerick, R.W.; Borroum, Y.; & Pedri, J.E., 2006. California and National Toxics Rule Implementation and Development of Protective Hardness Based Metal Effluent Limitations. WEFTEC, Chicago, Ill.

[2] The ECA is defined in Appendix 1 of the SIP (page Apendix 1-2). The ECA is used to calculate water quality-based effluent limitations in accordance with Section 1.4 of the SIP

The equation describing the total recoverable regulatory criterion, as established in the CTR, is as follows:

$CTR\ Criterion = WER \times (e^{m[\ln(H)]+b})$ (Equation 1)

Where:

$H$ = hardness (as $CaCO_3$)
*WER* = water-effect ratio
*m, b* = metal- and criterion-specific constants

In accordance with the CTR, the default value for the WER is 1. A WER study must be conducted to use a value other than 1. The constants "m" and "b" are specific to both the metal under consideration, and the type of total recoverable criterion (i.e., acute or chronic). The metal-specific values for these constants are provided in the CTR at paragraph (b)(2), Table 1.

The equation for the ECA is defined in Section 1.4, Step 2, of the SIP and is as follows:

$ECA = C + D(C-B);\ when\ C > B$ (Equation 2)

$ECA = C;\ when\ C \leq B$ (Equation 3)

Where:

| | | |
|---|---|---|
| *C* | = | the priority pollutant criterion/objective, adjusted for hardness (see Equation 1, above) |
| *B* | = | the ambient background concentration |
| D | = | the dilution credit |

The 2006 Study demonstrated that the relationship between hardness and the calculated criteria is the same for some metals, so the same procedure for calculating the ECA may be used for these metals. The same procedure can be used for chronic cadmium, chromium III, copper, nickel, and zinc. These metals are hereinafter referred to as "Concave Down Metals". "Concave Down" refers to the shape of the curve represented by the relationship between hardness and the CTR criteria in Equation 1. Another similar procedure can be used for determining the ECA for acute cadmium, lead, and acute silver, which are referred to hereafter as "Concave Up Metals".

<u>ECA for Concave Down Metals</u> – For Concave Down Metals (i.e., chronic cadmium, chromium III, copper, nickel, and zinc) the 2006 Study demonstrates that when the effluent is in compliance with the CTR criteria and the upstream receiving water is in compliance with the CTR criteria, any mixture of the effluent and receiving water will always be in compliance with the CTR criteria. Therefore, based on any observed ambient background

hardness, no receiving water assimilative capacity for metals (i.e., ambient background metals concentrations are at their respective CTR criterion) and the minimum effluent hardness, the ECA calculated using Equation 1 with a hardness equivalent to the minimum effluent hardness is protective under all discharge conditions (i.e., high and low dilution conditions and under all mixtures of effluent and receiving water as the effluent mixes with the receiving water). This is applicable whether the effluent hardness is less than or greater than the ambient background receiving water hardness.

The effluent hardness ranged from 211 mg/L to 248 mg/L (as $CaCO_3$), based on 5 samples from July 2007 to November 2011. The upstream receiving water hardness varied from 52 mg/L to 240 mg/L (as $CaCO_3$), based on 5 samples from July 2007 to November 2011. Using a hardness of 211 mg/L (as $CaCO_3$) to calculate the ECA for all Concave Down Metals will result in water quality-based effluent limitations that are protective under all potential effluent/receiving water mixing scenarios and under all known hardness conditions, as demonstrated in the example using copper shown in Table F-4, below. This example assumes the following conservative conditions for the upstream receiving water:

- Upstream receiving water always at the lowest observed upstream receiving water hardness (i.e., 52 mg/L as $CaCO_3$)

- Upstream receiving water copper concentration always at the CTR criteria (i.e., no assimilative capacity). Based on available data, the receiving water never exceeded the CTR criteria for any metal with hardness-dependent criteria.

Using these reasonable worst-case conditions, the discharge can be mixed with the receiving water and a resulting downstream mixed hardness (or metals concentration) can be calculated for all discharge and mixing conditions (e.g., 0% effluent to 100% effluent) based on a simple mass balance as shown in Equation 3, below. By evaluating all discharge conditions the reasonable worst-case downstream hardness can be determined for adjusting the CTR criteria.

$$C_{MIX} = C_{RW} \times (1\text{-}EF) + C_{Eff} \times (EF) \qquad \text{(Equation 4)}$$

Where:

$C_{MIX}$ = Mixed concentration (e.g. metals or hardness)
$C_{RW}$ = Upstream receiving water concentration
$C_{Eff}$ = Effluent concentration
$EF$ = Effluent Fraction

As demonstrated in Table F-4, using a hardness of 211 mg/L (as $CaCO_3$) to calculate the ECA for Concave Down Metals ensures the discharge is protective under all discharge and mixing conditions. In this example, the effluent is in compliance with the CTR criteria and any mixture of the effluent and receiving water is in compliance with the CTR criteria. An ECA based on a lower hardness (e.g. lowest upstream receiving water hardness) would also be protective, but would result in unnecessarily stringent effluent limits considering the known conditions. Therefore, in this Order the ECA for all Concave Down Metals has been calculated using Equation 1 with a hardness of 211 mg/L (as $CaCO_3$).

**Table F-5. Copper ECA Evaluation**

| | | | |
|---|---|---|---|
| **Minimum Observed Effluent Hardness** | | | **211 mg/L** (as $CaCO_3$) |
| **Minimum Observed Upstream Receiving Water Hardness** | | | **52 mg/L** (as $CaCO_3$) |
| **Maximum Assumed Upstream Receiving Water Copper Concentration** | | | **5.3 µg/L**[1] |
| **Copper ECA$_{chronic}$**[2] | | | **17.7 µg/L** |
| **Effluent Fraction** | **Mixed Downstream Ambient Concentration** | | |
| | **Hardness**[3] **(mg/L)** (as $CaCO_3$) | **CTR Criteria**[4] **(µg/L)** | **Copper**[5] **(µg/L)** |
| 1% | 53.59 | 5.5 | 5.3 |
| 5% | 59.95 | 6.0 | 5.4 |
| 15% | 75.85 | 7.4 | 5.6 |
| 25% | 91.75 | 8.7 | 6.2 |
| 50% | 131.5 | 11.8 | 8.6 |
| 75% | 171.25 | 14.8 | 12.4 |
| 100% | 211 | 17.7 | 17.7 |

1 Maximum assumed upstream receiving water copper concentration calculated using Equation 1 for chronic criterion at a hardness of **52 mg/L (as $CaCO_3$)**.

2 ECA calculated using Equation 1 for chronic criterion at a hardness of **211 mg/L (as $CaCO_3$)**.

3 Mixed downstream ambient hardness is the mixture of the receiving water and effluent hardness at the applicable effluent fraction using Equation 3.

4 Mixed downstream ambient criteria are the chronic criteria calculated using Equation 1 at the mixed hardness.

5 Mixed downstream ambient copper concentration is the mixture of the receiving water and effluent copper concentrations at the applicable effluent fraction using Equation 3.

<u>ECA for Concave Up Metals</u> – For Concave Up Metals (i.e., acute cadmium, lead, and acute silver), the 2006 Study demonstrates that due to a different relationship between hardness and the metals criteria, the effluent and upstream receiving water can be in compliance with the CTR criteria, but the resulting mixture may be out of compliance. Therefore, the 2006 Study provides a mathematical approach to calculate the ECA to ensure that any mixture of effluent and receiving water is in compliance with the CTR criteria (see Equation 3, below). The ECA, as calculated using Equation 3, is based on the reasonable worst-case ambient background hardness, no receiving water assimilative capacity for metals (i.e., ambient background metals concentrations are at their respective CTR criterion), and the minimum observed effluent hardness. The reasonable worst-case ambient background

hardness depends on whether the effluent hardness is greater than or less than the upstream receiving water hardness. There are circumstances where the conservative ambient background hardness assumption is to assume that the upstream receiving water is at the highest observed hardness concentration. The conservative upstream receiving water condition as used in the Equation 4 below is defined by the term $H_{rw}$.

$$ECA = \left[\frac{m(H_e - H_{rw})(e^{m(\ln(H_{rw}))+b})}{H_{rw}}\right] + e^{m(\ln(H_{rw}))+b} \qquad \text{(Equation 5)}$$

Where:

| | | |
|---|---|---|
| *m, b* | = | criterion specific constants (from CTR) |
| $H_e$ | = | minimum observed effluent hardness |
| $H_{rw}$ | = | minimum observed upstream receiving water hardness when the minimum effluent hardness is always greater than observed upstream receiving water hardness ($H_{rw} < H_e$)<br>-or-<br>maximum observed upstream receiving water hardness when the minimum effluent hardness is always less than observed upstream receiving water hardness ($H_{rw} > H_e$)[1] |

A similar example as was done for the Concave Down Metals is shown for lead, a Concave Up Metal, in Table F-5, below. As previously mentioned, the minimum effluent hardness is 211 mg/L (as $CaCO_3$), while the upstream receiving water hardness ranged from 52 mg/L to 240 mg/L (as $CaCO_3$). In this case, the minimum effluent concentration is less than the range of observed upstream receiving water hardness concentrations. Thus, the ECA was calculated (Equation 4) based on the maximum observed upstream receiving water hardness, no receiving water assimilative capacity for lead (i.e., ambient background lead concentration is at the CTR chronic criterion) and the minimum effluent hardness.

Using Equation 4 to calculate the ECA for all Concave Up Metals will result in water quality-based effluent limitations that are protective under all potential effluent/receiving water mixing scenarios and under all known hardness conditions, as demonstrated in Table F-5, for lead. In this example, the

[1] When the minimum effluent hardness falls within the range of observed receiving water hardness concentrations, Equation 3 is used to calculate two ECAs, one based on the minimum observed upstream receiving water hardness and one based on the maximum observed upstream receiving water hardness. The minimum of the two calculated ECAs represents the ECA that ensures any mixture of effluent and receiving water is in compliance with the CTR criteria.

effluent is in compliance with the CTR criteria and any mixture of the effluent and receiving water is in compliance with the CTR criteria. Use of a lower ECA (e.g., calculated based solely on the lowest upstream receiving water hardness) is also protective, but would lead to unnecessarily stringent effluent limits considering the known conditions. Therefore, Equation 4 has been used to calculate the ECA for all Concave Up Metals in this Order.

**Table F-6. Lead ECA Evaluation**

| | | | |
|---|---|---|---|
| **Minimum Observed Effluent Hardness** | | | **211 mg/L (as $CaCO_3$)** |
| **Minimum Upstream Receiving Water Hardness** | | | **52 mg/L (as $CaCO_3$)** |
| **Maximum Assumed Upstream Receiving Water Lead Concentration** | | | **1.4 µg/L[1]** |
| **Lead $ECA_{chronic}$[2]** | | | **6.8 µg/L** |
| | **Mixed Downstream Ambient Concentration** | | |
| **Effluent Fraction** | **Hardness[3] (mg/L) (as $CaCO_3$)** | **CTR Criteria[4] (µg/L)** | **Lead[5] (µg/L)** |
| 1% | 53.6 | 1.4 | 1.4 |
| 5% | 60.0 | 1.7 | 1.7 |
| 15% | 75.9 | 2.2 | 2.2 |
| 25% | 91.8 | 2.9 | 2.7 |
| 50% | 131.5 | 4.5 | 4.1 |
| 75% | 171.3 | 6.3 | 5.4 |
| 100% | 211.0 | 8.2 | 6.8 |

1 Minimum assumed upstream receiving water lead concentration calculated using Equation 1 for chronic criterion at a hardness of **52 mg/L (as $CaCO_3$)**.

2 ECA calculated using Equation 3 for chronic criteria.

3 Mixed downstream ambient hardness is the mixture of the receiving water and effluent hardness at the applicable effluent fraction.

4 Mixed downstream ambient criteria are the chronic criteria calculated using Equation 1 at the mixed hardness.

5 Mixed downstream ambient lead concentration is the mixture of the receiving water and effluent lead concentrations at the applicable effluent fraction.

**Table F-7. Lead ECA Evaluation**

| | |
|---|---|
| **Minimum Observed Effluent Hardness** | **211 mg/L (as $CaCO_3$)** |
| **Maximum Observed Upstream Receiving Water Hardness** | **240 mg/L (as $CaCO_3$)** |
| **Maximum Assumed Upstream Receiving Water Lead Concentration** | **1.4 µg/L[1]** |
| **Lead $ECA_{chronic}$[2]** | **8.2 µg/L** |

| **Effluent Fraction** | **Mixed Downstream Ambient Concentration** | | |
|---|---|---|---|
| | **Hardness[3] (mg/L) (as $CaCO_3$)** | **CTR Criteria[4] (µg/L)** | **Lead[5] (µg/L)** |
| 1% | 239.7 | 9.7 | 9.7 |
| 5% | 238.6 | 9.6 | 9.6 |
| 15% | 235.7 | 9.5 | 9.5 |
| 25% | 232.8 | 9.3 | 9.3 |
| 50% | 225.5 | 9.0 | 9.0 |
| 75% | 218.3 | 8.6 | 8.6 |
| 100% | 211.0 | 8.2 | 8.2 |

1. Maximum assumed upstream receiving water lead concentration calculated using Equation 1 for chronic criterion at a hardness of **240 mg/L (as $CaCO_3$)**.
2. ECA calculated using Equation 3 for chronic criteria.
3. Mixed downstream ambient hardness is the mixture of the receiving water and effluent hardness at the applicable effluent fraction using Equation 3.
4. Mixed downstream ambient criteria are the chronic criteria calculated using Equation 1 at the mixed hardness.
5. Mixed downstream ambient lead concentration is the mixture of the receiving water and effluent lead concentrations at the applicable effluent fraction using Equation 3.

**Table F-8. Lead ECA Evaluation**

| | | | |
|---|---|---|---|
| **Minimum Observed Effluent Hardness** | | | **211 mg/L (as $CaCO_3$)** |
| **Minimum Observed Upstream Receiving Water Hardness** | | | **52 mg/L (as $CaCO_3$)** |
| **Maximum Assumed Upstream Receiving Water Lead Concentration** | | | **8.9 µg/L[1]** |
| **Lead $ECA_{chronic}$[2]** | | | **1.7 µg/L** |
| **Effluent Fraction** | **Mixed Downstream Ambient Concentration** | | |
| | **Hardness[3] (mg/L) (as $CaCO_3$)** | **CTR Criteria[4] (µg/L)** | **Lead[5] (µg/L)** |
| 1% | 222.6 | 8.8 | 8.8 |
| 5% | 216.9 | 8.5 | 8.5 |
| 15% | 202.7 | 7.8 | 7.8 |
| 25% | 188.5 | 7.1 | 7.1 |
| 50% | 153.0 | 5.5 | 5.3 |
| 75% | 117.5 | 3.9 | 3.5 |
| 100% | 82 | 2.5 | 1.7 |

1 Maximum assumed upstream receiving water lead concentration based on maximum observed upstream receiving water lead concentration.

2 ECA determined iteratively until all mixtures of effluent and receiving water are in compliance with the CTR criteria.

3 Mixed downstream ambient hardness is the mixture of the receiving water and effluent hardness at the applicable effluent fraction using Equation 3.

4 Mixed downstream ambient criteria are the chronic criteria calculated using Equation 1 at the mixed hardness.

5 Mixed downstream ambient lead concentration is the mixture of the receiving water and effluent lead concentrations at the applicable effluent fraction using Equation 3.

**ii. Conversion Factors.** The CTR contains aquatic life criteria for arsenic, cadmium, chromium III, chromium VI, copper, lead, nickel, silver, and zinc which are presented in dissolved concentrations. USEPA recommends conversion factors to translate dissolved concentrations to total concentrations. The default USEPA conversion factors contained in Appendix 3 of the SIP were used to convert the applicable dissolved criteria to total recoverable criteria.

### d. Assimilative Capacity/Mixing Zone

Based on the available information, the worst-case dilution is assumed to be zero to provide protection for the receiving water beneficial uses. The impact of assuming zero assimilative capacity within the receiving water is that discharge limitations are end-of-pipe limits with no allowance for dilution within the receiving water. At times, the Lateral K (constructed agricultural drainage channel) has no receiving water, and is composed of only wastewater effluent. The Discharger did not present a Mixing Zone study for the City of Biggs WWTP.

### 3. Determining the Need for WQBELs

**a.** The Regional Water Board conducted the RPA in accordance with section 1.3 of the SIP. Although the SIP applies directly to the control of CTR priority pollutants, the State Water Board has held that the Regional Water Board may use the SIP as guidance for water quality-based toxics control.[1] The SIP states in the introduction "*The goal of this Policy is to establish a standardized approach for permitting discharges of toxic pollutants to non-ocean surface waters in a manner that promotes statewide consistency*." Therefore, in this Order the RPA procedures from the SIP were used to evaluate reasonable potential for both CTR and non-CTR constituents based on information submitted as part of the application, in studies, and as directed by monitoring and reporting programs.

**b. Constituents with Limited Data.** Reasonable potential cannot be determined for the following constituents because effluent data are limited or ambient background concentrations are not available. The Discharger is required to continue to monitor for these constituents in the effluent using analytical methods that provide the best feasible detection limits. When additional data become available, further analysis will be conducted to determine whether to add numeric effluent limitations or to continue monitoring.

**i. Bis (2-ethylhexyl) Phthalate.** Effluent data provided by the Discharger indicates that bis (2-ethylhexyl) phthalate was below analytical reporting limit; therefore results are estimated concentrations, which the laboratory equates to the DNQ estimated concentration flag. Also, because bis (2-ethylhexyl) phthalate is a common contaminant of sample containers, sampling apparatus, and analytical equipment, and sources of the detected bis (2-ethylhexyl) phthalate may be from plastics used for sampling or analytical equipment, the Regional Water Board is not establishing effluent limitations for bis (2-ethylhexyl) phthalate at this time. Instead of limitations, additional monitoring has been established for bis (2-ethylhexyl) phthalate; should monitoring results indicate that the discharge has the reasonable potential to cause or contribute to an exceedance of a water quality standard, then this Order may be reopened and modified by adding an appropriate effluent limitation. Furthermore, the SIP Section 2.4.2 states that the Minimum Level (ML) is the lowest quantifiable concentration in a sample based on the proper application of all method-based analytical procedures and the absence of any matrix interferences.

a) Required MLs are listed in Appendix 4 of the SIP. Where more than one ML is listed in Appendix 4, the discharger may select any one of the cited analytical methods for compliance determination. The selected ML used for compliance determination is referred to as the Reporting Level (RL).

[1] See Order WQO 2001-16 (Napa) and Order WQO 2004-0013 (Yuba City).

b) A Reporting Level can be lower than the Minimum Level in Appendix 4 only when the discharger agrees to use a Reporting Level that is lower than the Minimum Level listed in Appendix 4. The Regional Board and the discharger have no agreement to use a Reporting Limit lower than the listed Minimum Levels.
c) SIP Section 1.2 requires that the Regional Board use all available, valid, relevant, representative data and information, as determined by the Regional Board, to implement the SIP. SIP Section 1.2 further states that the Regional Board has the discretion to consider if any data are inappropriate or insufficient for use in implementing the SIP.
d) Data reported below the Minimum Level indicates the data may not be valid due to possible matrix interferences during the analytical procedure.
e) Further, SIP Section 2.4.5 (Compliance Determination) supports the insufficiency of data reported below the Minimum Level or Reporting Level. In part it states, "Dischargers shall be deemed out of compliance with an effluent limitation, for reporting and administrative enforcement purposes, if the concentration of the priority pollutant in the monitoring sample is greater than the effluent limitation and greater than or equal to the RL." Thus, if submitted data is below the Reporting Limit, that data cannot be used to determine compliance with effluent limitations.
f) Data reported below the Minimum Level is not considered valid data for use in determining Reasonable Potential. Therefore, in accordance with Section 1.2 of the SIP, the Board has determined that data reported below the Minimum Level is inappropriate and insufficient to be used to determine Reasonable Potential.
g) In implementing its discretion, the Board is not finding that Reasonable Potential does not exist; rather the Board cannot make such a determination given the invalid data. Therefore, the Board will require additional monitoring for such constituents until such time a determination can be made in accordance with the SIP policy.

SIP Appendix 4 cites several Minimum Levels (ML) for Bis(2-ethylhexyl)phthalate. The lowest ML cited for Bis(2-ethylhexyl)phthalate is 5 µg/L. The Discharger used an analytical method that was more sensitive than the minimum level required by the SIP. The effluent results were all non-detect or estimated values at a concentration well below the lowest ML, as shown in the table below. Therefore the submitted effluent Bis(2-ethylhexyl)phthalate data is inappropriate and insufficient to determine reasonable potential under the SIP.

**Biggs Effluent and Receiving Water Bis(2-ethylhexyl)phthalate Data (2007-2010)**

| Date | Effluent (ug/L) | Receiving Water (ug/L) | SIP Minimum Level (ug/L) |
|---|---|---|---|
| 07/17/2007 | 1 DNQ | <0.9 | 5 |
| 09/18/2007 | <0.9 | <0.9 | 5 |
| 11/6/2007 | 2 DNQ | <0.9 | 5 |
| 01/09/2008 | 2 DNQ | <0.9 | 5 |
| 03/12/2008 | <0.9 | <0.9 | 5 |
| 05/06/2008 | 1 DNQ | <0.9 | 5 |
| 01/20/2010 | 2 DNQ | <0.9 | 5 |

c. **Constituents with No Reasonable Potential.** WQBELs are not included in this Order for constituents that do not demonstrate reasonable potential; however, monitoring for those pollutants is established in this Order as required by the SIP. If the results of effluent monitoring demonstrate reasonable potential, this Order may be reopened and modified by adding an appropriate effluent limitation.

i. **Aluminum**

(a) **WQO.** USEPA developed National Recommended Ambient Water Quality Criteria (NAWQC) for protection of freshwater aquatic life for aluminum. The recommended 4-day average (chronic) and 1-hour average (acute) criteria for aluminum are 87 µg/L and 750 µg/L, respectively, for waters with a pH of 6.5 to 9.0. The Secondary Maximum Contaminant Level (MCL) - Consumer Acceptance Limit for aluminum is 200 µg/L. USEPA recommends that the ambient criteria are protective of the aquatic beneficial uses of receiving waters in lieu of site-specific criteria. The receiving stream has been measured to have a low hardness—typically between 52 and 240 mg/L as $CaCO_3$. This condition is not supportive of the applicability of the NAWQC for aluminum, according to USEPA's development document.

(b) **RPA Results.** The maximum effluent concentration (MEC) for aluminum was <24.5 µg/L while the maximum observed upstream receiving water concentration was 693 µg/L. Therefore, aluminum in the discharge does not have the reasonable potential to cause or contribute to an in-stream excursion above the 87 µg/L chronic criterion for the protection of aquatic life, the 750 µg/L acute criterion for the protection of aquatic life, or the Secondary MCL.

Due to no assimilative capacity, dilution credits are not allowed for development of the WQBELs for aluminum. This Order does not contain a final average monthly effluent limitation (AMEL) nor maximum daily effluent limitation (MDEL) for Aluminum, but requires the discharger to conduct effluent and receiving water monitoring for aluminum to provide sufficient data for an RPA and development of applicable numeric criteria.

**d. Constituents with Reasonable Potential.** The Regional Water Board finds that the discharge has a reasonable potential to cause or contribute to an in-stream excursion above a water quality standard for ammonia and copper. WQBELs for these constituents are included in this Order. A summary of the RPA is provided in Attachment G, and a detailed discussion of the RPA for each constituent is provided below.

**i. Ammonia**

**(a) WQO.** The NAWQC for the protection of freshwater aquatic life for total ammonia, recommends acute (1-hour average; criteria maximum concentration or CMC) standards based on pH and chronic (30-day average; criteria continuous concentration or CCC) standards based on pH and temperature. USEPA also recommends that no 4-day average concentration should exceed 2.5 times the 30-day CCC. USEPA found that as pH increased, both the acute and chronic toxicity of ammonia increased. Salmonids were more sensitive to acute toxicity effects than other species. However, while the acute toxicity of ammonia was not influenced by temperature, it was found that invertebrates and young fish experienced increasing chronic toxicity effects with increasing temperature. Because the Lateral K has a beneficial use of cold freshwater habitat, the recommended criteria for waters where salmonids and early life stages are present were used.

The maximum permitted effluent pH is 8.5, as the Basin Plan objective for pH in the receiving stream is the range of 6.5 to 8.5. In order to protect against the worst-case short-term exposure of an organism, a pH value of 8.5 was used to derive the acute criterion. The resulting acute criterion is 2.14 mg/L.

The maximum observed 30-day rolling average temperature and pH of the receiving water were used to calculate the 30-day CCC. The lowest 1/10$^{th}$ percentile 30-day CCC was selected as the applicable CCC. This CCC was calculated as 4.4 mg/L.

The 4-day average concentration is derived in accordance with the USEPA criterion as 2.5 times the 30-day CCC. Based on the 30-day CCC of 4.4 mg/L (as N), the 4-day average concentration that should not be exceeded is 11 mg/L (as N).

**(b) RPA Results.** Untreated domestic wastewater contains ammonia. Nitrification is a biological process that converts ammonia to nitrite and nitrite to nitrate. Denitrification is a process that converts nitrate to nitrite or nitric oxide and then to nitrous oxide or nitrogen gas, which is then released to the atmosphere. The Discharger does not currently use nitrification to remove ammonia from the waste stream. Inadequate or incomplete nitrification may result in the discharge of ammonia to the receiving stream. Ammonia is known to cause toxicity to aquatic organisms in surface waters. Discharges of ammonia in concentrations that produce detrimental physiological responses in human, plant, animal, or aquatic life or in toxic amounts would violate the Basin Plan narrative toxicity objective. The maximum effluent concentration (MEC) for ammonia was 18 mg/L while the maximum observed upstream receiving water concentration was 0.3 mg/L. Therefore, ammonia in the discharge has a reasonable potential to cause or contribute to an in-stream excursion above the calculated acute, 30-day average, and 4-day average NAWQC.

**(c) WQBELs.** The Regional Water Board calculates WQBELs in accordance with SIP procedures for non-CTR constituents, and ammonia is a non-CTR constituent. The SIP procedure assumes a 4-day averaging period for calculating the long-term average discharge condition (LTA). However, USEPA recommends modifying the procedure for calculating permit limits for ammonia using a 30-day averaging period for the calculation of the LTA corresponding to the 30-day CCC. Therefore, while the LTAs corresponding to the acute and 4-day chronic criteria were calculated according to SIP procedures, the LTA corresponding to the 30-day CCC was calculated assuming a 30-day averaging period. The lowest LTA representing the acute, 4-day CCC, and 30-day CCC is then selected for deriving the average monthly effluent limitation (AMEL) and the maximum daily effluent limitation (MDEL). The remainder of the WQBEL calculation for ammonia was performed according to the SIP procedures. This Order contains a final average monthly effluent limitation (AMEL) and maximum daily effluent limitation (MDEL) for ammonia of 1.23 µg/L and 2.15 µg/L, respectively, based on the NAWQC standard.

**(d) Plant Performance and Attainability.** The treatment plant is currently unable to treat wastewater to the established final effluent limits for ammonia. New or modified control measures are necessary in order to comply with the effluent limitations, and the new or modified control measures cannot be designed, installed and put into operation within 30 calendar days. A time schedule for compliance with the ammonia effluent limitations is established in **Time Schedule Order No. R5-2012-0084** in accordance with Water Code section 13300, that requires preparation and implementation of a pollution prevention plan in compliance with Water Code section 13263.3.

**ii. Copper**

**(a) WQO.** The CTR includes hardness dependent criteria for the protection of freshwater aquatic life for copper. Using the default conversion factors and reasonable worst-case measured hardness, as described in section VI.C.2.c of this Fact Sheet, the applicable acute (1-hour average) criterion is **28** µg/L and the applicable chronic (4-day average) criterion is **18** µg/L, as total recoverable.

**(b) RPA Results.** The maximum effluent concentration (MEC) for **copper** was **9.8** µg/L (as total recoverable) while the maximum observed upstream receiving water concentration was **20.8** µg/L (as total recoverable). Therefore, **copper** in the discharge has a reasonable potential to cause or contribute to an in-stream excursion above the CTR criterion for the protection of freshwater aquatic life.

**(c) WQBELs.** Due to no assimilative capacity, dilution credits are not allowed for development of the WQBELs for copper. This Order contains a final average monthly effluent limitation (AMEL) and maximum daily effluent limitation (MDEL) for **copper** of **15.1** µg/L and **28.0** µg/L, respectively, based on the CTR criterion for the protection of freshwater aquatic life.

**(d) Plant Performance and Attainability.** Analysis of the effluent data shows that the MEC of 9.8 µg/L is less than applicable WQBELs. The Discharger is expected to be able to immediately comply with the new effluent limit.

**iii. Chlorine Residual**

**(a) WQO.** USEPA developed NAWQC for protection of freshwater aquatic life for chlorine residual. The recommended 4-day average (chronic) and 1-hour average (acute) criteria for chlorine residual are 0.011 µg/L and 0.019 µg/L, respectively. These criteria are protective of the Basin Plan's narrative toxicity objective.

**(b) RPA Results.** The Discharger uses chlorine for disinfection, which is extremely toxic to aquatic organisms. The Discharger uses a sulfur dioxide process to dechlorinate the effluent prior to discharge to Lateral K. Due to the existing chlorine use and the potential for chlorine to be discharged, the discharge has a reasonable potential to cause or contribute to an in-stream excursion above the NAWQC.

**(c) WQBELs.** The USEPA *Technical Support Document for Water Quality-Based Toxics Control* [EPA/505/2-90-001] contains statistical methods for converting chronic (4-day) and acute (1-hour) aquatic life criteria to average monthly and maximum daily effluent limitations based on the variability of the existing data and the expected frequency of monitoring. However, because chlorine is an acutely toxic constituent that can and will be monitored continuously, an average 1-hour limitation is considered more appropriate than an average daily limitation. This Order contains a 4-day average effluent limitation and 1-hour average effluent limitation for chlorine residual of

0.011 µg/L and 0.019 µg/L, respectively, based on USEPA's NAWQC, which implements the Basin Plan's narrative toxicity objective for protection of aquatic life.

(d) **Plant Performance and Attainability.** Based on plant performance history, the treatment plant is capable of achieving effluent limits for chlorine residual.

**iv. Pathogens**

**WQO.** DPH has developed reclamation criteria, CCR, Division 4, Chapter 3 (Title 22), for the reuse of wastewater. Title 22 requires that for spray irrigation of food crops, parks, playgrounds, schoolyards, and other areas of similar public access, wastewater be adequately disinfected, oxidized, coagulated, clarified, and filtered, and that the effluent total coliform levels not exceed 2.2 MPN/100 mL as a 7-day median. As coliform organisms are living and mobile, it is impracticable to quantify an exact number of coliform organisms and to establish weekly average limitations. Instead, coliform organisms are measured as a most probable number and regulated based on a 7-day median limitation.

Title 22 also requires that recycled water used as a source of water supply for non-restricted recreational impoundments be disinfected tertiary recycled water that has been subjected to conventional treatment. A non-restricted recreational impoundment is defined as *"…an impoundment of recycled water, in which no limitations are imposed on body-contact water recreational activities."* Title 22 is not directly applicable to surface waters; however, the Regional Water Board finds that it is appropriate to apply an equivalent level of treatment to that required by the Department of Public Health's reclamation criteria because the receiving water is used for irrigation of agricultural land and for contact recreation purposes. The stringent disinfection criteria of Title 22 are appropriate since the undiluted effluent may be used for the irrigation of food crops and/or for body-contact water recreation. Coliform organisms are intended as an indicator of the effectiveness of the entire treatment train and the effectiveness of removing other pathogens.

(a) **RPA Results.** The beneficial uses of the Lateral K include municipal and domestic supply, water contact recreation, and agricultural irrigation supply, and there is, at times, less than 20:1 dilution. To protect these beneficial uses, the Regional Water Board finds that the wastewater must be disinfected and adequately treated to prevent disease. The method of treatment is not prescribed by this Order; however, wastewater must be treated to a level equivalent to that recommended by DPH.

(b) **WQBELs.** As recommended by DPH, based on the requirements of Title 22, this Order includes total coliform organism effluent limitations, when dilution is <20:1, of 2.2 MPN/100 mL as a 7-day median; 23 MPN/100 mL, not to be exceeded more than once in a 30-day period; and 240 MPN/100 mL as an instantaneous maximum. Where the wastewater receives dilution of 20:1 or more, this Order includes effluent limitations for coliform concentrations not to exceed 23 MPN/100 mL as a 7-day median or 240 MPN/100 mL more than once in any 30 day period.

In addition to coliform limitations, a turbidity operational specification has been included as a second indicator of the effectiveness of the treatment process and to assure compliance with the required level of treatment. A tertiary treatment process, or equivalent, is capable of reliably meeting a turbidity limitation of 2 nephelometric turbidity units (NTU) as a daily average. Failure of the filtration system such that virus removal is impaired would normally result in increased particles in the effluent, which result in higher effluent turbidity. Turbidity has a major advantage for monitoring filter performance, allowing immediate detection of filter failure and rapid corrective action. Coliform testing, by comparison, is not conducted continuously and requires several hours, to days, to identify high coliform concentrations. Therefore, to ensure compliance with DPH recommended Title 22 disinfection criteria, weekly average effluent limitations are impracticable for turbidity. This Order includes operational specifications for turbidity of 2 NTU as a daily average; 5 NTU, not to be exceeded more than 5% of the time within a 24-hour period; and 10 NTU as an instantaneous maximum, when dilution is <20:1.

(c) **Plant Performance and Attainability.** Based on plant performance history, the treatment plant appears to be capable of achieving effluent limits for total coliform organisms. Effluent monitoring data (since July 2010) has consistently resulted in total coliform bacteria levels below 2.2 MPN/100 ml. However, in the event the Discharger cannot comply with the total coliform organism effluent limitations or turbidity operational requirements when dilution is <20:1, this permit has a reopener provision to establish a compliance schedule to allow time for the Discharger to make necessary improvements to achieve compliance.

**v. pH**

(a) **WQO.** The Basin Plan includes a water quality objective for surface waters (except for Goose Lake) that the *"…pH shall not be depressed below 6.5 nor raised above 8.5. Changes in normal ambient pH levels shall not exceed 0.5 in fresh waters with designated COLD or WARM beneficial uses.*"

**(b) RPA Results.** The discharge of secondary treated wastewater has a reasonable potential to cause or contribute to an excursion above the Basin Plan's numeric objectives for pH.

**(c) WQBELs.** Effluent limitations for pH of 6.5 as an instantaneous minimum and 8.5 as an instantaneous maximum are included in this Order based on protection of the Basin Plan objectives for pH.

**(d) Plant Performance and Attainability**. Based on plant performance history, the treatment plant is capable of meeting effluent limits for pH.

**vi. Salinity**

**(a) WQO.** The Basin Plan contains chemical constituent objectives that incorporates state MCLs, contains narrative objective, and contains numeric water quality objectives for certain specified water bodies for electrical conductivity, total dissolved solids, sulfate, and chloride. The USEPA Ambient Water Quality Criteria for Chloride recommends acute and chronic criteria for the protection of aquatic life. There are no USEPA water quality criteria for the protection of aquatic life for electrical conductivity, total dissolved solids, and sulfate. Additionally, there are no USEPA numeric water quality criteria for the protection of agricultural, live-stock, and industrial uses. Numeric values for the protection of these uses are typically based on site specific conditions and evaluations to determine the appropriate constituent threshold necessary to interpret the narrative chemical constituent Basin Plan objective.

**Table F-9. Salinity Water Quality Criteria/Objectives**

| Parameter | Agricultural WQ Objective[1] | Secondary MCL[3] | USEPA NAWQC | Effluent | |
|---|---|---|---|---|---|
| | | | | Average | Maximum |
| EC (µmhos/cm) | Varies[2] | 900, 1600, 2200 | N/A | 735.2 | 920.0 |
| TDS (mg/L) | Varies | 500, 1000, 1500 | N/A | | |
| Sulfate (mg/L) | Varies | 250, 500, 600 | N/A | | |
| Chloride (mg/L) | Varies | 250, 500, 600 | 860 1-hr<br>230 4-day | | |

1 Narrative chemical constituent objective of the Basin Plan. Procedures for establishing the applicable numeric limitation to implement the narrative objective can be found in the Policy for Application of Water Quality, Chapter IV, Section 8 of the Basin Plan. However, the Basin Plan does not require improvement over naturally occurring background concentrations. In cases where the natural background concentration of a particular constituent exceeds an applicable water quality objective, the natural background concentration will be considered to comply with the objective.

2 The EC level in irrigation water that harms crop production depends on the crop type, soil type, irrigation methods, rainfall, and other factors.

3 The secondary MCLs are stated as a recommended level, upper level, and a short-term maximum level.

(1) **Chloride.** The Secondary MCL for chloride is 250 mg/L, as a recommended level, 500 mg/L as an upper level, and 600 mg/L as a short-term maximum. The Central Valley Water Board must determine the applicable numeric limit to implement the narrative objective for the protection of agricultural supply. The most limiting agricultural water quality goal to interpret the narrative chemical constituents objective, is 106 mg/L as a long-term average based on Water Quality for Agriculture, Food and Agriculture Organization of the United Nations—Irrigation and Drainage Paper No. 29, Rev. 1 (R.S. Ayers and D.W. Westcot, Rome, 1985). The water quality goal is intended to protect against adverse effects on sensitive crops when irrigated via sprinklers. However, the agricultural water quality goal is not a site-specific goal or objective, but rather a general measure to protect salt-sensitive crops. Site-specific levels of chloride for the receiving waters are necessary to interpret the narrative chemical constituents objective for protection of agricultural supply.

The Central Valley Water Board is currently implementing the CV-SALTS initiative to develop a Basin Plan Amendment that will establish a salt and nitrate Management Plan for the Central Valley. Through this effort the Basin Plan will be amended to define how the narrative water quality objective is to be interpreted for the protection of agricultural use. All studies conducted through this Order to establish an agricultural limit to implement the narrative objective will be reviewed by and consistent with the efforts currently underway by CV-SALTS.

(2) **Electrical Conductivity.** The Secondary MCL for EC is 900 µmhos/cm as a recommended level, 1600 µmhos/cm as an upper level, and 2200 µmhos/cm as a short-term maximum. The Central Valley Water Board must determine the applicable numeric limit to implement the narrative objective for the protection of agricultural supply. The most limiting agricultural water quality goal may be as low as 700 µmhos/cm as a long-term average based on Water Quality for Agriculture, Food and Agriculture Organization of the United Nations—Irrigation and Drainage Paper No. 29, Rev. 1 (R.S. Ayers and D.W. Westcot, Rome, 1985). However, the 700 µmhos/cm agricultural water quality goal is not a site-specific goal or objective, but rather a general measure of electrical conductivity that was determined to protect salt-sensitive crops, such as beans, carrots, turnips, and strawberries under certain soil and climate conditions. Most other crops can tolerate higher EC concentrations without harm. Site specific levels of EC for the receiving waters to interpret the narrative chemical constituents objective in the Basin Plan for protection of agricultural supply are necessary. Overall, salinity of agricultural irrigation water must be maintained at levels in which growers do not need to take extra measures to minimize or eliminate any harmful impacts.

The Central Valley Water Board is currently implementing the CV-SALTS initiative to develop a Basin Plan Amendment that will establish a salt and nitrate Management Plan for the Central Valley. Through this effort the Basin Plan will be amended to define how the narrative water quality objective is to be interpreted for the protection of agricultural use. All studies conducted through this Order to evaluate salinity in the discharge will be reviewed by and be consistent with the efforts currently underway by CV-SALTS.

(3) **Sulfate.** The secondary MCL for sulfate is 250 mg/L as a recommended level, 500 mg/L as an upper level, and 600 mg/L as a short-term maximum.

(4) **Total Dissolved Solids.** The Secondary MCL for TDS is 500 mg/L as a recommended level, 1000 mg/L as an upper level, and 1500 mg/L as a short-term maximum. The Central Valley Water Board must determine the applicable numeric limit to implement the narrative objective for the protection of agricultural supply. The most limiting agricultural water quality goal may be as low as 450 mg/L as a long-term average based on Water Quality for Agriculture, Food and Agriculture Organization of the United Nations—Irrigation and Drainage Paper No. 29, Rev. 1 (R.S. Ayers and D.W. Westcot, Rome, 1985). Water Quality for Agriculture evaluates the impacts of salinity levels on crop tolerance and yield reduction, and establishes water quality goals that are protective of the agricultural uses. However, the water quality goal is not a site-specific goal, but rather a general measure of TDS that was determined to protect salt-sensitive crops. Only the most salt sensitive crops require irrigation water of 450 mg/L or less to prevent loss of yield. Most other crops can tolerate higher TDS concentrations without harm, Site-specific levels of TDS for the receiving waters to interpret the narrative chemical constituents objective are necessary.

The Central Valley Water Board is currently implementing the CV-SALTS initiative to develop a Basin Plan Amendment that will establish a salt and nitrate Management Plan for the Central Valley. Through this effort the Basin Plan will be amended to define how the narrative water quality objective is to be interpreted for the protection of agricultural use. All studies conducted through this Order to establish an agricultural limit to implement the narrative objective will be reviewed by and consistent with the efforts currently underway by CV-SALTS.

(b) **RPA Results.**

**(1) Chloride.** Data not available.

**(2) Electrical Conductivity.** A review of the Discharger's monitoring reports shows a range of EC values from 566.5 µmhos/cm to 920.0 µmhos/cm. Annual average EC ranged from 727 µmhos/cm to 775 µmhos/cm with an average of 738.3 µmhos/cm. Although the maximum EC value exceeds the Secondary MCL, the maximum annual average EC has not exceeded the Secondary MCL (900 µmhos/cm). The background receiving water EC averaged 242.1 µmhos/cm.

**(3) Sulfate.** Data not available.

**(4) Total Dissolved Solids.** Data not available.

**(c) WQBELs.** Effluent limitations based on the MCL or the Basin Plan would likely require construction and operation of a reverse osmosis treatment plant. The State Water Board, in Water Quality Order 2005-005 (for the City of Manteca), states, "*…the State Board takes official notice [pursuant to Title 23 of California Code of Regulations, Section 648.2] of the fact that operation of a large-scale reverse osmosis treatment plant would result in production of highly saline brine for which an acceptable method of disposal would have to be developed. Consequently, any decision that would require use of reverse osmosis to treat the City's municipal wastewater effluent on a large scale should involve thorough consideration of the expected environmental effects.*" The State Water Board states in that Order, "*Although the ultimate solution to southern Delta salinity problems have not yet been determined, previous actions establish that the State Board intended for permit limitations to play a limited role with respect to achieving compliance with the EC water quality objectives in the southern Delta.*" The State Water Board goes on to say, "*Construction and operation of reverse osmosis facilities to treat discharges…prior to implementation of other measures to reduce the salt load in the southern Delta, would not be a reasonable approach*."

The Central Valley Water Board, with cooperation of the State Water Board, has begun the process to develop a new policy for the regulation of salinity in the Central Valley. In a statement issued at the 16 March 2006, Central Valley Water Board meeting, Board Member Dr. Karl Longley recommended that the Central Valley Water Board continue to exercise its authority to regulate discharges of salt to minimize salinity increases within the Central Valley. Dr. Longley stated, "*The process of developing new salinity control policies does not, therefore, mean that we should stop regulating salt discharges until a salinity Policy is developed. In the meantime, the Board should consider all possible interim approaches to continue controlling and regulating salts in a reasonable manner, and encourage all stakeholder groups that may be affected by the Regional Board's policy to actively participate in policy development.*"

The discharge does not have reasonable potential to cause or contribute to an in-stream excursion over the secondary MCL for EC. However, since the Discharger discharges to Cherokee Canal a tributary of Butte Creek and eventually the Sacramento-San Joaquin Delta, of additional concern is the salt contribution to Delta waters. Allowing the Discharger to increase its current salt loading may be contrary to the Region-wide effort to address salinity in the Central Valley. The effluent limit for EC in Order No. R5-2007-0032 is 900 µmhos/cm, as a monthly average. Therefore, this Order includes an effluent limitation of **900** µmhos/cm for EC to be applied as an annual average (Secondary MCL). The maximum annual average EC concentration is 775 µmhos/cm. It appears the Discharger can meet this limit.

In order to ensure that the Discharger will continue to control the discharge of salinity, this Order includes a requirement to develop and implement a salinity evaluation and minimization plan. Also water supply monitoring is required to evaluate the relative contribution of salinity from the source water to the effluent.

**(d) Plant Performance and Attainability**. Based on historical performance data, the treatment works is capable of meeting the effluent limit for EC.

**4. WQBEL Calculations**

**a.** This Order includes WQBELs for ammonia, and copper. The general methodology for calculating WQBELs based on the different criteria/objectives is described in subsections IV.C.4.b through e, below. See Attachment H for the WQBEL calculations.

**b. Effluent Concentration Allowance.** For each water quality criterion/objective, the ECA is calculated using the following steady-state mass balance equation from Section 1.4 of the SIP:

*Equations 2 and 3 again:*

$$ECA = C + D(C - B); \text{ when } C > B, \text{ and}$$
$$ECA = C; \text{ when } C \leq B$$

<u>where:</u>
*ECA* = effluent concentration allowance
*D* = dilution credit
*C* = the priority pollutant criterion/objective
*B* = the ambient background concentration.

According to the SIP, the ambient background concentration (B) in the equation above shall be the observed maximum with the exception that an ECA calculated from a priority pollutant criterion/objective that is intended to protect human health from carcinogenic effects shall use the arithmetic mean concentration of the ambient background samples. For ECAs based on MCLs, which implement the Basin Plan's chemical constituents objective and are applied as annual averages, an arithmetic mean is also used for B due to the long-term basis of the criteria.

c. **Basin Plan Objectives and MCLs.** For WQBELs based on site-specific numeric Basin Plan objectives or MCLs, the effluent limitations are applied directly as the ECA as either an MDEL, AMEL, or average annual effluent limitations, depending on the averaging period of the objective.

d. **Aquatic Toxicity Criteria.** WQBELs based on acute and chronic aquatic toxicity criteria are calculated in accordance with Section 1.4 of the SIP. The ECAs are converted to equivalent long-term averages (i.e. LTAacute and LTAchronic) using statistical multipliers and the lowest LTA is used to calculate the AMEL and MDEL using additional statistical multipliers.

e. **Human Health Criteria.** WQBELs based on human health criteria, are also calculated in accordance with Section 1.4 of the SIP. The ECAs are set equal to the AMEL and a statistical multiplier was used to calculate the MDEL.

$$AMEL = mult_{AMEL}\left[\min\left(\overbrace{M_A ECA_{acute}}^{LTA_{acute}}, M_C ECA_{chronic}\right)\right] \quad \text{(Equation 6)}$$

$$MDEL = mult_{MDEL}\left[\min\left(M_A ECA_{acute}, \underbrace{M_C ECA_{chronic}}_{LTA_{chronic}}\right)\right] \quad \text{(Equation 7)}$$

$$MDEL_{HH} = \left(\frac{mult_{MDEL}}{mult_{AMEL}}\right) AMEL_{HH} \quad \text{(Equation 8)}$$

<u>where:</u>
*multAMEL* = statistical multiplier converting minimum LTA to AMEL
*multMDEL* = statistical multiplier converting minimum LTA to MDEL
*MA* = statistical multiplier converting acute ECA to $LTA_{acute}$
*MC* = statistical multiplier converting chronic ECA to $LTA_{chronic}$

**Summary of Water Quality-Based Effluent Limitations**
**Discharge Point No. EFF-001**

**Table F-10. Summary of Water Quality-Based Effluent Limitations**

| Parameter | Units | Effluent Limitations | | | | |
|---|---|---|---|---|---|---|
| | | Average Monthly | Average Weekly | Maximum Daily | Instantaneous Minimum | Instantaneous Maximum |
| Ammonia | mg/L | 1.23 | 2.15 | n/a | n/a | n/a |
| Copper, Total Recoverable | mg/L | 15.1 | 28.0 | n/a | n/a | n/a |
| Total Chlorine Residual | mg/L | | 0.011 (4-day average) | 0.019 (1-hr average) | | |
| pH | S.U. | | | | 6.5 | 8.5 |
| Electrical Conductivity (25° C) | µmhos/cm | 900 **annual average** | | | | |

a. **Total Coliform Organisms.** When the wastewater receives dilution of less than 20:1, effluent total coliform organisms shall not exceed:

**i.** 2.2 most probable number (MPN) per 100 mL, as a 7-day median;
**ii.** 23 MPN/100mL no more than once in any 30-day period; and
**iii.** 240 MPN/100 mL, as an instantaneous maximum.

When the wastewater receives dilution of 20:1 or more, effluent total coliform organisms shall not exceed:

**i.** 23 MPN/100mL, as a 7-day median, and
**ii.** 240 MPN/100 mL, no more than once in any 30-day period.

**5. Whole Effluent Toxicity (WET)**

For compliance with the Basin Plan's narrative toxicity objective, this Order requires the Discharger to conduct whole effluent toxicity testing for acute and chronic toxicity, as specified in the Monitoring and Reporting Program (Attachment E section V.). This Order also contains effluent limitations for acute toxicity and requires the Discharger to implement best management practices to investigate the causes of, and identify corrective actions to reduce or eliminate effluent toxicity.

a. **Acute Aquatic Toxicity.** The Basin Plan contains a narrative toxicity objective that states, "*All waters shall be maintained free of toxic substances in concentrations that produce detrimental physiological responses in human, plant, animal, or aquatic life.*" (Basin Plan at page III-8.00) The Basin Plan also states that, *"…effluent limits based upon acute biotoxicity tests of effluents will be prescribed where appropriate…*". USEPA Region 9 provided guidance for the development of acute toxicity effluent limitations in the absence of numeric water quality objectives for toxicity in its document titled "Guidance for NPDES Permit Issuance", dated February 1994. In section B.2. "Toxicity Requirements" (pgs.

14-15) it states that, "*In the absence of specific numeric water quality objectives for acute and chronic toxicity, the narrative criterion 'no toxics in toxic amounts' applies. Achievement of the narrative criterion, as applied herein, means that ambient waters shall not demonstrate for acute toxicity: 1) less than 90% survival, 50% of the time, based on the monthly median, or 2) less than 70% survival, 10% of the time, based on any monthly median. For chronic toxicity, ambient waters shall not demonstrate a test result of greater than 1 TUc.*" Accordingly, effluent limitations for acute toxicity have been included in this Order as follows:

**Acute Toxicity.** Survival of aquatic organisms in 96-hour bioassays of undiluted waste shall be no less than:

Minimum for any one bioassay-- ------------------------------------ 70%
Median for any three consecutive bioassays --------- 90%

b. **Chronic Aquatic Toxicity.** The Basin Plan contains a narrative toxicity objective that states, "*All waters shall be maintained free of toxic substances in concentrations that produce detrimental physiological responses in human, plant, animal, or aquatic life.*" (Basin Plan at page III-8.00). Based on chronic WET testing performed by the Discharger from July 2007 through November 2011, the discharge does not have reasonable potential to cause or contribute to an in-stream excursion above of the Basin Plan's narrative toxicity objective as shown in the table below.

**Table F-11. Whole Effluent Chronic Toxicity Testing Results**

| Date | **Fathead Minnow** *Pimephales promelas* Survival (TUc) | Growth (TUc) | **Water Flea** *Ceriodaphnia dubia* Survival (TUc) | Reproduction (TUc) | **Green Algae** *Selenastrum capricornutum* Growth (TUc) |
|---|---|---|---|---|---|
| 7-1-07 | 1 | 1 | 1 | 1 | 1 |
| 1-1-10 | 1 | 1 | 1 | 1 | 1 |
| 3-3-11 | 1 | 1 | 1 | 1 | 1 |

The Monitoring and Reporting Program of this Order requires <u>annual</u> chronic WET monitoring for demonstration of compliance with the narrative toxicity objective. In addition to WET monitoring, the Special Provision in section VI.C.2.a. of the Order requires the Discharger to submit to the Regional Water Board an Initial Investigative TRE Workplan for approval by the Executive Officer, to ensure the Discharger has a plan to immediately move forward with the initial tiers of a TRE, in the event effluent toxicity is encountered in the future. The provision also includes a numeric toxicity monitoring trigger, requirements for accelerated monitoring, and requirements for TRE initiation if toxicity is demonstrated.

Numeric chronic WET effluent limitations have not been included in this Order. The SIP contains implementation gaps regarding the appropriate form and implementation of chronic toxicity limits. This has resulted in the petitioning of a NPDES permit in the Los Angeles Region[1] that contained numeric chronic toxicity effluent limitations. To address the petition, the State Water Board adopted WQO 2003-012 directing its staff to revise the toxicity control provisions in the SIP. The State Water Board states the following in WQO 2003-012, "*In reviewing this petition and receiving comments from numerous interested persons on the propriety of including numeric effluent limitations for chronic toxicity in NPDES permits for publicly-owned treatment works that discharge to inland waters, we have determined that this issue should be considered in a regulatory setting, in order to allow for full public discussion and deliberation. We intend to modify the SIP to specifically address the issue. We anticipate that review will occur within the next year. We therefore decline to make a determination here regarding the propriety of the final numeric effluent limitations for chronic toxicity contained in these permits.*" The process to revise the SIP is currently underway. Proposed changes include clarifying the appropriate form of effluent toxicity limits in NPDES permits and general expansion and standardization of toxicity control implementation related to the NPDES permitting process. Since the toxicity control provisions in the SIP are under revision it is infeasible to develop numeric effluent limitations for chronic toxicity. Therefore, this Order requires that the Discharger meet best management practices for compliance with the Basin Plan's narrative toxicity objective, as allowed under 40 CFR 122.44(k).

To ensure compliance with the Basin Plan's narrative toxicity objective, the Discharger is required to conduct chronic WET testing, as specified in the Monitoring and Reporting Program (Attachment E section V.). Furthermore, the Special Provision contained at VI.C.2.a. of this Order requires the Discharger to investigate the causes of, and identify and implement corrective actions to reduce or eliminate effluent toxicity. If the discharge demonstrates toxicity exceeding the numeric toxicity monitoring trigger, the Discharger is required to initiate a Toxicity Reduction Evaluation (TRE) in accordance with an approved TRE workplan. The numeric toxicity monitoring trigger is not an effluent limitation; it is the toxicity threshold at which the Discharger is required to perform accelerated chronic toxicity monitoring, as well as, the threshold to initiate a TRE if effluent toxicity has been demonstrated.

## D. Final Effluent Limitations

### 1. Mass-based Effluent Limitations

40 CFR 122.45(f)(1) requires effluent limitations be expressed in terms of mass, with some exceptions, and 40 CFR 122.45(f)(2) allows pollutants that are limited in terms of mass to additionally be limited in terms of other units of measurement. This Order

[1] In the Matter of the Review of Own Motion of Waste Discharge Requirements Order Nos. R4-2002-0121 [NPDES No. CA0054011] and R4-2002-0123 [NPDES NO. CA0055119] and Time Schedule Order Nos. R4-2002-0122 and R4-2002-0124 for Los Coyotes and Long Beach Wastewater Reclamation Plants Issued by the California Regional Water Quality Control Board, Los Angeles Region SWRCB/OCC FILES A-1496 AND 1496(a)

includes effluent limitations expressed in terms of mass and concentration. In addition, pursuant to the exceptions to mass limitations provided in 40 CFR 122.45(f)(1), some effluent limitations are not expressed in terms of mass, such as pH and temperature, and when the applicable standards are expressed in terms of concentration (e.g., CTR criteria and MCLs) and mass limitations are not necessary to protect the beneficial uses of the receiving water.

Mass-based effluent limitations were calculated based upon the design flow (Average Dry Weather Flow) permitted in section IV.A.1. of this Order.

**2. Averaging Periods for Effluent Limitations**

40 CFR 122.45 (d) requires average weekly and average monthly discharge limitations for publicly owned treatment works (POTWs) unless impracticable. However, for toxic pollutants and pollutant parameters in water quality permitting, USEPA recommends the use of a maximum daily effluent limitation in lieu of average weekly effluent limitations for two reasons. "*First, the basis for the 7-day average for POTWs derives from the secondary treatment requirements. This basis is not related to the need for assuring achievement of water quality standards. Second, a 7-day average, which could comprise up to seven or more daily samples, could average out peak toxic concentrations and therefore the discharge's potential for causing acute toxic effects would be missed.*" (TSD, pg. 96) This Order utilizes maximum daily effluent limitations in lieu of average weekly effluent limitations for **ammonia and copper** as recommended by the TSD for the achievement of water quality standards and for the protection of the beneficial uses of the receiving stream. Furthermore, for **ammonia and copper** weekly average effluent limitations have been replaced or supplemented with effluent limitations utilizing shorter averaging periods. The rationale for using shorter averaging periods for these constituents is discussed in section IV.C.3. of this Fact Sheet.

For effluent limitations based on Primary and Secondary MCLs, except nitrate, this Order includes annual average effluent limitations. The Primary and Secondary MCLs are drinking water standards contained in Title 22 of the California Code of Regulations. Title 22 requires compliance with these standards on an annual average basis (except for nitrate), when sampling at least quarterly. Since it is necessary to determine compliance on an annual average basis, it is impracticable to calculate average weekly and average monthly effluent limitations.

**3. Satisfaction of Anti-Backsliding Requirements**

The Clean Water Act specifies that a revised permit may not include effluent limitations that are less stringent than the previous permit unless a less stringent limitation is justified based on exceptions to the anti-backsliding provisions contained in Clean Water Act sections 402(o) or 303(d)(4), or, where applicable, 40 CFR 122.44(l).

The effluent limitations in this Order are at least as stringent as the effluent limitations in the existing Order, with the exception of effluent limitations for

**Electrical Conductivity**. The effluent limitations for these pollutants are less stringent than those in Order No. R5-2007-0032, based on the averaging period. This relaxation of effluent limitations is consistent with the anti-backsliding requirements of the CWA and federal regulations. Order No. R5-2007-0032 contained a monthly averaging period for Electrical Conductivity, and this permit contains an annual averaging period. The water quality criteria for electrical conductivity is based on a Secondary MCL. For Secondary MCLs, the Central Valley Water Board has concluded in prior permitting actions that the appropriate frequency basis for a limitation is an annual average, because Secondary MCLs are drinking water standards contained in Title 22 and Title 22 requires compliance with these standards on an annual average. The previous Order No. R5-2007-0032 incorrectly set the electrical conductivity effluent limitations on a monthly average, rather than an annual averaging period.

Changing the averaging period from monthly to annually is consistent with the antidegradation provisions of 40 CFR 131.12 and State Water Board Resolution No. 68-16. Any impact on existing water quality will be insignificant.

**4. Satisfaction of Antidegradation Policy**

This Order does not allow for an increase in flow or mass of pollutants to the receiving water. Therefore, a complete antidegradation analysis is not necessary. The Order requires compliance with applicable federal technology-based standards and with WQBELs where the discharge could have the reasonable potential to cause or contribute to an exceedance of water quality standards. The permitted discharge is consistent with the antidegradation provisions of 40 CFR 131.12 and State Water Board Resolution No. 68-16. Compliance with these requirements will result in the use of best practicable treatment or control of the discharge. The impact on existing water quality will be insignificant.

a. **Surface Water.** The permitted surface water discharge is consistent with the antidegradation provisions of 40 CFR 131.12 and State Water Board Resolution No. 68-16. Compliance with these requirements will result in the use of best practicable treatment or control of the discharge. The impact on existing water quality will be insignificant.

b. **Groundwater.** The Discharger utilizes aeration lagoons. Domestic wastewater contains constituents such as total dissolved solids (TDS), specific conductivity, pathogens, nitrates, organics, metals and oxygen demanding substances (BOD). Percolation from the lagoons may result in an increase in the concentration of these constituents in groundwater. The increase in the concentration of these constituents in groundwater must be consistent with Resolution No. 68-16. Any increase in pollutant concentrations in groundwater must be shown to be necessary to allow wastewater utility service necessary to accommodate housing and economic expansion in the area and must be consistent with maximum benefit to the people of the State of California. Some degradation of groundwater by the Discharger is consistent with Resolution No. 68-16 provided that:

- **i.** the degradation is limited in extent;
- **ii.** the degradation after effective source control, treatment, and control is limited to waste constituents typically encountered in municipal wastewater as specified in the groundwater limitations in this Order;
- **iii.** the Discharger minimizes the degradation by fully implementing, regularly maintaining, and optimally operating best practicable treatment and control (BPTC) measures; and
- **iv.** the degradation does not result in water quality less than that prescribed in the Basin Plan.

**5. Stringency of Requirements for Individual Pollutants**

This Order contains both technology-based effluent limitations and WQBELs for individual pollutants. The technology-based effluent limitations consist of restrictions on BOD, pH, and TSS. The WQBELs consist of restrictions on ammonia, copper, chlorine residual, pH, and total coliform organisms. This Order's technology-based pollutant restrictions implement the minimum, applicable federal technology-based requirements.

WQBELs have been scientifically derived to implement water quality objectives that protect beneficial uses. Both the beneficial uses and the water quality objectives have been approved pursuant to federal law and are the applicable federal water quality standards. To the extent that toxic pollutant WQBELs were derived from the CTR, the CTR is the applicable standard pursuant to 40 CFR 131.38. The scientific procedures for calculating the individual WQBELs for priority pollutants are based on the CTR-SIP, which was approved by USEPA on 18 May 2000. All beneficial uses and water quality objectives contained in the Basin Plan were approved under state law and submitted to and approved by USEPA prior to 30 May 2000. Any water quality objectives and beneficial uses submitted to USEPA prior to 30 May 2000, but not approved by USEPA before that date, are nonetheless "applicable water quality standards for purposes of the CWA" pursuant to 40 CFR 131.21(c)(1). Collectively, this Order's restrictions on individual pollutants are no more stringent than required to implement the requirements of the CWA.

On 25 January 2012 the Discharger submitted economic information indicating that the cost of complying with this Order would be approximately eight million dollars ($8,000,000). The Regional Water Board has considered the specific costs identified in the Discharger's submittal. As discussed in the Fact Sheet, the individual pollutant restrictions are no more stringent than necessary to implement applicable federal requirements or standards under the CWA. Relaxation of the effluent limitations is not permissible. Where appropriate, the accompanying Time Schedule Order provides additional time to achieve the pollutant-specific restriction.

## Summary of Final Effluent Limitations
## Discharge Point No. EFF-001

**Table F-12. Summary of Final Effluent Limitations**

| Parameter | Units | Effluent Limitations | | | | |
|---|---|---|---|---|---|---|
| | | **Average Monthly** | **Average Weekly** | **Maximum Daily** | **Instantaneous Minimum** | **Instantaneous Maximum** |
| Flow | MGD | 0.38 | | | | |
| Biochemical Oxygen Demand 5-day @ 20°C | mg/L | 30 | 45 | 60 | | |
| | lbs/day[1] | 95 | 143 | 190 | | |
| Total Suspended Solids | mg/L | 30 | 45 | 60 | | |
| | lbs/day[1] | 95 | 143 | 190 | | |
| pH | standard units | | | | 6.5 | 8.5 |
| Ammonia | mg/L | 1.23 | | 2.15 | | |
| Electrical Conductivity (25° C) | µmhos/cm | 900 **annual average** | | | | |
| Copper, Total Recoverable | µg/L | 15.1 | | 28.0 | | |

a. **Percent Removal.** The average monthly percent removal of 5-day biochemical oxygen demand ($BOD_5$) and total suspended solids (TSS) shall not be less than 65 percent.

b. **Acute Whole Effluent Toxicity.** Survival of aquatic organisms in 96-hour bioassays of undiluted waste shall be no less than:

   **i.** 70%, minimum for any one bioassay; and
   **ii.** 90%, median for any three consecutive bioassays.

c. **Total Residual Chlorine.** Effluent total residual chlorine shall not exceed:

   **i.** 0.011 mg/L, as a 4-day average; and
   **ii.** 0.019 mg/L, as a 1-hour average.

d. **Total Coliform Organisms.** When the wastewater receives dilution of less than 20:1, effluent total coliform organisms shall not exceed:

   **i.** 2.2 most probable number (MPN) per 100 mL, as a 7-day median;
   **ii.** 23 MPN/100mL no more than once in any 30-day period; and
   **iii.** 240 MPN/100 mL, as an instantaneous maximum.

   When the wastewater receives dilution of 20:1 or more, effluent total coliform organisms shall not exceed:

   **i.** 23 MPN/100mL, as a 7-day median, and

**ii.** 240 MPN/100 mL, no more than once in any 30-day period.

**e. Average Dry Weather Flow.** The average dry weather discharge flow shall not exceed **0.38 MGD**

### E. Interim Effluent Limitations – Not Applicable

### F. Reclamation Specifications – Not Applicable

## V. RATIONALE FOR RECEIVING WATER LIMITATIONS

Basin Plan water quality objectives to protect the beneficial uses of surface water and groundwater include numeric objectives and narrative objectives, including objectives for chemical constituents, toxicity, and tastes and odors. The toxicity objective requires that surface water and groundwater be maintained free of toxic substances in concentrations that produce detrimental physiological responses in humans, plants, animals, or aquatic life. The chemical constituent objective requires that surface water and groundwater shall not contain chemical constituents in concentrations that adversely affect any beneficial use or that exceed the maximum contaminant levels (MCLs) in Title 22, CCR. The tastes and odors objective states that surface water and groundwater shall not contain taste- or odor-producing substances in concentrations that cause nuisance or adversely affect beneficial uses. The Basin Plan requires the application of the most stringent objective necessary to ensure that surface water and groundwater do not contain chemical constituents, toxic substances, radionuclides, or taste and odor producing substances in concentrations that adversely affect domestic drinking water supply, agricultural supply, or any other beneficial use.

### A. Surface Water

**1.** CWA section 303(a-c), requires states to adopt water quality standards, including criteria where they are necessary to protect beneficial uses. The Regional Water Board adopted water quality criteria as water quality objectives in the Basin Plan. The Basin Plan states that "[t]*he numerical and narrative water quality objectives define the least stringent standards that the Regional Water Board will apply to regional waters in order to protect the beneficial uses.*" The Basin Plan includes numeric and narrative water quality objectives for various beneficial uses and water bodies. This Order contains receiving surface water limitations based on the Basin Plan numerical and narrative water quality objectives for ammonia, bacteria, biostimulatory substances, color, chemical constituents, dissolved oxygen, floating material, oil and grease, pH, pesticides, radioactivity, suspended sediment, settleable substances, suspended material, tastes and odors, temperature, toxicity, and turbidity.

# VI. RATIONALE FOR MONITORING AND REPORTING REQUIREMENTS

40 CFR 122.48 requires that all NPDES permits specify requirements for recording and reporting monitoring results. Water Code sections 13267 and 13383 authorizes the Regional Water Board to require technical and monitoring reports. The Monitoring and Reporting Program (Attachment E) of this Order, establishes monitoring and reporting requirements to implement federal and state requirements. The following provides the rationale for the monitoring and reporting requirements contained in the Monitoring and Reporting Program for the Facility.

## A. Influent Monitoring

**1.** Influent monitoring is required to collect data on the characteristics of the wastewater and to assess compliance with effluent limitations (e.g., $BOD_5$ and TSS reduction requirements). The monitoring frequencies for BOD, TSS, and pH (weekly) have been retained from Order No. R5-2007-0032.

## B. Effluent Monitoring

**1.** Pursuant to the requirements of 40 CFR 122.44(i)(2) effluent monitoring is required for all constituents with effluent limitations. Effluent monitoring is necessary to assess compliance with effluent limitations, assess the effectiveness of the treatment process, and to assess the impacts of the discharge on the receiving stream and groundwater.

**2.** Effluent monitoring frequencies and sample types for BOD, TSS, chlorine residual, ammonia, and temperature have been retained from Order No. R5-2007-0032 to determine compliance with effluent limitations for these parameters.

## C. Whole Effluent Toxicity Testing Requirements

**1. Acute Toxicity.** Quarterly 96-hour bioassay testing is required to demonstrate compliance with the effluent limitation for acute toxicity.

**2. Chronic Toxicity.** Annual chronic whole effluent toxicity testing is required in order to demonstrate compliance with the Basin Plan’s narrative toxicity objective.

## D. Receiving Water Monitoring

### 1. Surface Water

**a.** Receiving water monitoring is necessary to assess compliance with receiving water limitations and to assess the impacts of the discharge on the receiving stream.

### 2. Groundwater

**a.** CWC section 13267 states, in part, “*(a) A Regional Water Board, in establishing…waste discharge requirements… may investigate the quality* *of any waters of the state within its region" and "(b) (1) In conducting an investigation…, the Regional Water Board may require that any person who… discharges… waste…that could affect the quality of waters within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the Regional Water Board requires. The burden, including costs, of these reports shall bear a reasonable relationship to the need for the report and the benefits to be obtained from the reports.*" The burden, including costs, of these reports shall bear a reasonable relationship to the need for the report and the benefits to be obtained from the reports. In requiring those reports, the Regional Water Board shall provide the person with a written explanation with regard to the need for the reports, and shall identify the evidence that supports requiring that person to provide the reports. The Monitoring and Reporting Program is issued pursuant to CWC section 13267. The groundwater monitoring and reporting program required by this Order and the Monitoring and Reporting Program are necessary to assure compliance with these waste discharge requirements. The Discharger is responsible for the discharges of waste at the facility subject to this Order.

**b.** Monitoring of the groundwater must be conducted to determine if the discharge has caused an increase in constituent concentrations, when compared to background. The monitoring must, at a minimum, require a complete assessment of groundwater impacts including the vertical and lateral extent of degradation, an assessment of all wastewater-related constituents which may have migrated to groundwater, an analysis of whether additional or different methods of treatment or control of the discharge are necessary to provide best practicable treatment or control to comply with Resolution No. 68-16. Economic analysis is only one of many factors considered in determining best practicable treatment or control. If monitoring indicates that the discharge has incrementally increased constituent concentrations in groundwater above background, this permit may be reopened and modified. Until groundwater monitoring is sufficient, this Order contains Groundwater Limitations that allow groundwater quality to be degraded for certain constituents when compared to background groundwater quality, but not to exceed water quality objectives. If groundwater quality has been degraded by the discharge, the incremental change in pollutant concentration (when compared with background) may not be increased. If groundwater quality has been or may be degraded by the discharge, this Order may be reopened and specific numeric limitations established consistent with Resolution No. 68-16 and the Basin Plan.

**c.** This Order requires the Discharger to continue groundwater monitoring and includes a regular schedule of groundwater monitoring in the attached Monitoring and Reporting Program. The groundwater monitoring reports are necessary to evaluate impacts to waters of the State to assure protection of beneficial uses and compliance with Regional Water Board plans and policies, including Resolution No. 68-16. Evidence in the record includes effluent monitoring data that indicates the presence of constituents that may degrade groundwater and surface water.

### E. Other Monitoring Requirements

#### 1. Biosolids Monitoring

Biosolids monitoring is required to ensure compliance with the biosolids disposal requirements contained in the Special Provision contained in section VI.C.6.a. of this Order. Biosolids disposal requirements are imposed pursuant to 40 CFR Part 503 to protect public health and prevent groundwater degradation.

#### 2. Water Supply Monitoring

Water supply monitoring is required to evaluate the source of constituents in the wastewater.

## VII. RATIONALE FOR PROVISIONS

### A. Standard Provisions

Standard Provisions, which apply to all NPDES permits in accordance with 40 CFR 122.41, and additional conditions applicable to specified categories of permits in accordance with 40 CFR 122.42, are provided in Attachment D. The discharger must comply with all standard provisions and with those additional conditions that are applicable under 40 CFR 122.42.

40 CFR 122.41(a)(1) and (b) through (n) establish conditions that apply to all State-issued NPDES permits. These conditions must be incorporated into the permits either expressly or by reference. If incorporated by reference, a specific citation to the regulations must be included in the Order. 40 CFR 123.25(a)(12) allows the state to omit or modify conditions to impose more stringent requirements. In accordance with 40 CFR 123.25, this Order omits federal conditions that address enforcement authority specified in 40 CFR 122.41(j)(5) and (k)(2) because the enforcement authority under the CWC is more stringent. In lieu of these conditions, this Order incorporates by reference CWC section 13387(e).

### B. Special Provisions

#### 1. Reopener Provisions

**a.** Conditions that necessitate a major modification of a permit are described in 40 CFR 122.62, including:

**i.** If new or amended applicable water quality standards are promulgated or approved pursuant to section 303 of the CWA, or amendments thereto, this permit may be reopened and modified in accordance with the new or amended standards.

**ii.** When new information, that was not available at the time of permit issuance, would have justified different permit conditions at the time of issuance.

**b.** This Order may be reopened for modification, or revocation and reissuance, as a result of the detection of a reportable priority pollutant generated by special conditions included in this Order. These special conditions may be, but are not limited to, fish tissue sampling, whole effluent toxicity, monitoring requirements on internal waste stream(s), and monitoring for surrogate parameters. Additional requirements may be included in this Order as a result of the special condition monitoring data.

**c. Mercury.** This provision allows the Regional Water Board to reopen this Order in the event mercury is found to be causing toxicity based on acute or chronic toxicity test results, or if a TMDL program is adopted. In addition, this Order may be reopened if the Regional Water Board determines that a mercury offset program is feasible for dischargers subject to NPDES permits.

**d. Whole Effluent Toxicity.** This Order requires the Discharger to investigate the causes of, and identify corrective actions to reduce or eliminate effluent toxicity through a Toxicity Reduction Evaluation (TRE). This Order may be reopened to include a numeric chronic toxicity limitation, a new acute toxicity limitation, and/or a limitation for a specific toxicant identified in the TRE. Additionally, if a numeric chronic toxicity water quality objective is adopted by the State Water Board, this Order may be reopened to include a numeric chronic toxicity limitation based on that objective.

**e. Water Effects Ratio (WER) and Metal Translators.** A default WER of 1.0 has been used in this Order for calculating CTR criteria for applicable priority pollutant inorganic constituents. If the Discharger performs studies to determine site-specific WERs and/or site-specific dissolved-to-total metal translators, this Order may be reopened to modify the effluent limitations for the applicable inorganic constituents.

**f. Disinfection Requirements.** This reopener provision allows the Regional Water Board to reopen this Order to include a compliance schedule if the discharge does not meet the disinfection effluent limitations when dilution is <20:1.

**2. Special Studies and Additional Monitoring Requirements**

**a. Chronic Whole Effluent Toxicity Requirements.** The Basin Plan contains a narrative toxicity objective that states, "*All waters shall be maintained free of toxic substances in concentrations that produce detrimental physiological responses in human, plant, animal, or aquatic life.*" (Basin Plan at page III-8.00) Based on whole effluent chronic toxicity testing performed by the Discharger from July 2007 through November 2011, the discharge does not have reasonable potential to cause or contribute to an in-stream excursion above of the Basin Plan's narrative toxicity objective.

**Monitoring Trigger.** A numeric toxicity monitoring trigger of **> 1** TUc (where TUc = 100/NOEC) is applied in the provision, because this Order does not allow any dilution for the chronic condition. Therefore, a TRE is triggered when the effluent exhibits toxicity at 100% effluent.

**Accelerated Monitoring.** The provision requires accelerated WET testing when a regular WET test result exceeds the monitoring trigger. The purpose of accelerated monitoring is to determine, in an expedient manner, whether there is toxicity before requiring the implementation of a TRE. Due to possible seasonality of the toxicity, the accelerated monitoring should be performed in a timely manner, preferably taking no more than 2 to 3 months to complete.

The provision requires accelerated monitoring consisting of four chronic toxicity tests in a six-week period (i.e., one test every two weeks) using the species that exhibited toxicity. Guidance regarding accelerated monitoring and TRE initiation is provided in the *Technical Support Document for Water Quality-based Toxics Control,* EPA/505/2-90-001, March 1991 (TSD). The TSD at page 118 states, "*EPA recommends if toxicity is repeatedly or periodically present at levels above effluent limits more than 20 percent of the time, a TRE should be required.*" Therefore, four accelerated monitoring tests are required in this provision. If no toxicity is demonstrated in the four accelerated tests, then it demonstrates that toxicity is not present at levels above the monitoring trigger more than 20 percent of the time (only 1 of 5 tests are toxic, including the initial test). However, notwithstanding the accelerated monitoring results, if there is adequate evidence of effluent toxicity (i.e. toxicity present exceeding the monitoring trigger more than 20 percent of the time), the Executive Officer may require that the Discharger initiate a TRE.

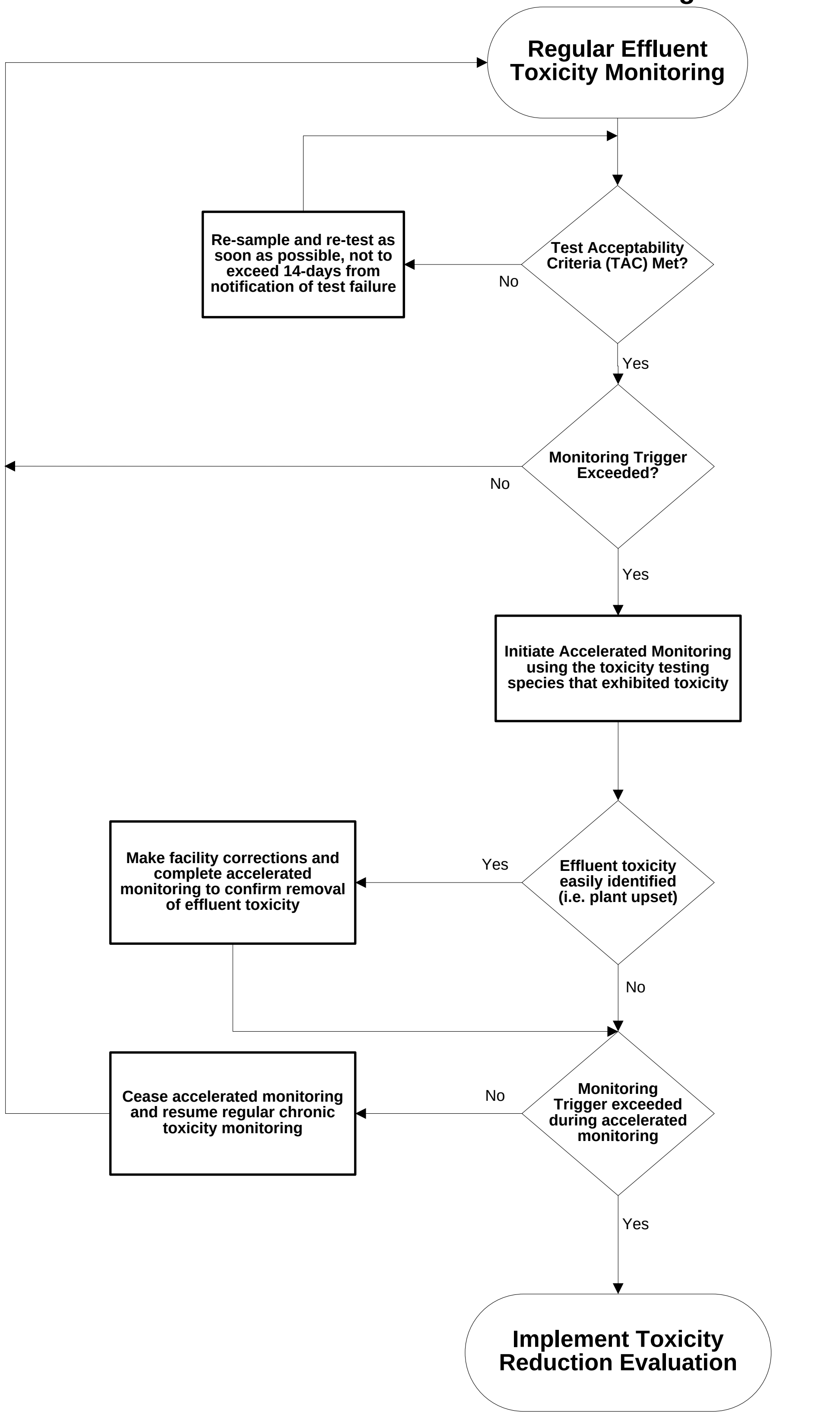
See the WET Accelerated Monitoring Flow Chart (Figure F-1), below, for further clarification of the accelerated monitoring requirements and for the decision points for determining the need for TRE initiation.

**TRE Guidance.** The Discharger is required to prepare a TRE Workplan in accordance with USEPA guidance. Numerous guidance documents are available, as identified below:

- Toxicity Reduction Evaluation Guidance for Municipal Wastewater Treatment Plants, EPA/833-B-99/002, August 1999.
- Generalized Methodology for Conducting Industrial Toxicity Reduction Evaluations (TREs), EPA/600/2-88/070, April 1989.
- Methods for Aquatic Toxicity Identification Evaluations: Phase I Toxicity Characterization Procedures, Second Edition, EPA 600/6-91/003, February 1991.

- Toxicity Identification Evaluation: Characterization of Chronically Toxic Effluents, Phase I, EPA/600/6-91/005F, May 1992.
- Methods for Aquatic Toxicity Identification Evaluations: Phase II Toxicity Identification Procedures for Samples Exhibiting Acute and Chronic Toxicity, Second Edition, EPA/600/R-92/080, September 1993.
- Methods for Aquatic Toxicity Identification Evaluations: Phase III Toxicity Confirmation Procedures for Samples Exhibiting Acute and Chronic Toxicity, Second Edition, EPA 600/R-92/081, September 1993.
- Methods for Measuring the Acute Toxicity of Effluents and Receiving Waters to Freshwater and Marine Organisms, Fifth Edition, EPA-821-R-02-012, October 2002.
- Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms, Fourth Edition, EPA-821-R-02-013, October 2002.
- Technical Support Document for Water Quality-based Toxics Control, EPA/505/2-90-001, March 1991.

## Figure F-1
## WET Accelerated Monitoring Flow Chart



**b. Effluent and Receiving Water Characterization Study.** An effluent and receiving water monitoring study is required to ensure adequate information is available for the next permit renewal. During the third year of this permit term, the Discharger is required to conduct **quarterly** monitoring of the effluent at EFF-001 and of the receiving water at RSW-001 for all priority pollutants and other constituents of concern as described in Attachment I.

**3. Best Management Practices and Pollution Prevention**

**a. Salinity Evaluation and Minimization Plan.** An Evaluation and Minimization Plan for salinity is required in this Order to ensure adequate measures are developed and implemented by the Discharger to reduce the discharge of salinity to Lateral K.

**4. Construction, Operation, and Maintenance Specifications**

**a.** The operation and maintenance specifications for the aeration and ballast ponds are necessary to protect the beneficial uses of the groundwater. The specifications included in this Order are retained from R5-2007-0032. In addition, reporting requirements related to use of the aeration and ballast ponds are required to monitor their use and the potential impact on groundwater.

**b. Turbidity Operational Requirements.** Turbidity is included as an operational specification as an indicator of the effectiveness of the treatment process and to assure compliance with effluent limitations for total coliform organisms. Failure of the treatment system such that virus removal is impaired would normally result in increased particles in the effluent, which result in higher effluent turbidity. Turbidity has a major advantage for monitoring filter performance, allowing immediate detection of filter failure and rapid corrective action. The operational specification requires that turbidity shall not exceed 2 NTU as a daily average; 5 NTU not to be exceeded more than 5% of the time within a 24-hour period, and 10 NTU as an instantaneous maximum, when dilution is <20:1. Turbidity specifications are included as operating criteria in section VI.C.4.b of this Order to ensure that adequate disinfection of wastewater is achieved.

**5. Special Provisions for Municipal Facilities (POTWs Only)**

**a. Pretreatment Requirements.**

**i.** The federal CWA section 307(b), and federal regulations, 40 CFR Part 403, require publicly owned treatment works to develop an acceptable industrial pretreatment program. A pretreatment program is required to prevent the introduction of pollutants, which will interfere with treatment plant operations or sludge disposal, and prevent pass through of pollutants that exceed water quality objectives, standards or permit limitations. Pretreatment requirements are imposed pursuant to 40 CFR Part 403.

**ii.** The Discharger shall implement and enforce its approved pretreatment program and is an enforceable condition of this Order. If the Discharger fails to perform the pretreatment functions, the Regional Water Board, the State Water Board or USEPA may take enforcement actions against the Discharger as authorized by the CWA.

**b.** The State Water Board issued General Waste Discharge Requirements for Sanitary Sewer Systems, Water Quality Order No. 2006-0003-DWQ (General Order) on 2 May 2006. The General Order requires public agencies that own or operate sanitary sewer systems with greater than one mile of pipes or sewer lines to enroll for coverage under the General Order. The General Order requires agencies to develop sanitary sewer management plans (SSMPs) and report all sanitary sewer overflows (SSOs), among other requirements and prohibitions.

Furthermore, the General Order contains requirements for operation and maintenance of collection systems and for reporting and mitigating sanitary sewer overflows. Inasmuch that the Discharger's collection system is part of the system that is subject to this Order, certain standard provisions are applicable as specified in Provisions, section VI.C.5. For instance, the 24-hour reporting requirements in this Order are not included in the General Order. The Discharger must comply with both the General Order and this Order. The Discharger and public agencies that are discharging wastewater into the facility were required to obtain enrollment for regulation under the General Order by 1 December 2006.

**6. Other Special Provisions**

**a. Ownership Change.** To maintain the accountability of the operation of the Facility, the Discharger is required to notify the succeeding owner or operator of the existence of this Order by letter if, and when, there is any change in control or ownership of land or waste discharge facilities presently owned or controlled by the Discharger.

**7. Compliance Schedules**

The Discharger submitted a request, and justification (dated 31 August 2012), for a compliance schedule for compliance with filtration and turbidity requirements. The compliance schedule justification included all items specified in the Compliance Schedule Policy. This Order establishes a compliance schedule for the new filtration and turbidity requirements and requires full compliance by **5 years from the effective date of this Order.** Because this permit requires filtration, and the wastewater treatment plant does not currently provide filtration, this permit includes a compliance schedule to allow time for the Discharger to make necessary improvements.

# VIII. PUBLIC PARTICIPATION

The Regional Water Board is considering the issuance of WDRs that will serve as an NPDES permit for the Facility. As a step in the WDR adoption process, the Regional Water Board staff has developed tentative WDRs. The Regional Water Board encourages public participation in the WDR adoption process.

## A. Notification of Interested Parties

The Regional Water Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for the discharge and has provided them with an opportunity to submit their written comments and recommendations. Notification was provided through direct mailings and internet posting.

## B. Written Comments

The staff determinations are tentative. Interested persons are invited to submit written comments concerning these tentative WDRs. Comments must be submitted either in person or by mail to the Executive Office at the Regional Water Board at the address above on the cover page of this Order.

To be fully responded to by staff and considered by the Regional Water Board, written comments must be received at the Regional Water Board offices by 5:00 p.m. on 27 August 2012.

## C. Public Hearing

The Regional Water Board will hold a public hearing on the tentative WDRs during its regular Board meeting on the following date and time and at the following location:

Date: 4/5 October 2012
Time: 8:30 a.m.
Location: Regional Water Quality Control Board, Central Valley Region
11020 Sun Center Dr., Suite #200
Rancho Cordova, CA 95670

Interested persons are invited to attend. At the public hearing, the Regional Water Board will hear testimony, if any, pertinent to the discharge, WDRs, and permit. Oral testimony will be heard; however, for accuracy of the record, important testimony should be in writing.

Please be aware that dates and venues may change. Our Web address is www.waterboards.ca.gov/centralvalley where you can access the current agenda for changes in dates and locations.

### D. Waste Discharge Requirements Petitions

Any aggrieved person may petition the State Water Board to review the decision of the Regional Water Board regarding the final WDRs. The petition must be submitted within 30 days of the Regional Water Board's action to the following address:

State Water Resources Control Board
Office of Chief Counsel
P.O. Box 100, 1001 I Street
Sacramento, CA 95812-0100

### E. Information and Copying

The Report of Waste Discharge, related documents, tentative effluent limitations and special provisions, comments received, and other information are on file and may be inspected at the address above at any time between 8:30 a.m. and 4:45 p.m., Monday through Friday. Copying of documents may be arranged through the Regional Water Board by calling (530) 224-4845.

### F. Register of Interested Persons

Any person interested in being placed on the mailing list for information regarding the WDRs and NPDES permit should contact the Regional Water Board, reference this Facility, and provide a name, address, and phone number.

### G. Additional Information

Requests for additional information or questions regarding this order should be directed to Katie Bowman at (530) 226-3458.

## ATTACHMENT G – SUMMARY OF REASONABLE POTENTIAL ANALYSIS

| Constituent | Units | MEC | B | C | CMC | CCC | Water & Org | Org. Only | Basin Plan | MCL | Reasonable Potential |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Ammonia | mg/L | 18 | NA | 4.4 | 2.14 | 4.4 | | | | | Yes |
| Total Recoverable Aluminum | µg/L | <24.5 | 693 | NA | 750 | 87 | | | | | No |
| Total Recoverable Copper | µg/L | 9.8 | 20.8 | 18 | 28 | 18 | | | | | Yes |
| Nitrate (as N) | mg/L | 0.5 | NA | | | | | | | 10 | No |
| Bis(2-Ethylhexyl)Phthalate | µg/L | <2 (DNQ) | ND | 1.8 | | | 1.8 | | | | No |

General Note: All inorganic concentrations are given as a total recoverable.
MEC = Maximum Effluent Concentration
B = Maximum Receiving Water Concentration or lowest detection level, if non-detect
C = Criterion used for Reasonable Potential Analysis
CMC = Criterion Maximum Concentration (CTR or NTR)
CCC = Criterion Continuous Concentration (CTR or NTR)
Water & Org = Human Health Criterion for Consumption of Water & Organisms (CTR or NTR)
Org. Only = Human Health Criterion for Consumption of Organisms Only (CTR or NTR)
Basin Plan = Numeric Site-specific Basin Plan Water Quality Objective
MCL = Drinking Water Standards Maximum Contaminant Level
NA = Not Available
ND = Non-detect

# ATTACHMENT H – CALCULATION OF WQBELS

| Parameter | Units | Most Stringent Criteria | | | Dilution Factors | | | HH Calculations | | | Aquatic Life Calculations | | | | | | | | | Final Effluent Limitations | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | HH | CMC | CCC | HH | CMC | CCC | $ECA_{HH}$ = $AMEL_{HH}$ | AMEL/MDEL $Multiplier_{HH}$ | $MDEL_{HH}$ | ECA $Multiplier_{acute}$ | $LTA_{acute}$ | ECA $Multiplier_{chronic}$ | $LTA_{chronic}$ | Lowest LTA | AMEL $Multiplier_{95}$ | $AMEL_{AL}$ | MDEL $Multiplier_{99}$ | $MDEL_{AL}$ | Lowest AMEL | Lowest MDEL |
| Ammonia Nitrogen, Total (as N) | mg/L | -- | 2.14[2] | 4.4 | -- | -- | -- | -- | -- | -- | 0.527 | 1.13 | 0.715 | 3.14 | 1.13 | 1.09 | 1.23 | 1.9 | 2.15 | 1.23 | 2.15 |
| Copper, Total Recoverable | µg/L | -- | 28 | 18[1] | -- | -- | -- | -- | -- | -- | 0.373 | 10.44 | 0.581 | 10.46 | 10.44 | 1.45 | 15.1 | 2.68 | 28.0 | 15.1 | 28.0 |

1 USEPA Ambient Water Quality Criteria.
2 The lowest LTA representing the acute, 4-day CCC, and 30-day CCC is selected for deriving the AMEL and MDEL.

# ATTACHMENT I – EFFLUENT AND RECEIVING WATER CHARACTERIZATION STUDY

**I. Background.** Sections 2.4.1 through 2.4.4 of the SIP provide minimum standards for analyses and reporting. (Copies of the SIP may be obtained from the State Water Resources Control Board, or downloaded from http://www.waterboards.ca.gov/iswp/index.html). To implement the SIP, effluent and receiving water data are needed for all priority pollutants. Effluent and receiving water pH and hardness are required to evaluate the toxicity of certain priority pollutants (such as heavy metals) where the toxicity of the constituents varies with pH and/or hardness. Section 3 of the SIP prescribes mandatory monitoring of dioxin congeners. In addition to specific requirements of the SIP, the Regional Water Board is requiring the following monitoring:

**A. Drinking water constituents**. Constituents for which drinking water Maximum Contaminant Levels (MCLs) have been prescribed in the California Code of Regulation are included in the *Water Quality Control Plan, Fourth Edition, for the Sacramento and San Joaquin River Basins* (Basin Plan). The Basin Plan defines virtually all surface waters within the Central Valley Region as having existing or potential beneficial uses for municipal and domestic supply. The Basin Plan further requires that, at a minimum, water designated for use as domestic or municipal supply shall not contain concentrations of chemical constituents in excess of the MCLs contained in the California Code of Regulations.

**B. Effluent and receiving water temperature.** This is both a concern for application of certain temperature-sensitive constituents, such as fluoride, and for compliance with the Basin Plan's thermal discharge requirements.

**C. Effluent and receiving water hardness and pH.** These are necessary because several of the CTR constituents are hardness and pH dependent.

**II. Monitoring Requirements.**

**A. Quarterly Monitoring**. Quarterly priority pollutant samples shall be collected from the effluent and upstream receiving water (EFF-001 and RSW-001) and analyzed for the constituents listed in Table I-1 **during the third year of the permit term**. The results of such monitoring shall be submitted to the Regional Water Board, during the fourth year of the permit term. Each individual monitoring event shall provide representative sample results for the effluent and upstream receiving water.

**B. Concurrent Sampling.** Effluent and receiving water sampling shall be performed at approximately the same time, on the same date.

**C. Sample type.** All effluent samples shall be taken as 24-hour flow proportioned composite samples. All receiving water samples shall be taken as grab samples.

## Table I-1. Priority Pollutants

| CTR # | Constituent | CAS Number | Controlling Water Quality Criterion for Surface Waters: Basis | Controlling Water Quality Criterion for Surface Waters: Criterion Concentration ug/L or noted[1] | Criterion Quantitation Limit ug/L or noted | Suggested Test Methods |
|---|---|---|---|---|---|---|
| **VOLATILE ORGANICS** | | | | | | |
| 28 | 1,1-Dichloroethane | 75343 | Primary MCL | 5 | 0.5 | EPA 8260B |
| 30 | 1,1-Dichloroethene | 75354 | National Toxics Rule | 0.057 | 0.5 | EPA 8260B |
| 41 | 1,1,1-Trichloroethane | 71556 | Primary MCL | 200 | 0.5 | EPA 8260B |
| 42 | 1,1,2-Trichloroethane | 79005 | National Toxics Rule | 0.6 | 0.5 | EPA 8260B |
| 37 | 1,1,2,2-Tetrachloroethane | 79345 | National Toxics Rule | 0.17 | 0.5 | EPA 8260B |
| 75 | 1,2-Dichlorobenzene | 95501 | Taste & Odor | 10 | 0.5 | EPA 8260B |
| 29 | 1,2-Dichloroethane | 107062 | National Toxics Rule | 0.38 | 0.5 | EPA 8260B |
| | cis-1,2-Dichloroethene | 156592 | Primary MCL | 6 | 0.5 | EPA 8260B |
| 31 | 1,2-Dichloropropane | 78875 | Calif. Toxics Rule | 0.52 | 0.5 | EPA 8260B |
| 101 | 1,2,4-Trichlorobenzene | 120821 | Public Health Goal | 5 | 0.5 | EPA 8260B |
| 76 | 1,3-Dichlorobenzene | 541731 | Taste & Odor | 10 | 0.5 | EPA 8260B |
| 32 | 1,3-Dichloropropene | 542756 | Primary MCL | 0.5 | 0.5 | EPA 8260B |
| 77 | 1,4-Dichlorobenzene | 106467 | Primary MCL | 5 | 0.5 | EPA 8260B |
| 17 | Acrolein | 107028 | Aquatic Toxicity | 21 | 2 | EPA 8260B |
| 18 | Acrylonitrile | 107131 | National Toxics Rule | 0.059 | 2 | EPA 8260B |
| 19 | Benzene | 71432 | Primary MCL | 1 | 0.5 | EPA 8260B |
| 20 | Bromoform | 75252 | Calif. Toxics Rule | 4.3 | 0.5 | EPA 8260B |
| 34 | Bromomethane | 74839 | Calif. Toxics Rule | 48 | 1 | EPA 8260B |
| 21 | Carbon tetrachloride | 56235 | National Toxics Rule | 0.25 | 0.5 | EPA 8260B |
| 22 | Chlorobenzene (mono chlorobenzene) | 108907 | Taste & Odor | 50 | 0.5 | EPA 8260B |
| 24 | Chloroethane | 75003 | Taste & Odor | 16 | 0.5 | EPA 8260B |
| 25 | 2- Chloroethyl vinyl ether | 110758 | Aquatic Toxicity | 122 (3) | 1 | EPA 8260B |
| 26 | Chloroform | 67663 | OEHHA Cancer Risk | 1.1 | 0.5 | EPA 8260B |
| 35 | Chloromethane | 74873 | USEPA Health Advisory | 3 | 0.5 | EPA 8260B |
| 23 | Dibromochloromethane | 124481 | Calif. Toxics Rule | 0.41 | 0.5 | EPA 8260B |
| 27 | Dichlorobromomethane | 75274 | Calif. Toxics Rule | 0.56 | 0.5 | EPA 8260B |
| 36 | Dichloromethane | 75092 | Calif. Toxics Rule | 4.7 | 0.5 | EPA 8260B |
| 33 | Ethylbenzene | 100414 | Taste & Odor | 29 | 0.5 | EPA 8260B |
| 88 | Hexachlorobenzene | 118741 | Calif. Toxics Rule | 0.00075 | 1 | EPA 8260B |
| 89 | Hexachlorobutadiene | 87683 | National Toxics Rule | 0.44 | 1 | EPA 8260B |
| 91 | Hexachloroethane | 67721 | National Toxics Rule | 1.9 | 1 | EPA 8260B |
| 94 | Naphthalene | 91203 | USEPA IRIS | 14 | 10 | EPA 8260B |
| 38 | Tetrachloroethene | 127184 | National Toxics Rule | 0.8 | 0.5 | EPA 8260B |
| 39 | Toluene | 108883 | Taste & Odor | 42 | 0.5 | EPA 8260B |

| CTR # | Constituent | CAS Number | Controlling Water Quality Criterion for Surface Waters | | Criterion Quantitation Limit ug/L or noted | Suggested Test Methods |
|---|---|---|---|---|---|---|
| | | | Basis | Criterion Concentration ug/L or noted[1] | | |
| 40 | trans-1,2-Dichloroethylene | 156605 | Primary MCL | 10 | 0.5 | EPA 8260B |
| 43 | Trichloroethene | 79016 | National Toxics Rule | 2.7 | 0.5 | EPA 8260B |
| 44 | Vinyl chloride | 75014 | Primary MCL | 0.5 | 0.5 | EPA 8260B |
| | Methyl-tert-butyl ether (MTBE) | 1634044 | Secondary MCL | 5 | 0.5 | EPA 8260B |
| | Trichlorofluoromethane | 75694 | Primary MCL | 150 | 5 | EPA 8260B |
| | 1,1,2-Trichloro-1,2,2-Trifluoroethane | 76131 | Primary MCL | 1200 | 10 | EPA 8260B |
| | Styrene | 100425 | Taste & Odor | 11 | 0.5 | EPA 8260B |
| | Xylenes | 1330207 | Taste & Odor | 17 | 0.5 | EPA 8260B |
| **SEMI-VOLATILE ORGANICS** | | | | | | |
| 60 | 1,2-Benzanthracene | 56553 | Calif. Toxics Rule | 0.0044 | 5 | EPA 8270C |
| 85 | 1,2-Diphenylhydrazine | 122667 | National Toxics Rule | 0.04 | 1 | EPA 8270C |
| 45 | 2-Chlorophenol | 95578 | Taste and Odor | 0.1 | 2 | EPA 8270C |
| 46 | 2,4-Dichlorophenol | 120832 | Taste and Odor | 0.3 | 1 | EPA 8270C |
| 47 | 2,4-Dimethylphenol | 105679 | Calif. Toxics Rule | 540 | 2 | EPA 8270C |
| 49 | 2,4-Dinitrophenol | 51285 | National Toxics Rule | 70 | 5 | EPA 8270C |
| 82 | 2,4-Dinitrotoluene | 121142 | National Toxics Rule | 0.11 | 5 | EPA 8270C |
| 55 | 2,4,6-Trichlorophenol | 88062 | Taste and Odor | 2 | 10 | EPA 8270C |
| 83 | 2,6-Dinitrotoluene | 606202 | USEPA IRIS | 0.05 | 5 | EPA 8270C |
| 50 | 2-Nitrophenol | 25154557 | Aquatic Toxicity | 150 (5) | 10 | EPA 8270C |
| 71 | 2-Chloronaphthalene | 91587 | Aquatic Toxicity | 1600 (6) | 10 | EPA 8270C |
| 78 | 3,3'-Dichlorobenzidine | 91941 | National Toxics Rule | 0.04 | 5 | EPA 8270C |
| 62 | 3,4-Benzofluoranthene | 205992 | Calif. Toxics Rule | 0.0044 | 10 | EPA 8270C |
| 52 | 4-Chloro-3-methylphenol | 59507 | Aquatic Toxicity | 30 | 5 | EPA 8270C |
| 48 | 4,6-Dinitro-2-methylphenol | 534521 | National Toxics Rule | 13.4 | 10 | EPA 8270C |
| 51 | 4-Nitrophenol | 100027 | USEPA Health Advisory | 60 | 5 | EPA 8270C |
| 69 | 4-Bromophenyl phenyl ether | 101553 | Aquatic Toxicity | 122 | 10 | EPA 8270C |
| 72 | 4-Chlorophenyl phenyl ether | 7005723 | Aquatic Toxicity | 122 (3) | 5 | EPA 8270C |
| 56 | Acenaphthene | 83329 | Taste and Odor | 20 | 1 | EPA 8270C |
| 57 | Acenaphthylene | 208968 | No Criteria Available | | 10 | EPA 8270C |
| 58 | Anthracene | 120127 | Calif. Toxics Rule | 9,600 | 10 | EPA 8270C |
| 59 | Benzidine | 92875 | National Toxics Rule | 0.00012 | 5 | EPA 8270C |
| 61 | Benzo(a)pyrene (3,4-Benzopyrene) | 50328 | Calif. Toxics Rule | 0.0044 | 0.1 | EPA 8270C |
| 63 | Benzo(g,h,i)perylene | 191242 | No Criteria Available | | 5 | EPA 8270C |
| 64 | Benzo(k)fluoranthene | 207089 | Calif. Toxics Rule | 0.0044 | 2 | EPA 8270C |
| 65 | Bis(2-chloroethoxy) methane | 111911 | No Criteria Available | | 5 | EPA 8270C |
| 66 | Bis(2-chloroethyl) ether | 111444 | National Toxics Rule | 0.031 | 1 | EPA 8270C |

| CTR # | Constituent | CAS Number | Controlling Water Quality Criterion for Surface Waters: Basis | Controlling Water Quality Criterion for Surface Waters: Criterion Concentration ug/L or noted[1] | Criterion Quantitation Limit ug/L or noted | Suggested Test Methods |
|---|---|---|---|---|---|---|
| 67 | Bis(2-chloroisopropyl) ether | 39638329 | Aquatic Toxicity | 122 (3) | 10 | EPA 8270C |
| 68 | Bis(2-ethylhexyl) phthalate | 117817 | National Toxics Rule | 1.8 | 3 | EPA 8270C |
| 70 | Butyl benzyl phthalate | 85687 | Aquatic Toxicity | 3 (7) | 10 | EPA 8270C |
| 73 | Chrysene | 218019 | Calif. Toxics Rule | 0.0044 | 5 | EPA 8270C |
| 81 | Di-n-butylphthalate | 84742 | Aquatic Toxicity | 3 (7) | 10 | EPA 8270C |
| 84 | Di-n-octylphthalate | 117840 | Aquatic Toxicity | 3 (7) | 10 | EPA 8270C |
| 74 | Dibenzo(a,h)-anthracene | 53703 | Calif. Toxics Rule | 0.0044 | 0.1 | EPA 8270C |
| 79 | Diethyl phthalate | 84662 | Aquatic Toxicity | 3 (7) | 2 | EPA 8270C |
| 80 | Dimethyl phthalate | 131113 | Aquatic Toxicity | 3 (7) | 2 | EPA 8270C |
| 86 | Fluoranthene | 206440 | Calif. Toxics Rule | 300 | 10 | EPA 8270C |
| 87 | Fluorene | 86737 | Calif. Toxics Rule | 1300 | 10 | EPA 8270C |
| 90 | Hexachlorocyclopentadiene | 77474 | Taste and Odor | 1 | 1 | EPA 8270C |
| 92 | Indeno(1,2,3-c,d)pyrene | 193395 | Calif. Toxics Rule | 0.0044 | 0.05 | EPA 8270C |
| 93 | Isophorone | 78591 | National Toxics Rule | 8.4 | 1 | EPA 8270C |
| 98 | N-Nitrosodiphenylamine | 86306 | National Toxics Rule | 5 | 1 | EPA 8270C |
| 96 | N-Nitrosodimethylamine | 62759 | National Toxics Rule | 0.00069 | 5 | EPA 8270C |
| 97 | N-Nitrosodi-n-propylamine | 621647 | Calif. Toxics Rule | 0.005 | 5 | EPA 8270C |
| 95 | Nitrobenzene | 98953 | National Toxics Rule | 17 | 10 | EPA 8270C |
| 53 | Pentachlorophenol | 87865 | Calif. Toxics Rule | 0.28 | 0.2 | EPA 8270C |
| 99 | Phenanthrene | 85018 | No Criteria Available | | 5 | EPA 8270C |
| 54 | Phenol | 108952 | Taste and Odor | 5 | 1 | EPA 8270C |
| 100 | Pyrene | 129000 | Calif. Toxics Rule | 960 | 10 | EPA 8270C |
| **INORGANICS** | | | | | | |
| | Aluminum | 7429905 | Ambient Water Quality | 87 | 50 | EPA 6020/200.8 |
| 1 | Antimony | 7440360 | Primary MCL | 6 | 5 | EPA 6020/200.8 |
| 2 | Arsenic | 7440382 | Ambient Water Quality | 0.018 | 0.01 | EPA 1632 |
| 15 | Asbestos | 1332214 | National Toxics Rule/ Primary MCL | 7 MFL | 0.2 MFL >10um | EPA/600/R-93/116(PCM) |
| | Barium | 7440393 | Basin Plan Objective | 100 | 100 | EPA 6020/200.8 |
| 3 | Beryllium | 7440417 | Primary MCL | 4 | 1 | EPA 6020/200.8 |
| 4 | Cadmium | 7440439 | Public Health Goal | 0.07 | 0.25 | EPA 1638/200.8 |
| 5a | Chromium (total) | 7440473 | Primary MCL | 50 | 2 | EPA 6020/200.8 |
| 5b | Chromium (VI) | 18540299 | Public Health Goal | 0.2 | 0.5 | EPA 7199/1636 |
| 6 | Copper | 7440508 | National Toxics Rule | 4.1 (2) | 0.5 | EPA 6020/200.8 |
| 14 | Cyanide | 57125 | National Toxics Rule | 5.2 | 5 | EPA 9012A |
| | Fluoride | 7782414 | Public Health Goal | 1000 | 0.1 | EPA 300 |
| | Iron | 7439896 | Secondary MCL | 300 | 100 | EPA 6020/200.8 |

| CTR # | Constituent | CAS Number | Controlling Water Quality Criterion for Surface Waters | | Criterion Quantitation Limit ug/L or noted | Suggested Test Methods |
|---|---|---|---|---|---|---|
| | | | Basis | Criterion Concentration ug/L or noted[1] | | |
| 7 | Lead | 7439921 | Calif. Toxics Rule | 0.92 (2) | 0.5 | EPA 1638 |
| 8 | Mercury | 7439976 | TMDL Development | | 0.0002 (11) | EPA 1669/1631 |
| | Manganese | 7439965 | Secondary MCL/ Basin Plan Objective | 50 | 20 | EPA 6020/200.8 |
| 9 | Nickel | 7440020 | Calif. Toxics Rule | 24 (2) | 5 | EPA 6020/200.8 |
| 10 | Selenium | 7782492 | Calif. Toxics Rule | 5 (8) | 5 | EPA 6020/200.8 |
| 11 | Silver | 7440224 | Calif. Toxics Rule | 0.71 (2) | 1 | EPA 6020/200.8 |
| 12 | Thallium | 7440280 | National Toxics Rule | 1.7 | 1 | EPA 6020/200.8 |
| | Tributyltin | 688733 | Ambient Water Quality | 0.063 | 0.002 | EV-024/025 |
| 13 | Zinc | 7440666 | Calif. Toxics Rule/ Basin Plan Objective | 54/ 16 (2) | 10 | EPA 6020/200.8 |
| **PESTICIDES - PCBs** | | | | | | |
| 110 | 4,4'-DDD | 72548 | Calif. Toxics Rule | 0.00083 | 0.02 | EPA 8081A |
| 109 | 4,4'-DDE | 72559 | Calif. Toxics Rule | 0.00059 | 0.01 | EPA 8081A |
| 108 | 4,4'-DDT | 50293 | Calif. Toxics Rule | 0.00059 | 0.01 | EPA 8081A |
| 112 | alpha-Endosulfan | 959988 | National Toxics Rule | 0.056 (9) | 0.02 | EPA 8081A |
| 103 | alpha-Hexachlorocyclohexane (BHC) | 319846 | Calif. Toxics Rule | 0.0039 | 0.01 | EPA 8081A |
| | Alachlor | 15972608 | Primary MCL | 2 | 1 | EPA 8081A |
| 102 | Aldrin | 309002 | Calif. Toxics Rule | 0.00013 | 0.005 | EPA 8081A |
| 113 | beta-Endosulfan | 33213659 | Calif. Toxics Rule | 0.056 (9) | 0.01 | EPA 8081A |
| 104 | beta-Hexachlorocyclohexane | 319857 | Calif. Toxics Rule | 0.014 | 0.005 | EPA 8081A |
| 107 | Chlordane | 57749 | Calif. Toxics Rule | 0.00057 | 0.1 | EPA 8081A |
| 106 | delta-Hexachlorocyclohexane | 319868 | No Criteria Available | | 0.005 | EPA 8081A |
| 111 | Dieldrin | 60571 | Calif. Toxics Rule | 0.00014 | 0.01 | EPA 8081A |
| 114 | Endosulfan sulfate | 1031078 | Ambient Water Quality | 0.056 | 0.05 | EPA 8081A |
| 115 | Endrin | 72208 | Calif. Toxics Rule | 0.036 | 0.01 | EPA 8081A |
| 116 | Endrin Aldehyde | 7421934 | Calif. Toxics Rule | 0.76 | 0.01 | EPA 8081A |
| 117 | Heptachlor | 76448 | Calif. Toxics Rule | 0.00021 | 0.01 | EPA 8081A |
| 118 | Heptachlor Epoxide | 1024573 | Calif. Toxics Rule | 0.0001 | 0.01 | EPA 8081A |
| 105 | Lindane (gamma-Hexachlorocyclohexane) | 58899 | Calif. Toxics Rule | 0.019 | 0.019 | EPA 8081A |
| 119 | PCB-1016 | 12674112 | Calif. Toxics Rule | 0.00017 (10) | 0.5 | EPA 8082 |
| 120 | PCB-1221 | 11104282 | Calif. Toxics Rule | 0.00017 (10) | 0.5 | EPA 8082 |
| 121 | PCB-1232 | 11141165 | Calif. Toxics Rule | 0.00017 (10) | 0.5 | EPA 8082 |
| 122 | PCB-1242 | 53469219 | Calif. Toxics Rule | 0.00017 (10) | 0.5 | EPA 8082 |
| 123 | PCB-1248 | 12672296 | Calif. Toxics Rule | 0.00017 (10) | 0.5 | EPA 8082 |
| 124 | PCB-1254 | 11097691 | Calif. Toxics Rule | 0.00017 (10) | 0.5 | EPA 8082 |
| 125 | PCB-1260 | 11096825 | Calif. Toxics Rule | 0.00017 (10) | 0.5 | EPA 8082 |

| CTR # | Constituent | CAS Number | Controlling Water Quality Criterion for Surface Waters | | Criterion Quantitation Limit ug/L or noted | Suggested Test Methods |
|---|---|---|---|---|---|---|
| | | | Basis | Criterion Concentration ug/L or noted[1] | | |
| 126 | Toxaphene | 8001352 | Calif. Toxics Rule | 0.0002 | 0.5 | EPA 8081A |
| | Atrazine | 1912249 | Public Health Goal | 0.15 | 1 | EPA 8141A |
| | Bentazon | 25057890 | Primary MCL | 18 | 2 | EPA 643/ 515.2 |
| | Carbofuran | 1563662 | CDFG Hazard Assess. | 0.5 | 5 | EPA 8318 |
| | 2,4-D | 94757 | Primary MCL | 70 | 10 | EPA 8151A |
| | Dalapon | 75990 | Ambient Water Quality | 110 | 10 | EPA 8151A |
| | 1,2-Dibromo-3-chloropropane (DBCP) | 96128 | Public Health Goal | 0.0017 | 0.01 | EPA 8260B |
| | Di(2-ethylhexyl)adipate | 103231 | USEPA IRIS | 30 | 5 | EPA 8270C |
| | Dinoseb | 88857 | Primary MCL | 7 | 2 | EPA 8151A |
| | Diquat | 85007 | Ambient Water Quality | 0.5 | 4 | EPA 8340/ 549.1/HPLC |
| | Endothal | 145733 | Primary MCL | 100 | 45 | EPA 548.1 |
| | Ethylene Dibromide | 106934 | OEHHA Cancer Risk | 0.0097 | 0.02 | EPA 8260B/504 |
| | Glyphosate | 1071836 | Primary MCL | 700 | 25 | HPLC/EPA 547 |
| | Methoxychlor | 72435 | Public Health Goal | 30 | 10 | EPA 8081A |
| | Molinate (Ordram) | 2212671 | CDFG Hazard Assess. | 13 | 2 | EPA 634 |
| | Oxamyl | 23135220 | Public Health Goal | 50 | 20 | EPA 8318/632 |
| | Picloram | 1918021 | Primary MCL | 500 | 1 | EPA 8151A |
| | Simazine (Princep) | 122349 | USEPA IRIS | 3.4 | 1 | EPA 8141A |
| | Thiobencarb | 28249776 | Basin Plan Objective/ Secondary MCL | 1 | 1 | HPLC/EPA 639 |
| 16 | 2,3,7,8-TCDD (Dioxin) | 1746016 | Calif. Toxics Rule | 1.30E-08 | 5.00E-06 | EPA 8290 (HRGC) MS |
| | 2,4,5-TP (Silvex) | 93765 | Ambient Water Quality | 10 | 1 | EPA 8151A |
| | Diazinon | 333415 | CDFG Hazard Assess. | 0.05 | 0.25 | EPA 8141A/GCMS |
| | Chlorpyrifos | 2921882 | CDFG Hazard Assess. | 0.014 | 1 | EPA 8141A/GCMS |
| **OTHER CONSTITUENTS** | | | | | | |
| | Ammonia (as N) | 7664417 | Ambient Water Quality | 1500 (4) | | EPA 350.1 |
| | Chloride | 16887006 | Agricultural Use | 106,000 | | EPA 300.0 |
| | Flow | | | 1 CFS | | |
| | Hardness (as $CaCO_3$) | | | 5000 | | EPA 130.2 |
| | Foaming Agents (MBAS) | | Secondary MCL | 500 | | SM5540C |
| | Nitrate (as N) | 14797558 | Primary MCL | 10,000 | 2,000 | EPA 300.0 |
| | Nitrite (as N) | 14797650 | Primary MCL | 1000 | 400 | EPA 300.0 |
| | pH | | Basin Plan Objective | 6.5-8.5 | 0.1 | EPA 150.1 |
| | Phosphorus, Total (as P) | 7723140 | USEPA IRIS | 0.14 | | EPA 365.3 |
| | Specific conductance (EC) | | Agricultural Use | 700 umhos/cm | | EPA 120.1 |
| | Sulfate | | Secondary MCL | 250,000 | 500 | EPA 300.0 |

| CTR # | Constituent | CAS Number | Controlling Water Quality Criterion for Surface Waters | | Criterion Quantitation Limit ug/L or noted | Suggested Test Methods |
|---|---|---|---|---|---|---|
| | | | Basis | Criterion Concentration ug/L or noted[1] | | |
| | Sulfide (as S) | | Taste and Odor | 0.029 | | EPA 376.2 |
| | Sulfite (as $SO_3$) | | No Criteria Available | | | SM4500-SO3 |
| | Temperature | | Basin Plan Objective | $^oF$ | | |
| | Total Disolved Solids (TDS) | | Agricultural Use | 450,000 | | EPA 160.1 |

FOOTNOTES:

(1) - The Criterion Concentrations serve only as a point of reference for the selection of the appropriate analytical method. They do not indicate a regulatory decision that the cited concentration is either necessary or sufficient for full protection of beneficial uses. Available technology may require that effluent limits be set lower than these values.

(2) - Freshwater aquatic life criteria for metals are expressed as a function of total hardness (mg/L) in the water body. Values displayed correspond to a total hardness of 40 mg/L.

(3) - For haloethers

(4) - Freshwater aquatic life criteria for ammonia are expressed as a function of pH and temperature of the water body. Values displayed correspond to pH 8.0 and temperature of 22°C.

(5) - For nitrophenols.

(6) - For chlorinated naphthalenes.

(7) - For phthalate esters.

(8) - Basin Plan objective = 2 ug/L for Salt Slough and specific constructed channels in the Grassland watershed.

(9) - Criteria for sum of alpha- and beta- forms.

(10) - Criteria for sum of all PCBs.

(11) - Mercury monitoring shall utilize "ultra-clean" sampling and analytical methods. These methods include:

Method 1669: Sampling Ambient Water for Trace Metals at USEPA Water Quality Criteria Levels, USEPA; and

Method 1631: Mercury in Water by Oxidation, Purge and Trap, and Cold Vapor Atomic Fluoresence, USEPA

## III. Additional Study Requirements

**A. Laboratory Requirements.** The laboratory analyzing the monitoring samples shall be certified by the Department of Health Services in accordance with the provisions of Water Code 13176 and must include quality assurance/quality control data with their reports (ELAP certified).

**B. Criterion Quantitation Limit (CQL).** The criterion quantitation limits will be equal to or lower than the minimum levels (MLs) in Appendix 4 of the SIP or the detection limits for purposes of reporting (DLRs) below the controlling water quality criterion concentrations summarized in Table I-1 of this Order. In cases where the controlling water quality criteria concentrations are below the detection limits of all approved analytical methods, the best available procedure will be utilized that meets the lowest of the MLs and DLR. Table I-1 contains suggested analytical procedures. The Discharger is not required to use these specific procedures as long as the procedure selected achieves the desired minimum detection level.

**C. Method Detection Limit (MDL**). The method detection limit for the laboratory shall be determined by the procedure found in 40 CFR Part 136, Appendix B (revised as of May 14, 1999).

**D. Reporting Limit (RL).** The reporting limit for the laboratory. This is the lowest quantifiable concentration that the laboratory can determine. Ideally, the RL should be equal to or lower than the CQL to meet the purposes of this monitoring.

**E. Reporting Protocols.** The results of analytical determinations for the presence of chemical constituents in a sample shall use the following reporting protocols:

1. Sample results greater than or equal to the reported RL shall be reported as measured by the laboratory (i.e., the measured chemical concentration in the sample).

2. Sample results less than the reported RL, but greater than or equal to the laboratory's MDL, shall be reported as "Detected, but Not Quantified," or DNQ. The estimated chemical concentration of the sample shall also be reported.

3. For the purposes of data collection, the laboratory shall write the estimated chemical concentration next to DNQ as well as the words "Estimated Concentration" (may shortened to "Est. Conc.). The laboratory, if such information is available, may include numerical estimates of the data quantity for the reported result. Numerical estimates of data quality may be percent accuracy (+ or – a percentage of the reported value), numerical ranges (low and high), or any other means considered appropriate by the laboratory.

4. Sample results that are less than the laboratory's MDL shall be reported as "Not Detected" or ND.

**F. Data Format.** The monitoring report shall contain the following information for each pollutant:

1. The name of the constituent.
2. Sampling location.
3. The date the sample was collected.
4. The time the sample was collected.
5. The date the sample was analyzed. For organic analyses, the extraction data will also be indicated to assure that hold times are not exceeded for prepared samples.
6. The analytical method utilized.
7. The measured or estimated concentration.
8. The required Criterion Quantitation Limit (CQL).
9. The laboratory's current Method Detection Limit (MDL), as determined by the procedure found in 40 CFR Part 136, Appendix B (revised as of May 14, 1999).
10. The laboratory's lowest reporting limit (RL).
11. Any additional comments.